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SusAlgaeFuel

Exploring the synergies between direct carbon-capture, nutrient recovery and next-generation purification technologies for cost-competitive and sustainable microalgal aviation fuels

List of Participants #@APP-FORM-HERIAIA@#

	Participant Name	Country	Type	Main Expertise/Contributions
1	University College Dublin (UCD)	IE	UNI	Coordinator, microalgae cultivation, biorefinery, simulations, pilot-scale system
2	A4F – Algafuel SA (A4F)	PT	SME	Microalgae cultivation, pilot-scale system
3	Bia Energy Ltd (BIAE)	IE	SME	Anaerobic digestion, pilot-scale system
4	IRIS Technology Solutions SL (IRIS)	ES	SME	Sensor technology, digital twin
5	Karlsruhe Institute for Technology (KIT)	DE	UNI	Biorefinery fractionation of microalgae
6	University of Cambridge (UCAM)	UK	UNI	Lipid-to-aviation-fuel conversion and purification, pilot-scale system
7	Fraunhofer Gesellschaft (UMSICHT)	DE	RTO	Residue-to-aviation-fuel conversion and purification, pilot-scale system
8	Sustainable Flight Solutions Ltd (SFS)	IE	SME	Aviation sustainability, social life-cycle, policy
9	European Science Communication Institute gGmbH (ESCI)	DE	SME	Bioeconomy science communication and dissemination
Confirmed Members of the Advisory Board: Boeing, ES; CityJet, DK.				

Executive Summary

Under the Fit-for-55 Mandate, the EU has set a highly ambitious goal to increase the share of Sustainable Aviation Fuels (SAFs) in the aviation industry from **2% in 2025 to 64% by 2050.** Microalgae can play a critical role in meeting this fuel target, directly capturing carbon dioxide and contributing to the development of circular bioeconomy through the recovery of waste nutrients from agri-food-bioenergy sectors (such as Anaerobic Digestion (AD)). **SusAlgaeFuel** aims to develop a series of innovative approaches that work in tandem to enable a more cost-competitive, efficient, and circular production of microalgae SAFs**.** These approaches include **a)** the **direct capture of CO2** emissions from biogas upgrading and utilisation of waste liquid digestate as a **low-cost nutrient source to support algae growth**, expanding the feedstock basis for microalgae while enabling the valorisation of AD by-products, b) development of a novel in-line process analytical technology (PAT) complemented with machine learning and selective UV irradiation **to monitor and purify bacterial contamination from algae culture**, preventing costly culture crash, **c)** the development of **a cascading biorefinery that relies on energy-saving autolysis (self-disruption) and maximises solvent recycling** in order to fractionate biomass into lipids (for jet fuel), protein serum (for feed) and cellulose-rich biomass residue (for further fuel conversion) at low energy and solvent requirements, and **d)** the development of algae-specific thermocatalytic pathways for **efficient conversion** of algae-lipids to Hydroprocessed Esters Fatty Acids-Synthetic Paraffinic Kerosene (HEFA-SPK) and residue to kerosene followed by a range of **purification methods** for fuel upgrade to meet **international aviation standards**.

Technology development will iteratively be complemented with process simulations, techno-economic and life-cycle analyses to assess scalability from economic, social and environmental perspectives and identify process improvements, while a dedicated commercialisation plan and policy recommendations will be produced to guide future technology transfer from the lab to the industry. SusAlgaeFuel will culminate in the **building and operation of a pilot-scale algal facility on an AD operator site in Ireland** (based on the approaches developed in the project) with the capacity to directly capture CO_2 from AD flue gas, utilise waste digestate and produce up to 10 kg of algae lipids per year. SusAlgaeFuel can have a transformative and multifaceted impact on **EU energy independence**; if successfully scaled at EU+UK level, the process has the potential to annually treat **172 Mt digestate per annum** and deliver **1 Mt of HEFA-SPK per annum (or 20% of Europe's projected SAF requirements of 5 Mt in 2030)** at a **reduced cost** per kg of algal HEFA-SPK of 6.30$ (a decrease of 49% from estimated 12.30$). This tremendous potential has attracted the attention of **Boeing and CityJet**, who have committed to act as advisors to the project and position themselves as **early adopters**.

1. EXCELLENCE #@REL-EVA-RE@#

1.1. Objectives and Ambition #@PRJ-OBJ-PO@#

1.1.1. Rationale

The EU has set an ambitious target to increase the proportion of SAFs in the aviation industry from 2% in 2025 to 64% by 2050 (Fit for 55)[1]. This is in line with the goal outlined in the REDII Directive which aims to increase the

[1] EASA. Sustainable Aviation Fuels. 2022

share of 3.5% of advanced biofuels in transport to a minimum of 32% by 2030 and to reduce GHG emission by 40% by 2030[1]. However, currently only a mere 0.05% of aviation fuel in Europe are SAFs, the majority of which has been manufactured from second-generation feedstocks (such as Used Cooking Oil UCO) using a HEFA-SPK conversion pathway[1]. Limited availability of reliable domestic feedstock and a significant price difference compared to conventional jet fuel ($1.9–7.7/kg vs. $0.7–1.4/kg[1]) due to costly production system remain a critical entry barrier. To meet its goal in reducing reliance on jet fuel imports and achieving energy independence, the EU will need to significantly and rapidly expand its SAF production. Thus, there is a pressing need for the development of a domestic feedstock that can be produced within the EU without displacing agriculture resources and a more efficient production technology that can achieve cost competitiveness with conventional jet fuels at commercial levels.

Microalgae can play a critical role in meeting the EU's SAF target, having the ability to accumulate lipids that can be subjected to proven conversion pathways (such as HEFA-SPK) for production of drop-in fuels and the adaptability to be grown across different regions in the EU. In addition to having no Indirect-Land-Use-Change (ILUC) impact because they do not compete for agricultural resources (e.g. arable land and fresh water), microalgae can capture CO_2, treat wastewater, and synthesise valuable co-products (e.g. proteins) for the circular bioeconomy.

The production of microalgae advanced biofuels, however, has been hampered by a range of systemic and technological constraints which limit supply and increase production cost, with estimates of the cost of algae HEFA projected to be as high as 11.41 $/kg[2]. This is multiple fold higher than the 2030 target of 0.49 $/kg (35 €/MWh) set by the European Commission's Strategic Energy Technology (SET) plan Action 8. To be commercially competitive, the cost of microalgae-based jet fuel will have to at least match those of commercial SAFs in the range of 1.9 – 7.7 $/kg SAF[1]. Some of the most critical constraints preventing the commercialisation of microalgae biofuels are:

a) In cultivation: the lack of access to cheap nutrients and readily accessible CO_2 for microalgae cultivation, leading to high cultivation cost and limited feedstock availability,
b) In culture monitoring and purification: difficulty in monitoring and managing the level of bacterial contamination in algal culture, particularly when using wastewater as a medium for large-scale growth, leading to culture crash,
c) In biorefinery and biomass purification: high energy requirements associated with breaking tough algal cell walls for releasing intracellular products and high solvent requirements for lipid extraction, leading to costly biorefinery,
d) In fuel conversion and purification: limited understanding of the thermochemical routes for the conversion of microalgae lipids to energy-dense aviation fuels and the required purification steps for production of high-quality fuels, leading to a low fuel conversion efficiency,

Disruptive approaches which address these critical challenges in each process step while also adopting a full systems perspective through integrated techno-economic and life-cycle assessments are urgently needed to reduce the cost of microalgae biofuels and increase feedstock basis of microalgal SAFs.

1.1.2. Objectives

SusAlgaeFuel aims to develop a set of innovative approaches in microalgae biofuel production to enable a more cost-competitive, efficient, and circular production of microalgae SAFs, overcoming the aforementioned constraints:

a) Cultivation of microalgae using cheap and abundantly available by-products from **anaerobic digestion (AD)**, including the utilisation of digestate as N and P source and direct capture of CO_2 emission from biogas upgrading.
b) Development of state-of-the-art optical sensors for in-line monitoring of bacterial contamination in the culture.
c) Development of autolytic cell disruption capable of inducing self-weakening in tough microalgae cell walls and reducing energy requirements associated with biorefinery fractionation, and development of solvent system that allows for facile separation and solvent recycling to reduce solvent requirements.
d) Development of multi-catalytic pathways for the efficient conversion of microalgae lipids to HEFA-SPK and a series of targeted purification methods of the resulting fuel molecules based on the fuel's carbon distribution.

The biorefinery approach used in SusAlgaeFuel will also result in the production of protein-rich serum and residue from microalgal biomass that can be utilised for feed application (e.g. as soy replacement/additive for pig farming) and further fuel conversion through thermo-catalytic reforming + hydro-deoxygenation (TCR-HDO) pathway respectively (**Figure 1.1**), adding both commercial and fuel value to the production train. Overall, the SusAlgaeFuel process will enable the production of four distinct product streams (**Figure 1.1**): HEFA-SPK from microalgal lipid (as primary product), a TCR-HDO fuel from biomass residue (as second jet fuel), protein serum from biorefinery (as feed by-product), and solid digestate from phase separation of digestate feed (as biofertilizer by-product). SusAlgaeFuel will validate technology development with integrated process simulations, techno-economic, and life-cycle analyses, identifying process shortcomings and enabling continuous feedback for process improvement.

The project will focus its investigation on three oleaginous photosynthetic microalgae species with proven ability to achieve high biomass concentration (up to 4.8 g biomass/L) and to accumulate high level of biofuel lipids (up to 37 wt% of biomass) when cultivation on digestate **(Table 1.2a)**: *Chlorella pyrenoidosa, Auxenochlorella protothecoides,* and *Nannochloropsis salina* (please refer to **Section 1.2.1** for rationale on species selection).

[2] DOI: 10.1016/j.biombioe.2020.105881

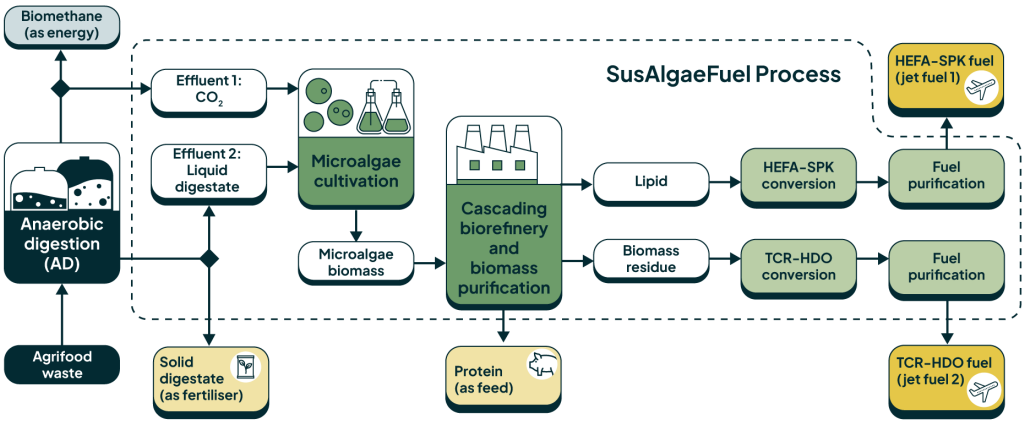


Figure 1.1: Schematics of SusAlgaeFuel Process

The objectives of SusAlgaeFuel (which will be fulfilled through our cohesive work plan) are:

1. **Objective 1 (WP1): To investigate strategies to capture biogenic CO_2 from biogas upgrading and recover waste nutrients from liquid digestate (N, P) for microalgae growth resulting in high biomass productivity and lipid accumulation.**

a) To characterise liquid digestate and develop necessary pre-treatments to formulate a cultivation medium that minimises/overrides the need for artificial nutrients (Task 1.1).
b) To develop and optimise high-density autotrophic cultivation strategies for oleaginous species *C. pyrenoidosa, A. protothecoides,* and *N. salina* on liquid digestate (Task 1.2).
c) To develop a novel membrane system for the delivery of CO_2 into aqueous solution, with the view of using the membrane system for effective direct capture of biogenic CO_2 from biogas upgrading (Task 1.3).
d) To integrate the membrane system (for CO_2 delivery) with cultivation strategies on digestate and optimise the hybrid system to attain biomass concentration > 3 g biomass/L and lipid content >30 wt% of biomass (Task 1.4).

2. **Objective 2 (WP2): To develop effective strategies for in-line monitoring and management of bacterial contamination/level in algal culture.**

a) To develop rapid, modular and affordable *in-situ* system that combine the principles of advanced spectroscopic methods in a single platform coupled with machine learning algorithm to distinguish and quantify the level of bacteria in microalgae culture (Task 2.1), **leveraging the multirange platform developed at IRIS.**
b) To develop a selective UV irradiation system as *in-situ* methods for enhancing microalgal growth and reducing/managing bacterial level (Task 2.2) as analysed by the spectroscopic methods developed in Task 2.1.
c) To study algae-bacteria interaction and refine cultivation strategies/UV irradiation system to favour the growth of symbiotic/beneficial bacteria and to suppress the growth of harmful bacteria (Task 2.3).

3. **Objective 3 (WP3): To develop a cascading biorefinery for low-energy and low-solvent fractionation of wet microalgae biomass into lipids for HEFA-SPK conversion, protein-rich serum for feed application, and biomass residue for thermos-catalytic fuel conversion.**

In WP3, we aim to significantly reduce energy and solvent requirements of biomass fractionation through autolysis cell disruption and biphasic extraction system (which maximises solvent recycling and mitigates emulsions) of wet microalgae slurry (100 – 200 g biomass/kg slurry).

a) To investigate and optimise low-energy cell disruption based on autolysis for direct fractionation of wet microalgae slurry into biofuel precursors and protein (target energy demand for cell disruption < 1 MJ/kg biomass). **The project will leverage recent discoveries in autolysis made by PI Halim at UCD (Task 3.1).**
b) To screen extraction solvents (including bio-based solvents) and optimise extraction conditions for maximum recovery of lipids (target lipid recovery > 80% of available lipids) at minimum solvent requirements (target solvent usage < 3.5 L/kg biomass). Strategies to mitigate solvent emulsion will be investigated (Task 3.2).

4. **Objective 4 (WP4): To develop and optimise catalytic pathways for high-efficiency conversion of microalgae biomass components (lipids and biomass residue) to kerosene mixture and to develop purification methods for the production of high-quality SAFs capable to achieving aviation specifications.**

a) To develop hydrotreatment and hydrocracking pathways for conversion of microalgae lipids into HEFA-SPK capable of achieving >50% conversion through the use of tailored catalytic composites comprising transition metals (Ni-Mo-Co-Cu-Fe) and mild/medium acidic supports (Task 4.1).
b) To couple the HEFA-SPK catalytic pathways with **a flagship chemical looping process developed at UCAM** to minimise hydrogen consumption and investigate heat-integration opportunities (Task 4.2).
c) To investigate the application of novel thermo-catalytic reforming + hydro-deoxygenation (TCR-HDO) pathways **patented by Fraunhofer UMSICHT** for conversion of microalgae biomass residue to kerosene fuel (Task 4.4).
d) To develop purification methods for the resulting HEFA-SPK and TCR-HDO fuel molecules to enable the production of high-quality aviation fuels meeting ASTM specifications (Task 4.3 and Task 4.5).

5. **Objective 5 (WP5): To continuously assess the commercial scalability, well-to-tank social, environmental & GHG-emission impacts of the investigated microalgae fuel production strategies to enable iterative optimisation**

a) To use process simulations for the investigated process to assess commercial scalability of the value chain, identification of major cost drivers and further optimization opportunities (Task 5.1).
b) To carry out a comprehensive Whole Life Cycle Analysis (WLCA) utilising innovative socio-technical system approach (STS) at the complete system level to fully assess the social effects, environmental impact and sustainability of the proposed algae-fuel value chain, and to identify socio-environmental hotspots (Task 5.2).
The techno-economic and LCA results will provide technology development in WPs 1-4 with iterative feedback and the SusAlgaeFuel team with the most optimal option for scaling up during pilot production in WP6.

6. **Objective 6: To demonstrate the technical scalability of SusAlgaeFuel process (at TRL5) through the build, commission, and operation of a pilot-scale system on BIAE site with the capacity for direct capture of CO_2 from biogas upgrade, treatment of liquid digestate and production of microalgal lipids.**

a) To build a pilot-scale algal production facility comprised of optimum cultivation and biorefinery unit operations developed in WP1-WP3 with the capacity of producing 10 kg of algal lipids p.a. (Task 6.1).
b) Using the optimum species found in WP1, to operate and optimise the pilot system under semi-continuous strategy (for 2 -3 months) to produce sufficient lipids for the manufacture of 1 kg of HEFA-SPK (Task 6.2).
c) Using the conversion and purification technologies developed in WP4, to produce 1 kg of HEFA-SPK from the lipids and 1 kg of TCR-HDO oil mixture from the cellulose-rich biomass residue obtained in Task 6.2, and to test the resulting fuels for compliance with ASTM standards. (Task 6.3).
d) To develop a digital twin of the integrated system as a decision toolbox to guide future stakeholders (Task 6.4).

7. **Objective 7: To implement successful strategies for dissemination, communication and exploitation.**

a) To develop and implement a comprehensive communication strategy to reach stakeholders (Task 7.1-7.3).
b) To disseminate impactful results, including workshops on pilot-scale infrastructure built in WP6 (Task 7.1-7.3).
c) To collaborate with other EU initiatives in the same area to share knowledge and boost outreach (Task 7.1-7.3).
d) To manage IPs and explore exploitation pathways for commercial utilization of results (Task 7.4).
e) To produce policy recommendations for algae usage in aviation fuels at national and EU levels (Task 7.5).

A primary innovation that SusAlgaeFuel will bring about to enhance the sustainability, circularity and feedstock basis of microalgae SAFs is the utilisation of AD by-products (liquid digestate and CO_2 emission from biogas upgrading process) as nutrients for microalgae growth, linking the two sectors together in a win-win synergy which provides mutual benefits (by-products valorisation for AD and a cheap and abundant nutrient supply for microalgae). AD harnesses the power of microorganisms to convert agricultural and food waste to biomethane. The process is expected to play a critical role in EU's long-term pursuit for energy independence and decarbonisation, as one of the few technologies able to supply drop-in replacement for natural gas, the supply for which the EU is so heavily reliant on imports. The AD process, however, generates two principal by-products for which there is either no outlet for utilisation (i.e. CO_2-rich gas effluent from biogas upgrade to form biomethane) or the current pathway for valorisation yields a low-value product (i.e. full digestate as very dilute fertilisers) with large distribution costs due to the EU Nitrates Directive (91/676/EEC) resulting in significant costs to AD operators. The process envisaged in SusAlgaeFuel will enable the development of a green strategy for nutrient recovery from both AD by-products, leading to the production of nutrient-dense biofertilisers (from digestate's solid fraction after phase separation) and microalgae biomass (from digestate's liquid fraction and CO_2 effluent).

The technologies investigated in SusAlgaeFuel have the potential to generate step reduction in the cost of algal SAFs. This will be achieved through a series of cost-cutting innovations in each processing step, including:

- cost of nutrients during cultivation (by using AD by-products instead of commercial CO2 and fertilisers)
- cost of energy during biorefinery (by investigating cell disruption that relies on microalgae self-lysing properties)
- cost of solvent during biorefinery (by using a solvent system that enables facile recovery and maximum recycling),
- cost of hydrogen during HEFA-SPK conversion (by developing a flagship chemical looping process).

SusAlgaeFuel will adopt a staggered approach, where technological development/optimisation at bench scale will take place in the first three years of the project before **being transferred into the building and commissioning of a pilot-scale system on an AD operator site** (production capacity = 10 kg lipids/year) on the BIAE site in Ireland in the final year. The scale up is intended to demonstrate technical scalability (up to TRL 5) while also generating sufficient final fuel products fuel-quality assessment and certification according to ASTM standards.

1.1.3. Beyond State-of-the-Art

SusAlgaeFuel will introduce **nine** innovative approaches/technologies which individually push beyond the current state-of-the-arts at different stages in microalgae value chain (cultivation WP1, culture purification WP2, biorefinery/biomass purification WP3 and fuel conversion/purification WP4).to produce a cohesive system for a more cost-competitive, efficient, and circular production of microalgae SAFs. These approaches will be developed with an integrated techno-economic and life-cycle analysis (WP5) to facilitate improvement and scale-up (WP6).

A novel membrane system for an effective delivery of CO_2 to microalgae culture with high carbon utilisation efficiency or CUE (>90%), enabling direct capture of CO_2 from biogas upgrade (Objective 1//WP1).
Current state: All conventional methods to directly deliver gaseous CO_2 into microalgae culture involve the conversion of CO_2 into bubbles (e.g. sparging, microbubblers, and sumps[3],[4]). The short residence time (0.022 – 55s at 0.3 m pond depth) associated with bubble delivery means that significant amounts of CO_2 escape into the atmosphere before they can be captured by microalgae via photosynthesis, leading to a low CUE and higher energy cost associated with CO_2 delivery. *Advancement:* In WP1, we will investigate a novel CO_2 delivery system using dense membrane to achieve bubbleless CO_2 delivery into microalgae culture (and thus high utilisation efficiency), enabling effective CO_2 capture from direct point source. System will be validated with CO_2 mass transfer model.

A cultivation system capable of direct capture of CO_2 emissions from biogas upgrade and digestate treatment to achieve high biomass concentration (Objective 1//WP1).
Current state: Microalgal cultivation on full digestate can often lead to a highly turbid medium which requires significant dilution to support autotrophic growth. Using the full digestate as growth medium also precludes the application of excess nutrients as valuable biofertilisers. SusAlgaeFuel, will focus on using only the supernatant obtained after phase separation (referred to as 'liquid digestate') for algal growth, enabling the formation of nutrient-dense biofertilizer from the solid fraction of digestate after phase separation (or 'solid digestate') as a co-product (**Figure 1.1**). *Advancement:* In WP1, we will establish a comprehensive strategy for supporting algal growth on liquid digestate by a) using a multi-step filtration process at different pore sizes to remove solid particles and bacteria from liquid digestate and b) employing extensive chemical characterisation to determine minimum dilution and requirements of micronutrients addition. Once growth on liquid digestate has been optimised, we will combine the growth strategy with the CO_2 membrane delivery system, enabling the development of a hybrid cultivation system that is able to directly capture CO_2 emission while recovering nutrients from digestate to achieve high biomass productivity (target final biomass concentration > 3 g biomass/L) and lipid content (target > 30wt% of biomass).

A-first-of-a-kind multirange spectroscopic sensor platform capable of bacterial detection and enumeration in microalgae culture (Objective 2//WP2).
Current state: The presence of undesired bacteria in microalgal culture can often lead to costly culture crashes. There is currently no known in-line monitoring system able to analyse bacterial population dynamics in microalgae culture *in-situ* and provide early warning for catastrophic contamination level to enable corrective actions; such a detection system is particularly critical for culture grown on wastewater or nutrient-rich waste products (e.g. liquid digestate). *Advancement:* In WP2, we will leverage an innovative Multirange Spectroscopic Platform developed at **IRIS** to investigate a modular, rapid, and affordable sensor system able to distinguish bacterial cells from microalgae cells and enumerate the level of bacterial contamination. The fusion platform will simultaneously utilise SWIR, NIR; VIS, Raman; and Fluorescence spectroscopy and be coupled with machine learning to enable future automation.

A bacterial management system that uses selective UV irradiation (Objective 2//WP2).
Current state: Recent studies have shown that microalgal cells and bacterial cells can react differently to UV irradiation, with the same treatment often resulting in bacterial inactivation while maintaining algal growth (and often boosting lipid accumulation)[5]. Such differential acclimatisation can potentially be exploited as means of managing bacterial population in microalgal culture, particularly those grown on waste products or nutrient-rich waste medium. *Advancement:* In WP2, we will develop a targeted 2-stage UV illumination regime for the investigated microalgae species under different wavelengths to a) sterilise growth medium (as a complement to filtration systems) and b) favour the growth of microalgae cells and symbiotic native bacteria in the culture over harmful bacteria to enable productivity boost. The effectiveness of the UV system will be validated with the developed spectroscopic platform.

A first-of-a-kind biorefinery which harnesses microalgal intrinsic capacity for <u>self-weakening (or autolysis)</u> for the development of low-energy biomass fractionation (Objective 3//WP3).
Current state: Biomass drying is highly energy intensive with the amount of energy required for complete water removal (10.9 MJ/kg biomass) exceeding the full lipid-energy potential of the biomass (10.1 MJ/kg biomass for biomass containing 30 wt% lipids). Drying must therefore be avoided to ensure a positive energy balance for fuel application. Wet biorefinery systems, however, require cell disruption which is either chemical intensive (e.g. enzymatic or chemical) or energy intensive (e.g. mechanical methods). Autolysis, a self-decomposition process where cells use their own endogenous enzymes to degrade/weaken their cell walls, can be coupled with externally applied cell disruption to reduce the intensity of these methods. *Advancement:* In WP3, we will leverage the discovery that **PI Halim (UCD)** recently made on the expression of autolysis in highly concentrated microalgae slurry (100 – 200 mg biomass/g slurry) to develop a novel biorefinery system able to fractionate biomass into lipid, protein and carbohydrate at low energy requirements (target energy demand for disruption < 1 MJ/kg biomass).

[3] DOI: 10.1016/j.biortech.2015.12.066
[4] DOI: 10.1016/j.cjche.2018.07.013
[5] DOI: 10.1007/s10811-018-1687-8

A lipid extraction strategy that minimises solvent usage through the use of non-polar solvents, emulsion mitigation and optimisation of solvent recycling (Objective 3/WP3)**.**
Current state: Our calculations show that the direct purchase cost of solvent and the indirect cost of energy associated with solvent recovery can account for up to 42% the operating cost of a microalgal biorefinery. Optimal solvent choice that allows for maximum lipid extraction at minimum solvent usage is therefore critical in achieving commercial viability of algal fuels. Polar solvents (e.g. ethanol, isopropanol) form monophase with water during extraction from wet biomass and are difficult to recover, leading to high solvent requirements. *Advancement:* In WP3, we will screen a range of non-polar. and low-polar solvents (e.g. hexane, cyclohexane, ethyl-acetate) for selective extraction of HEFA-SPK convertible lipids, and develop strategies to reduce solvent usage through mitigation of emulsion and optimisation of solvent recycling (target solvent usage < 3.5 L/kg biomass).

A flagship microalgal-specific multifunctional catalytic pathways using transition metals and H_2 looping route for efficient hydrotreatment conversion of microalgae lipids into HEFA-SPK (Objective 4/WP4)**.**
Current state: The conversion pathways for algae lipids to HEFA-SPK are not well understood, with early reports indicating accelerated catalyst aging compared to HEFA routes with vegetable oils and low SPK yields. Since microalgal lipids contain a higher fraction of oleic acids (C18:1) compared to plant oils, they are not very reactive to Pt-based catalysts. *Advancement:* In WP4, we will develop a novel algae-specific bifunctional catalyst using transition metals (Ni-Mo-Co-Cu) on mild/medium-acidic supports (e.g. mesoporous zeolites) to improve conversion efficiency and introduce **a flagship chemical-looping process** (recently developed at **UCAM**) to minimise hydrogen.

A flagship microalgal-specific thermocatalytic reforming + hydro-deoxygenation (TCR-HDO) treatment for the conversion of microalgae biomass residue to kerosene mixture suitable for aviation (Objective 4/WP4).
Current state: Pyrolysis of biomass (like algae) generates a dark brown oil consisting of more than 100 different compounds, including oxygenated compounds (e.g. aldehydes or carbonic acids), nitrogenated cyclic compounds, and sulfonated compounds. This oil mixture is not suitable for engine application, in addition to the exhausts of sulphur and nitrogen oxides. In contrast to conventional pyrolysis techniques, TCR produces a thermally stable oil that can then be subjected to hydrogenation or HDO (at 400 °C and 120 bars) to remove the hetero atoms O, N, S in form as H_2O, H_2S, NH_3 and generate saturated olefines. The resulting product is a mixture of hydrocarbons comprised of diesel, kerosene and petrol fractions that can conform to ASTM standards upon purification. *Advancement:* In WP4, we will demonstrate for the first time that the **TCR-HDO configurations patented by Fraunhofer UMSICHT (**WO2015158732A1) can be applied for the conversion of cellulose-rich algae biomass residue to fuel.

Tailored purification methods for algal fuel molecules to generate SAFs that meet international standards. (Objective 4/WP4).
Current state: SAFs must fulfil ASTM standards to ensure safe operations during flight. Fuel properties directly correlate to composition and the level of impurities. The amounts of impurities can be minimised through refining steps and fuel purification. *Advancement:* In WP4, we will investigate the combination of possible fuel upgrading steps that are most optimal for the algal-derived SAFs. Our goal is to correlate the key physical properties (thermal stability, volatility) with carbon distribution in the fuel components and the content of inorganics to define a minimum set of purification steps. Separate purification pathways for HEFA-SPK and TCR-HDO oils will be investigated

Continuous assessment of the commercial scalability and environmental impact of technologies being developed using process simulations and the latest approach in Whole Life Cycle Analysis (Objective 5/WP5)
In WP5, we will complement technology development in WPs1-4 with continuous and iterative use of process simulations (e.g. Aspen Hysys) to assess commercial scalability of the value chain and identify major cost driver and a comprehensive Whole Life Cycle Analysis (WLCA) utilising innovative socio-technical system approach (STS) at the complete system level to fully assess the social and environmental sustainability of the value chain.

A fully integrated prototype pilot-scale system on an AD operator site as demonstration of technical scalability for SusAlgaeFuel process with the capacity for the production of 10 kg lipids/year and direct carbon capture and digestate remediation from the AD plant (TRL4/5) (Objective 6/WP6)**.**
SusAlgaeFuel will culminate with the building and commissioning of a pilot-scale facility on BIAE site that integrates the technologies developed/investigated at bench scale throughout WP1-WP4 as a proof-of-concept of process scalability beyond project lifetime. The pilot-scale facility will be operated for a minimum of 3 months to produce sufficient algal HEFA-SPK and TCR-HDO fuels for testing against international fuel standards. The pilot-scale facility will serve as an exploitation tool for the project; where we will showcase the technologies to stakeholders and investors via on-site workshops.

INDUSTRY4.0 for the manufacture of algal SAFs: The concepts of data-driven analysis and next generation computer technology are deeply embedded in SusAlgaeFuel technological development, with the view of preparing microalgae productions for an autonomous and AI-driven manufacturing in the future. These include the integration of machine learning algorithms in the development of spectroscopic biosensor platform (WP2), use of multivariant correlations supported by machine learning algorithm for deconvolution of the possible fuel conversion pathways (WP4) and the development of a digital twin inegrating real-time data from pilot-scale prototype (WP6).

1.1.4. Project Maturity

Key Technologies	WP	Pre-project TRL	Post-project TRL
Autotrophic growth strategy on liquid digestate	1	3	4 -5 (via pilot scale up)
Membrane system for CO_2 delivery to microalgae	1	2	4 - 5 (via pilot scale up)
In-line sensor to monitor bacteria in culture	2	2	4 - 5
Selective UV irradiation for culture purification	2	2	4 - 5 (via pilot scale up)
Autolytic cell disruption for low-energy fractionation of wet microalgae biomass	3	2	4 - 5 (via pilot scale up)
Lipid extraction strategy to minimise solvent	3	2 – 3	4 – 5 (via pilot scale up)
Multifunctional catalytic process with H_2 looping route for conversion of algal lipids to HEFA-SPK	4	1-2	4 – 5 (via pilot scale up)
Algae-specific HEFA-SPK purification method	4	1	4
Process for the conversion of microalgae biomass residue to TCR-HDO oil	4	1 – 2	4 – 5 (via pilot scale up)
Algae-specific TCR-HDO purification method	4	1	4

Table 1.1: The Maturity of SusAlgaeFuel Technologies #§PRJ-OBJ-PO§#

1.2. Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

1.2.1. Approach

HEFA-SPK pathway vs FT-SPK or ATJ-SPK pathways
Microalgae biomass can be converted to jet fuels through various pathways, including full-biomass gasification routes (FT-SPK or ATJ-SPK). The HEFA pathway offers a commercially mature route for short-term deployment to address immediate fuel shortage and has thus been selected as SusAlgaeFuel's primary approach.
Route to Market: Existing commercial-scale HEFA facilities can readily accept raw algae-derived lipids, streamlining the manufacturing process. **EU SAF Mandates:** The European Commission recognizes HEFA for Sustainable Aviation Fuel (SAF), where algae-derived lipids could supplement constrained UCO and tallow supply. **Valuable Co-products:** Algae HEFA-SPK biorefinery route leads to generation of valuable co-products, such as high-quality proteins and biofertilisers, that can offset cost of fuel production rather. **Fuel quality:** HEFA-SPK pathway produces jet fuel with high cetane number and low sulphur content.

Overview: SusAlgaeFuel is divided into **eight WPs** designed to complement each other in delivering an integrated solution for microalgae jet fuel production. In the first phase of the project (M1–36), WPs 1-4 will focus on overcoming critical bottlenecks in microalgae fuel production (as outlined in Section 1.1) through the development of scalable core technologies for three lipid-rich microalgae species (*C. pyrenoidosa, A. protothecoides, N. salina*). The results from bench experiments in WPs 1-4 will then be transferred to the build of a pilot-scale facility in the second phase of the project (WP6 'Pilot scale-up, fuel testing and digital twin', M30-48) to provide demonstration of the technical scalability of the investigated process and generate high-quality fuels for ASTM quality analysis. The number of species and technological options deployed in WP6 will be narrowed down according to the results of WPs1-4 and the feedback provided by WP5 'Process integration and whole life-cycle analysis'. WP5 runs in parallel to core technology development and pilot scale-up throughout project lifetime, using results from the respective WPs to continuously carry out process integration and environmental assessments and relaying these results back to the WPs in a circular loop to enable iterative optimisation. Consolidated results obtained in WP5 will then be used to guide decisions and inform best practice/parameters for scale-up in WP6. To provide clear exploitation pathways from the lab to industry, WP7 'Communication, policy and exploitation roadmap' will develop a business/commercialisation plan (that takes into account inputs from Stakeholders within the consortium and wider sectors) and policy guidance, both rooted in the EU market. WP8'Project coordination' will tie all scientific, administrative and financial activities together while identifying risk and implementing mitigation measures.

WP1: Membrane system for efficient CO_2 delivery to microalgae, enabling direct CO_2 capture AD

Challenges facing direct CO_2 capture from point emission by microalgae. The capacity for carbon capture from a point emission source is one of the key benefits of microalgae cultivation. Conventional methods to directly deliver gaseous CO_2 into aqueous microalgae suspension (e.g. sparging, microbubblers, and sumps), however, involve the conversion of the CO_2 into bubbles (1–5 mm in diameter) or microbubbles (0.02–0.1 mm). A primary drawback of bubble-based delivery is its short residence time in aqueous suspension (0.022–55s at 0.3 m pond depth)[3,4] meaning that the majority of CO_2 delivered into the culture from point source is lost to the atmosphere before it can be captured by the microalgae via photosynthesis. This leads to a low Carbon Utilisation Efficiency (defined as the proportion of delivered C_{CO2} that is converted to biomass) and high energy cost associated with gas compression to deliver excess CO_2. CUE averages at 40% for bubbles and 70% for microbubbles[4].

Membrane system for CO_2 delivery. Recent studies have reported an alternative delivery system using non-porous membranes to increase CUE[3],[4],[6],[7]. In these studies, a composite membrane with a thin dense core membrane layer was employed to enable the diffusion of individual CO_2 molecules from vapour state ($CO_{2(g)}$) into the aqueous microalgae suspension in bubbleless form ($CO_{2(aq)}$)[3], [7], [8]. (**Figure 1.2a**) The approach was able to achieve >90% CUE and resulted in 15-20x increase in biomass productivity compared to sparging. Membrane carbonation, however, is still in its infancy and has so far only been investigated for the delivery of pure CO_2 into microalgae culture. CO_2 effluent from AD is a mixture comprised not only of CO_2 (~85 vol%) but also other constituents (such as CH_4) and will thus require a different membrane system to retard CH_4 permeation and selectively transmit CO_2[9]. The lack of optimisation of the membrane system and mass-transfer models to describe the transport of CO_2 across the membrane and then diffusion into the microalgae cells have made it difficult to predict system performance and control CO_2 flux in the system.

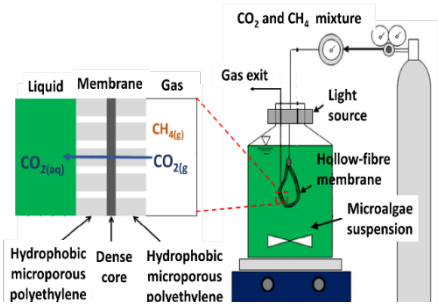


Figure 1.2a: Schematics of integrated membrane delivery / microalgae growth system

Development of membrane system for efficient CO_2 delivery to microalgae. In WP1 (Task 1.3), we will investigate the development of a novel membrane system for the efficient delivery of CO_2 into microalgae culture. The system will be optimised in terms of membrane types, membrane configurations, CO_2 partial pressure and pH of the solution to achieve high CUE (> 90%). The performance of the CO_2 delivery system will initially be assessed for aqueous solution (without microalgae) to understand the CO_2 dissolution/equilibrium process. In WP1 (Task 1.4), the CO_2 delivery system will be integrated into a microalgae cultivation on digestate, allowing for complex interaction between CO_2 delivery, CO_2 uptake by photosynthesis, and nutrient utilisation by microalgae to be characterised. The development will leverage existing research on microalgal CO_2 capture system at UCD for the project 'EirAlgae'. Mass transfer model to describe the kinetics of carbon transport across the membrane and subsequent utilisation by microalgae will be developed based on first principles, validated using experimental data and solved using MATLAB to facilitate performance forecast during scale up.

WP1: Autotrophic growth strategy on liquid digestate and direct capture of CO_2 emission from AD

Species selection. The three species we have selected, *C. pyrenoidosa, A. protothecoides, N. salina*, are well-known lipid accumulators, able to achieve a lipid content of up to 37 wt% of biomass when grown on nutrient-rich digestate. Findings obtained for *N.salina* are directly transferrable to other oleaginous species within the genus.

Critical barrier to achieving low-cost cultivation for biofuels. Large-scale autotrophic cultivation of microalgae for biofuel application continues to suffer from a critical barrier: a) the high cost of commercial nutrients which in turn increases cultivation cost (OPEX) and limits biomass availability, and b) slow onset of lipid accumulation resulting in sub-optimal biomass lipid contents which in turn increase subsequent costs of cultivation and biorefinery. Triglyceride accumulation is achieved under N starvation which forces cells to funnel photosynthetic carbon into lipid synthesis. Light attenuation due to self-shading effect in algal culture and CO_2 mass-transfer limitation associated with sparging, however, can negatively affect photosynthetic rate and thus limit lipid accumulation.

Table 1.2a: Performance of autotrophic cultivation of investigated species on digestate

Species	Dilution ratio	Final biomass concentration (g biomass/L)	Lipid content (wt% of dry biomass)	Refs
C. pyrenoidosa	1 – 8x	0.3 - 4.8	9.6 – 22	**13, 16**
A. protothecoides	1 – 4x	0.5 - 3.5	22.2 - 36.3	**11, 14, 15, 17**
N. salina	4 - 33x	0.7 - 0.9	20 – 35	**12**

Utilisation of AD effluents and overview of process development. To address the high cost of nutrients, SusAlgaeFuel will examine the utilisation of both AD effluent streams to support microalgae growth: digestate's liquid fraction (as N and P source) and biogenic CO_2 emission. The full scope of investigation on AD waste will adopt a multi-stage approach, with Task 1.1 and 1.2 (WP1) concentrating on cultivation strategies on digestate, and Task 1.3 (WP1) developing a membrane system for CO_2 delivery. Once the individual methods have been developed, they will both combined in single hybrid system to achieve both digestate utilisation and CO_2 capture (Task 1.4, WP1). All initial technological development in WP1 will take place at bench scale (and use commercial CO_2 supply as proxy for biogenic CO_2 emission). In WP6, the hybrid system will be scaled up through the build and operation of pilot-scale infrastructure on AD operator site for which real biogenic CO_2 emission from AD plant will be used for CO_2 capture.

[6] DOI: 10.1016/j.algal.2020.102070
[7] DOI: 10.1016/j.memsci.2019.117389
[8] DOI: 10.1016/j.jcou.2019.06.008.
[9] DOI: 10.1016/j.biortech.2019.121804

Development of growth strategies on digestate. Liquid digestate contains NH_4^+-N (139-3,456 mg/L), PO_4^{3-}-P (7-381 mg/L), and micronutrients (596-5,905 mg/L)[10] to support microalgae growth. Previous studies have shown that the three microalgae species investigated in the study can achieve high biomass concentration when cultivated in autotrophic mode (up to 4.8 g biomass/L, **Table 1.2a**) on liquid digestate with minimum nutrient supplementation[11],[12],[13],[14],[15],[16],[17]. Challenges associated with growing algae on liquid digestate include a) limited light penetration due to high turbidity and b) the presence of toxic components (e.g. heavy metals) and recalcitrant microorganisms from AD process exerting inhibitory effect on microalgae growth, both of which are generally resolved in previous studies through significant dilution with water. In Task 1.1 and 1.2., we plan to overcome these challenges through extensive characterisation of digestate using state-of-the-art methods (such as GC for Volatile Fatty Acids or VFA analysis and ICP-AES for heavy metals) before subjecting the digestate to a multi-step filtration process at different pore sizes[18]. The filtration process will allow for a) the removal of solids and microbes which consequentially reduce turbidity and b) reduction in the level of toxic components. We will then formulate growth medium at different dilution ratios for each species and test the effects on growth under lab-scale controlled conditions, with the objective to formulate optimal growth medium at minimum dilution ratio and nutrient supplementation. Nutrient mass balance will be carried out to determine level of required micronutrient supplementation. In addition, we will also track the assimilation of Volatile Fatty Acids (VFAs) during algae growth on digestate and their effect of the fatty acid profile of the biomass. If the filtration pre-treatment was found to be inadequate in reducing to toxicity level in the digestate, we will develop an appropriate detoxification strategy (e.g. fed-batch feeding) to make sure that the level of toxic contaminants remains below threshold at all times in the growth cycle. The development of growth strategies in WP1 is made possible only through the combination of A4F and UCD's experience in autotrophic microalgal growth and BIAE's expertise in AD characterisation.

Combination of algal growth on digestate and direct CO_2 capture. In WP1 (Task 1.4), the membrane system for CO_2 delivery will be integrated into microalgal growth strategies on digestate to attain simultaneous digestate treatment and CO_2 capture. Efficient CO_2 delivery by the membrane system is expected to intensify nutrient utilisation in the digestate-based medium, leading to rapid depletion of N in the medium which then allows for early onset of lipid accumulation in the biomass. The level of CO_2 in the system will be carefully controlled through pH and light optimisation. The performance of the hybrid system will be optimised to maximise final biomass concentration (>3 g biomass/L) and lipid accumulation (> 30wt% biomass) within short harvest cycle (7-14 days).

Pilot scale-up of hybrid system on AD site to demonstrate digestate remediation and direct capture of CO_2emission. In WP6 (Task 6.1), the hybrid system investigated in WP1, guided by techno-economic and LCA in WP5, will be scaled up to 1,000 L on partner BIAE site (AD operator) to demonstrate scalability. The facility will be *the first pilot microalgal infrastructure reported to-date able to simultaneously remediate digestate and directly capture CO_2 emission from biogas upgrading process.* The membrane system developed in SusAlgaeFuel is a platform technology that can easily be adapted for microalgae-based CO_2 capture system in other industries requiring CO_2 capture from a flue gas (e.g. breweries). In Task 6.2, the pilot infrastructure will be operated and further optimised to maximise final biomass concentration & lipid content. Sufficient biomass will be generated for the production of 1 kg HEFA-SPK.

WP2: In-line sensor to monitor bacterial population in culture

Costly culture contamination. A necessity for process commercialisation is the identification of **fast, precise, cost-effective, and easy-to-apply** methods to monitor the quality of the microalgal cultures. **Microbial contamination** in algal culture is a serious problem hampering the cultivation process, often resulting in culture crash and leading to considerable economic and time losses[19]. As such, early detection and culture monitoring are crucial. However, this is highly challenging due to the complexities of algae's interactions with other microorganisms.

In-line culture monitoring system. On-line and in-line monitoring using **Process Analytical Technology (PAT)** is a promising tool[20], with high-selectivity and the capability of monitoring several analytes and parameters (pH, dissolved oxygen, temperature, and biomass concentration, as well as certain pollutants which could jeopardize optimum conditions) simultaneously, for improving product quality, productivity, and process automation of microalgal culture and biorefinery. Moreover, as one of the classic indicators of low biomass quality is a high level of chlorophyll degradation leading to the formation of phaeophytins, chlorophyllides and pheaophorbides, a measure

[10] DOI: 10.1016/j.tibtech.2015.12.010
[11] DOI: 10.1016/j.biortech.2019.122445
[12] DOI: 10.1016/j.apenergy.2013.03.056
[13] DOI: 10.1016/j.biortech.2014.11.099
[14] DOI: 10.3390/atmos12091099
[15] DOI: 10.1016/j.jbiotec.2004.01.013
[16] DOI: 10.1016/j.chemosphere.2022.136177
[17] DOI: 10.1016/j.algal.2020.102129
[18] DOI: 10.1016/j.wasman.2020.08.037
[19] DOI: 10.1007/s00253-021-11396-7
[20] DOI: 10.1080/07388551.2017.1398132

of chlorophyll degradation could be used as an indicator of cellular health and damage in photosynthetic organisms. **Fluorescence-based techniques** can be used to monitor chlorophyll content, providing an indication of algal health.

Spectroscopic platform for bacterial analysis. In WP2 (Task 2.1), we will investigate the development of multirange spectroscopic methods for rapid and portable *in-situ* analysis and quantification of bacterial population dynamics in microalgae culture, enabling accurate monitoring of bacterial level in culture and the provision of early warning for catastrophic contamination level as well as effective bacterial management system. The approach that we will use is to leverage an innovative **Multirange Spectroscopic Platform** (recently developed at **IRIS**) whereby different spectroscopic PAT technologies have been engineered into a common framework to open up the possibility of tackling highly demanding monitoring applications by spectral data fusion in an effective and affordable way. This platform, called **VISUM Fusion**, is a disruptive framework that delivers the following sensing techniques (SWIR spectroscopy; NIR spectroscopy; VIS spectroscopy; Raman spectroscopy; and Fluorescence spectroscopy) through shared spectrometers (Raman uses a VIS or a NIR spectrometer and fluorescence a VIS spectrometer), as well as common illumination systems (VIS, NIR and SWIR can all use a common halogen lamp), and -in some cases- optical probes. This makes the platform **ideally suited for experimental work related to the measurement or detection of compounds at low concentration, often in complex matrices and with intrinsic barriers that do not match the ideal scenario of any of the traditional technologies**, such as the bacterial/microalgae system we are investigating in SusAlgaeFuel. Through Fusion, having several techniques in a common platform not only allows us to use the optimum technique for each measurement requirement but also enables **se**amless data fusion, to obtain dramatic improvements in precision, accuracy, and limit of detection. To further extend the framework capabilities, Fusion could also fuse data provided by off-the-shelf compatible temperature and electrochemical sensors. The combination of PAT and machine learning techniques has the potential to move microalgae production towards process automation through multivariate process control (MVPC) and software sensors trained on "big data".

WP2: Selective UV irradiation for culture purification

Impact of light quality on algae growth and composition. Light quality can have an effect on microalgae growth e.g. (i) the ability to modulate *Scenedesmus sp* nutrient removal from wastewater using different ratios of red and blue light (ii) Spirulina (*Arthrospira platensis*) growth increase under red light. Light quality was also shown to impact biomass composition in *Chlorella vulgaris*, by changing the relative proportions of C16:0 and C18:3[21].

Differential response between microalgae and bacteria cells to UV irradiation. Recent studies have shown that the application of UV-LED treatments (particularly UVB and UVC) can successfully inactivate bacteria in digestate[22],[23]. UVB irradiance at 52.5 $mJ.cm^{-2}$, for example, was found to be able to reduce the growth of several bacterial species by 36-58% in less than 1 minute, while UVC is routinely used in wastewater treatment (at multiple wavelengths), as a replacement for chlorine, to prevent bacteria reactivation. Interestingly, a study on phytoplankton inactivation highlighted the difference in bacteria and microalgae response to UV irradiation, showing that the same treatment that inactivates bacteria is ineffective for microalgae. Microalgae cells require a higher UV dose (60 mJ cm^{-2}) to reduce the initial concentration by 4 orders of magnitude (D4) compared to faecal bacteria and marine bacteria (19 and 44 mJ cm^{-2}) [5,23]. These differences in UV acclimatisation, if applied at the correct dosage, sequences, and wavelength, can be exploited as a low-cost means to purify culture during cultivation (particularly on waste products), selectively deactivating bacteria, while boosting the growth/lipid composition of microalgae cells. A more targeted UV approach can even potentially distinguish harmful bacteria from beneficial bacteria, allowing for adaptive growth that favours the proliferation of beneficial bacteria to symbiotically interact with microalgae cells.

Development of UV irradiation system for culture purification and lipid induction. In WP2 (Task 2.2), we plan to develop a 2-stage illumination system where we first expose the liquid digestate growth medium (before algal inoculation) to UV irradiation to inactivate bacteria in the medium (as a supplement to the filtration system developed in Task 1.1) and then subject microalgae culture grown on the sterilised digestate to a range light quality sources (e.g. artificial day light, standard white light) and UV irradiation under different wavelength ranges and regimens (e.g. red/blue light supplementation, continuous and pulsed) as in-line purification method to maintain low bacterial count arising from native bacteria in the algal inoculum, while boosting algal growth and lipid accumulation for biofuel production. The system thus has a dual purpose, enabling the sterilisation of bacteria on the digestate in the first stage while suppressing the growth of native bacteria present in the original/inoculum culture, and enhancing microalgae growth in the second stage. In WP2 (Task 2.3), we plan to take the system one step further. We will sequence native bacteria in each species (via 16S rDNA) and study their symbiosis through re-inoculation in axenic culture. Once a set of beneficial bacteria have been identified, the parameters of the UV irradiation system will be adapted to favour the growth of beneficial bacteria (over harmful ones), enabling further improvement in algal growth. WP2 will contribute novel science in algal-bacterial interaction and directed adaptation.

WP3: Autolytic cell disruption for low-energy fractionation of wet microalgae biomass

[21] DOI: 10.1016/j.biortech.2014.03.092
[22] DOI: 10.1111/1541-4337.12645.
[23] DOI: 10.1016/j.watres.2019.114875

The 'correct' water concentration of biomass for fuel production from algae: Biomass drying must absolutely be avoided when processing algae biomass for fuel production, with the amount of energy needed to completely remove water from algal concentrate (10.9 MJ/kg biomass) exceeding the full lipid-energy potential of the biomass (10.1 MJ/kg biomass at 30 wt% lipids). Dilute culture, however, must also be dewatered in order to avoid energy penalty of processing large water volume. The ideal biomass state is as slurry or concentrate at 100 – 200 g biomass/kg slurry. Wet processing route, however, leads to unique challenges associated with cell disruption and product liberation.

Energy-intensive cell disruption. Microalgal products, including lipids for HEFA conversion, are intracellular in nature, encapsulated within the biomass by multilayered cell walls which confer the cells with formidable defence against external processing (such as in the case of *Nannochloropsis)*. Microalgal slurry thus needs to be subjected to external cell disruption to liberate products for subsequent conversion. Enzymatic disruption is simply too expensive for low-value products (such as fuel). Chemical treatments using acids are possible options, but they often need to be applied at extreme pH and temperature (> 150°C) leading to degradation. Mechanical methods (such as bead milling) are effective and rapid, but generally accrue significant energy demand (**Table 1.2b**). Our calculations suggest that, even in optimistic scenario, the energy spent for cell disruption alone (using HPH) can account for up 25% of the total energy demand of algal biofuel production, often contributing to negative energy balance and an Energy Return on Investment (EROI) value < 1 (**Section 2.1** EO2 for calculations). For fuel production, a crucial criterion for the choice of cell disruption is the capacity to be deployed in an industrial scale at both reasonable cost and low energy requirements.

Common Mechanical Cell Disruption	Energy demand (kWh/kg biomass)	References
Ultrasonic	50-206.7	24
Microwave	1-117	25,26
Bead milling	0.4-8.9	27,28
Pulsed Electric Field (PEF)	0.5-2.5	29,30
High-Pressure Homogenisation (HPH)	0.3-5.5	27, 28
High Shear Mixing (HSM)	0.4-1.4	31

Table 1.2b: Energy demand for mechanical disruption

Autolysis as means to reduce the energy demand of cell disruption. Autolysis is a self-decomposition process whereby cells use their own endogenous (or internal) enzymes to degrade/weaken their cell walls, thus reducing the amount/intensity of external process that needs to be applied in order to achieve complete cell disruption. The underlying mechanism that drives autolysis is not well understood and, until recently, there were no known methods to induce autolysis in microalgae slurries suitable for large-scale processing. **PI Halim (UCD) recently reported a method to induce autolysis** in dense microalgae *(Nannochloropsis*) slurries under scalable conditions without the need of chemical agents[32]**.** The method involved incubating concentrated slurries under dark and anaerobic incubation at mildly elevated temperature (37°C) for 8-12 h. During the treatment, *Nannochloropsis* cells were found to activate self-fermentation pathways which thinned their cell walls by 40-50%, rendering the biomass more susceptible to subsequent externally applied cell disruption (e.g. by HPH) and eventually leading to 50% reduction in energy demand for attainment of complete cell disruption (from 0.7 to 0.35 MJ/kg biomass). Despite the groundbreaking potential of the treatment, we have not investigated autolysis on other species or with combination of other disruption.

Low-energy biorefinery fractionation based on autolysis. In WP3 (Task 3.1), we will leverage PI Halim's discovery in autolysis to develop a low-energy cascading biorefinery for the fractionation of microalgae biomass (as slurry) into HEFA-convertible lipids, protein serum and biomass residue. We will investigate the coupling of autolytic incubation with a range of mechanical and chemical disruption steps. By getting the cells to weaken themselves prior to externally applied cell disruption, we expect to be able to achieve complete disruption at milder operating conditions (i.e. lower pressure for HPH or lower speed for HSM), thus reducing the energy footprint and cost requirement for cell disruption. Since it will be unrealistic to screen every type of disruption, we will limit our investigation to processes known for having relatively low energy demand to start with (**Table 1.2b**): PEF, HPH and HSM. These selections are well aligned with KIT's findings as biorefinery WP leader in SABANA project, which shows bead milling, microwave and ultrasound to have an energy demand as high as 10x that of PEF and HPH. The combination of autolysis with dilute acid treatment will also be investigated because of its potential to attain very low energy demand as long as autolysis pre-treatment successfully reduced the pH requirements.

[24] DOI: 10.1016/j.algal.2023.103056.
[25] DOI: 10.3390/pr9020369
[26] DOI: 10.1016/j.procbio.2022.06.014
[27] DOI: 10.1016/j.biortech.2017.05.012
[28] DOI: 10.3390/fermentation8070301
[29] DOI: 10.1016/j.biortech.2015.12.012
[30] DOI: 10.1016/j.algal.2017.03.024
[31] DOI: 10.1016/j.renene.2019.10.133
[32] DOI: 10.1039/C8GC03186J

Energy target and process optimisation: Our target is to develop a biorefinery system capable of achieving 100% disruption (and thus complete fractionation) at energy demand of < **0.3 kWh/kg biomass (or 1 MJ/kg biomass)**. By capping the energy demand of cell disruption at 0.3 kWh/kg biomass, our calculations show that we will reduce the OPEX of cell disruption from ca. 1.61 to 0.56 $/kg HEFA, thus contributing to achieving our overall ambition of a HEFA cost of 6.30 $/kg (**Section 2.1** EO2). Even though this target is at the lower range of cell disruption energy demand, it is not unrealistic and can likely be achieved through parametric optimisation of the combined autolysis and mechanical/chemical disruption biorefinery, especially given that UCD has previously demonstrated the low energy demand of their autolysis HPH process and KIT has shown that they can reduce the energy consumption of PEF to 1-2 MJ/kg biomass even without autolysis coupling. *The biorefinery developed in WP3 will be the first reported cell disruption that has specifically been tailored to process wet biomass cultivated on digestate,* enabling novel development of sciences in the impact of growth conditions on cell-wall configurations and disruption.

Protein release: As part of the cascading biorefinery development, we will also analyse the amount of protein released by the biomass into the medium after cell disruption and phase separation as a separate product stream.

WP3: Lipid extraction strategy that allows for maximum solvent recycling and minimises solvent cost

High solvent demand during lipid extraction. Lipid extraction is a critical bottleneck in microalgae fuel production, requiring substantial solvent usage (or solvent-to-biomass ratio) ranging from 3 to 200 L/kg biomass (**Table 1.2c**). Many studies have selected their solvent usage arbitrarily, based on analytical amounts needed for total lipid extraction rather than consideration of scalability, thus contributing to the significant range observed in **Table 1.2c**. High solvent usage exerts double negative effect on OPEX, firstly through the direct cost of solvent purchase and, secondly, through the high energy cost associated with solvent evaporation (to separate solvent from extracted lipids). A critical strategy to reduce solvent usage is through effective solvent recycling. This, however, requires careful consideration to the solvent choice, as wet processing route entails complex interactions between water and solvents.

Table 1.2c: Solvent usage for lipid extraction

System	Solvents	Solvent usage [L/kg biomass]	References
Monophasic	Chloroform:methanol (2:1), ethanol, hexane:methanol, hexane:ethanol, hexane: IPA (3:2)	3 – 200	33,34,35,36, 37,38,39,40, 41
Biphasic	Hexane	1.5 – 60	

Solvent selection. Solvent forms complex interaction with water molecules surrounding the cells during lipid extraction from wet biomass. Solvent system can be divided into two categories: the 'monophasic' approach in which the solvents are perfectly miscible with water so that the whole extraction mixture forms a single phase and the 'biphasic' approach where non-water-miscible solvent i.e. a non-polar solvent (e.g. hexane, heptane) or solvent of very low polarity (ethyl-acetate, methyl tert-butyl ether MTBE) is used and the extraction mixture retains separate water and solvent phases. The monophasic approach generally uses a polar solvent which is then difficult to separate from water after extraction[42]. In the worst case, the solvent can form an azeotrope with the water (in the case of ethanol), which considerably increases the separation costs through fractional distillation. The difficulty associated with separating monophasic solvent from water means that solvent recycling will be very energy intensive, making it nearly impossible to effectively recycle solvent and thus to minimise solvent usage. On the other hand, for the biphasic approach, low-polar solvents do not form a single phase with water and can thus be separated from water in the slurry using a less energy-intensive phase separation step (e.g. centrifugation), facilitating effective solvent recycling and reduction in solvent usage. Additionally, biphasic solvent systems are more selective towards neutral lipids, the form desirable for HEFA conversion, preventing co-extraction of polar lipids and proteins.

Development of lipids extraction using biphasic solvent and solvent usage target. In WP3 (Task 3.2), we will investigate non-polar or low-polar solvent systems for biphasic extraction of lipids from treated/disrupted algae slurry obtained in Task 3.1. We will first screen different solvents for the three investigated species, determining which solvent or solvent system resulted in maximum lipid recovery from treated/disrupted microalgal slurry. Once the solvent system has been selected, we will systematically reduce the solvent usage for the specific species through parametric optimisation of the extraction step (e.g. agitation level, duration, number of equilibrium stages) and gradual increase in biomass concentration (from 100 to 200 g biomass / kg slurry) with the goal of extracting a

[33] DOI: 10.1016/j.rser.2018.05.012
[34] DOI: 10.1021/acs.jafc.5b00241
[35] DOI: 10.1016/j.algal.2016.11.005
[36] DOI: 10.1016/j.biombioe.2011.09.003
[37] DOI: 10.3844/ajbbsp.2011.70.73
[38] DOI: 10.1186/1475-2859-13-14
[39] DOI: 10.1007/s10811-013-0189-y
[40] DOI: 10.1007/s10811-020-02037-z
[41] DOI: 10.1016/j.biortech.2015.03.126.
[42] DOI: 10.1016/j.apenergy.2016.06.002

minimum of 80% available lipids. In Section 2.2, we have calculated the cost of algal HEFA based on an average solvent usage of 15 L/kg biomass and fuel target of 10 Mt/year. The cost of lipid extraction step was found to be 2.4$/kg HEFA, while the energy demand of lipid extraction was 20.6 kWh/kg HEFA, largely driven by the need for solvent evaporation. Our aim in WP3 is thus to cut down solvent usage to 3.5 L/kg biomass, enabling OPEX associated with lipid extraction to be reduced to 0.57$/kg HEFA and energy demand of solvent evaporation to decrease to 6.51 kWh/kg HEFA. This amount of reduction is needed in order to achieve our ambition of a HEFA cost of 6.30 $/kg and favourable EROI (>1). This target seems reasonable since both KIT and UCD have previously demonstrated that hexane extraction from PEF-treated *A. prototheorids* wet biomass and HPH-treated wet *Nannochloropsis* biomass[32] respectively can achieve 80% lipid extraction at 2-4 L solvent /kg biomass.

Emulsion mitigation to enable solvent recycling. In theory biphasic solvent system should lead to facile phase separation of solvents from water (via centrifugation), enabling effective recycling and reduced solvent usage. However, in reality, amphiphilic interactions between solvents, water and extracted lipids can often result in the formation of stable emulsion which traps solvents and leads to solvent losses. We have reported on this before for the hexane system[32] but literature remains scarce and data from soybean processing industry suggests that solvent losses between 0.05 and 0.3% of solvent is expected. The phenomenon, however, is difficult to observe and measure at bench scale. For this reason, once extraction conditions have been optimised at bench scale, we will test the best extraction systems at intermediate scale (ca. 300 g of wet slurry) and measure the level of emulsion formation. Methods to minimise emulsions will be investigated (e.g. pH adjustment) to achieve 95% solvent recycling.

Pilot scale-up of autolysis-based cell disruption and biphasic lipid extraction. In WP6 (Task 6.1), optimum cell disruption and lipid extraction processes (for the selected species) as discovered in WP3, and guided by techno-economic and LCA in WP5, will be scaled up on partner BIAE site. In WP6 (Task 6.2), the pilot-scale biorefinery will be operated for the extraction of sufficient lipids for conversion to 1 kg of HEFA-SPK

WP4: Multifunctional catalytic process for hydrotreatment conversion of microalgal lipids to HEFA-SPK

Lack of optimum catalyst for algal HEFA conversion. Algae oils are well positioned as HEFA feedstock thanks to the high content of high-carbon-length lipids suitable for HEFA conversion. However, implementation of algal oils in commercial HEFA installations has been shown to lead to accelerated catalyst deactivation and low yields of SPK-type alkanes; this is likely attributed to the lack of reactivity of oleic acid (C18:1), a major fatty acid in algal lipids, to Pt-based catalysts used in the conversion process[43].

Development of algae-specific catalysts and HEFA-conversion process. In WP4 (Task 4.1), we aim to achieve high HEFA conversion from algal lipids by developing novel multifunctional catalysts (based on transition metals) specifically tailored to handle algal lipid compositions. We will tap into **UCAM's extensive experience in fuel catalyst development** to first benchmark a two-step and single-step HEFA process using current understanding of hydroprocessing routes with hydrodeoxidation, isomerisation, hydrocracking reactions based on common catalysts; then, we will develop new, algal-specific catalysts, designed to work with high content of oleic acid in algal lipids. We will control the oxygen-removing mechanism by using transition metal catalysts (Ni, Mo, Co, Cu, Fe) and promote selective hydro-isomerisation with mild/medium-acidic supports (zeolites, alumina). Those two steps will be investigated using separate, single-function catalysts, optimised for their specific roles, and a combined bi-functional catalyst for a one-step HEFA treatment. Our goal is to significantly improve the yields of SPK components, with an overarching ambition to characterise algae lipid-to-HEFA conversion process and understand the catalytic mechanisms that pro_mote the production of SPK-desired C9-C15 alkanes from algal feedstocks. This will be achieved by analysing high-granularity experimental data obtained for varied feedstock composition/catalyst/support combinations in single- and two-step process approach.

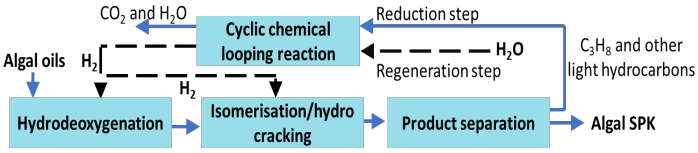


Figure 1.2b: HEFA route loop modification

Cost reduction during HEFA conversion. In addition to algae-specific catalyst development to facilitate high conversion, in WP4 (Task 4.2), we will be investigating innovative ways to reduce the cost associated with HEFA conversion process from algal lipid feedstock. In this case, we will develop a chemical looping system that allows for the production of green H_2 from the HEFA waste stream containing propane at ~500ºC and the recycling of the generated H_2 for chemical reaction (**Figure 1.2b**). Hydrogen is the key component in HEFA (required for hydrooxidation and hydrocracking), but most H_2 for the process is currently produced from energy-intensive steam-reforming, thus increasing the energy requirements and OPEX of HEFA conversion. Under the proposed system, UCAM will design a new, chemical looping reactor, filled with inexpensive Fe-based material, that will react with waste propane producing water and CO_2. The H_2-production step will be conducted by regenerating the Fe-material

[43] DOI: 10.1016/j.renene.2016.05.075

with water, leading to H_2O splitting in a thermochemical reaction. Adding in-house generation of H_2 will improve the economics of HEFA production; while contributing transferable innovation in HEFA processing beyond algae.

Pilot scale-up. The investigated route will be scaled up and validated in WP6 (Task 6.3) to produce 1 kg of HEFA.

WP4: Process for the conversion of biomass residue to TCR-HDO fuel

TCR-HDO as promising fuel conversion pathway. Cellulosic biomass can be converted to fuel intermediates through pyrolysis, hydrothermal technologies or gasification routes. However, there are still to-date no direct commercially mature routes for the application of pyrolysis oil (after modification) in fuel applications. Pyrolysis generates a dark brown oil consisting of 100+ different compounds (including oxygenated compounds, nitrogenated cyclic compounds, and sulfonated compounds) which lack thermal stability and are thus unsuitable for engine application. TCR, on the other hand, produces a thermally stable oil that can be subjected to subsequent hydrogenation step (HDO) for the removal of hetero atoms O, N, S (in form as H2O, H2S, NH3) and upgrade to saturated olefines. In WP4, we will investigate the full TCR-HDO pathway as a possible route for the production of hydrocarbons (diesel, kerosene and petrol fraction) from deproteinated and delipidated microalgae biomass residue. Even though TCR reforming is a relatively mature technology (TCR = 7), the subsequent hydrogenation step needed to upgrade the oil to petrol/diesel equivalence remains at a lower TRL level (TRL = 4). In the context of microalgae, no TCR investigation has ever been carried out on microalgae biomass or residue, underscoring the project novelty.

Development of TCR-HDO for microalgae residue. By investigating TRC+HDO conversion of algae biomass residue, additional energy from microalgae biomass can be generated beyond lipid extraction and conversion. This will increase the overall energy efficiency of SusAlgaeFuel process. In WP4 (Task 4.4), **Fraunhofer UMSICHT will adapt their patented TCR-HDO flagship process** (WO2015158732A1) previously developed for other biomass (e.g. sewage sludge) for algal application. A tailored hydrotreatment step (HDO) for algal biomass will be developed through optimisation of catalyst composition, temperature, pressure and gas velocity in order to maximise oil production (and minimise syngas and char production). The syngas generated from TCR process will be internally recycled as H_2 source for the hydrotreatment step, thus reducing overall conversion cost.

WP4: Purification of fuel molecules for the production of high-quality HEFA-SPK and TCR-HDO fuels

Development of purification methods for algae fuels. Sustainable aviation fuels must fulfil ASTM standards, which condition the minimum physical and chemical properties required to ensure safe operations during flight. Fuel properties directly correlate to their composition and the level of common impurities (usually water, metals and sulphur), the levels of which can in turn be controlled through refining and purification steps. The process needed to upgrade algal derived SAFs to achieve aviation specifications has never been previously reported. In WP4 (Task 4.3 and Task 4.5), we will leverage **UCAM's and UMSICHT's extensive experience and infrastructure in fuel development and upgrading** to investigate the combination of possible fuel upgrading steps that are most optimal for the algal-derived SAFs. Separate purification pathways will be investigated for HEFA-SPK (by UCAM) and TCR-HDO oil (by UMSICHT) to facilitate the development of customized purification for the specific fuel types. Our overarching goal for the purification process is to minimise the intensity and cost requirements of unit operations. This will be achieved by first correlating the key physical properties of the fuel (thermal stability, volatility) with carbon distribution in the fuel components and the content of inorganics. Then, we will use the correlation to define the minimum set of purification and refining steps, assessing possible experimental combinations. Dedicated microfluidic devices will be designed and built as necessary in order to reduce the amount of requisite fuel feedstock for purification steps (in the case that a large number of combinations needs to be investigated and we have limited fuel availability in the early phase of the project). For HEFA-SPK, the purification/upgrading strategies to be investigated include caustic treatment, filtration, and distillation. For TCR-HDO, we will investigate distillation, polar solvent extraction, and aromatic compound enrichment. Blending and additives will be considered if the proposed purification strategies appear insufficient. We will then create a decision toolbox (fed to WP5), connecting the purification steps with the achieved fuel composition and energy requirements.

WP5: Process simulations, techno-economic and social life-cycle analysis

Holistic techno-economic and life-cycle analysis. The integration tasks foreseen here are instrumental in developing a step-change solution for producing sustainable fuels from microalgae. From the start of the project, experimental research on algae cultivation, culture purification, biorefinery fractionation, and fuel production and purification will be supported by simulations, techno-economic and environmental assessments. All proposed processes will be studied separately in bench-scale setups before being integrated using simulations in a single process train, looking for synergies in process integrations (e.g. water recycling in biorefinery, heat transfer in HEFA reactions) for large-scale production under realistic set of process conditions. The design of the optimised configuration will be presented in a virtual biorefinery plant with the capacity of 10,000,000 kg SAF p.a. (WP5, Task 5.1), with the process and material flowsheeting directly linked with the environmental assessments using Life Cycle Analysis (LCA) (WP5, Task 5.2). Uncertainty analysis and techno-economic assessment will be carried out in WP5 to identify major cost drivers and assess commercial scalability of the proposed approaches. The LCA modelling in WP5 will require a number of new models to be developed for assessing the carbon and environmental footprint of SusAlgaeFuel

methods. The techno-economic and LCA results will be combined with experimental results from WPs1-4 to inform the SusAlgaeFuel team of the most optimal option for scaling up during pilot production in WP6.

Social analysis. Social LCA (sLCA) will also be carried out in WP5 (Task 5.2) to explore the social impacts of the developed products/technologies along the life cycle and derive holistic multi-actor assessments of process sustainability. We will identify the main social impact categories (e.g working conditions, health and safety impact), the key stakeholder groups that will be affected by value chain (e.g. employees, raw-material suppliers, people who live near AD plant) and evaluate the impact levels for each category/stakeholder group. The techno-economic, LCA and sLCA results from WP5 will be used to propose policy recommendations and business model in WP7.

1.2.2 Compliance with 'Do No Significant Harm' DNSH Principles

SusAlgaeFuel will comply with the DNSH Principles and align itself with the targets and mandates of major EU policies pertaining to the green bioeconomy (e.g. European Green Deal, Farm to Fork), waste management (e.g. EU 2019 New Waste Framework Directive), climate protection and renewable energy (e.g. RED II, Fit for 55) by: **a)** exploiting waste products from AD as nutrients for microalgae cultivation, thus providing bioenergy sectors with a more sustainable (and potentially more commercially viable) means of valorising their waste products while complying with EU-wide directives in waste management (e.g. Nitrate Vulnerable Zone legislation for AD)[18] and **b)** developing wet algal biorefinery which bypasses the need for drying and reduces GHG emission.

1.2.3. National and International Research Activities

SusAlgaeFuel will build on the technical know-hows and infrastructure acquired by project partners through participation/coordination in ongoing/completed EU and national research programmes (**Table 1.2d**).

Table 1.2d: Synergies of SusAlgaeFuel with other EU and national projects involving consortium partners

Project	Summary and Synergies with SusAlgaeFuel
MICRO-LYSIS: 'Cell-Wall Autolysis for the Scalable Fractionation of Microalgae into Biofuels and Novel Food Products', UCD, Irish Research Council, ongoing, 2022-2026	Micro-lysis plans to exploits autolysis for scalable and low-energy fractionation of several industrially important microalgae species for food and fuel applications. Autolytic incubation will be investigated in WP2.
EIRALGAE: 'Exploring the synergies between anaerobic digestion and microalgae for the development of energy independence, food-security and carbon neutrality in Ireland', UCD, Sustainable Energy Authority of Ireland, ongoing, 2023 – 2027.	EirAlgae aims to develop scalable biorefinery processes using *Nannochloropsis* to valorise AD waste into ω-3 PUFAs and protein for feed application. UCD will build on their knowledge of autotrophic growth on digestate and CO_2 capture system in WP1.
BIOFAT: 'Biofuels from Algae Technologies', A4F, FP7-ENERGY, completed, 2011-2016	A4F coordinated demonstration project with a farming area of 10-ha, gaining knowledge of autotrophic growth for WP1.
BioSPRINT. 'Improve biorefinery operations through process intensification and new end products', IRIS, H2020, BBI, Ongoing.	In this project, IRIS is developing a photonics-based monitoring solution to analyse compounds in the biomass residue from sugar biorefinery; this knowledge will be employed in WP2.
SABANA: 'Sustainable Algae Biorefinery for Agriculture and Aquaculture', KIT, H2020, completed, 2016-2021	SABANA compared conventional microalgae pretreatment technologies (HPH, bead milling) to PEF treatment. KIT has gained significant experience in microalgae processing to advance WP3.
TOSYNFUEL: 'Turning sewage sludge into fuels and hydrogen', UMSICHT, Horizon 2020, completed, 2017-2022.	UMSICHT coordinated TOSYNFUEL to demonstrate the conversion of sewage sludge to petrol using TCR. Methodologies and infrastructure will be transferred to WP4 & WP6.

1.2.4. Interdisciplinary Nature

SusAlgaeFuel will adopt a multidisciplinary approach in the intersection of microalgae biotechnology, bioprocess engineering/biorefinery, fuel conversion, economic, sustainability, and policy analysis. Indeed, the consortium will collaborate together in an interdisciplinary approach, converging knowledge clusters from the following fields across the work plan: microalgae, wastewater treatment, CO_2 capture, AD system (UCD, A4F, BIAE) for microalgae cultivation work (WP1); microalgae and fundamental biology, sensor development, machine learning (IRIS, UCD) for the real-time monitoring of bacterial contamination (WP2); bioprocess engineering, biorefinery system (KIT, UCD) for the biomass purification work (WP3); fuel engineering/chemical engineering, fuel chemistry, meta(data) analysis (UCAM, UMSICHT) for the fuel conversion/purification work (WP4); process simulations, LCA and sustainability (SFS, UCD) for the process integration (WP5); microalgae, bioprocess engineering, fuel engineering, fuel testing, digital twins (BIAE, A4F, UCD, UCAM, UMSICHT, IRIS) for the pilot scale up work (WP6), and finally expertise in science dissemination, research commercialisation, policy analysis (ESCI, UCD, SFS) necessary for the communication, exploitation and policy efforts that will be performed in WP7..

1.2.5. Social Sciences and Humanities Aspects

Our methodology integrates social sciences and humanities (SSH) aspects, whereby the technology development will be complemented with techno-economic and life-cycle analyses to assess scalability from **economic, social and environmental perspectives** and identify process improvements. As such, economic, environmental and social experts from UCD and SFS will work together to conduct a rigorous evaluation of the techno-economic and socio-environmental dimension of the results generated from the project and its impacts for the proposed value chain

(WP5), and the conclusions will in turn inform the technical and scientific R&D work in a circular loop to enable process improvements and iterative optimisations. Importantly, the **social LCA (sLCA)** that will be carried out to explore the social impacts and sustainability of the developed products and technologies, and their actual and potential positive as well as negative impacts along the life cycle, is still relatively novel but is becoming increasingly important in holistic sustainability assessments. In WP7, **business expertise** will be mobilised to develop a business/commercialisation model to map out potential exploitation pathways for different project results by the relevant stakeholders using methodologies such as one-to-one interviews and consultation with industry stakeholders in microalgae, AD, fuel production, aviation, and policy-making beyond the current consortium partners and lead to a joint publication outlining the roadmap for adoption by each of the relevant stakeholders (D7.4). During this process, GDPR-compliant data-collection principles on questionnaires preparation and unbiased interview techniques will be strictly implemented. Marketing and communications experts will also work with the consortium in the most effective way of engaging with, mobilising, and securing the buy-in of stakeholders.

1.2.6. Gender Dimension

The direct outcome of SusAlgaeFuel will be a more sustainable and cost-competitive algae biofuel. While this outcome will apply to both males and females indiscriminately, there are gender dimensions that impact various aspects of the technology and its implementation. Efforts to promote **gender diversity in our own team** can lead to a broader range of perspectives and innovative solutions. In our engagements beyond the consortium, in our one-to-one interviews and consultation with industry stakeholders, we will aim to secure inclusive and gender balanced participation to incorporate unbiased gender perspective in industry leadership, policy decision making, consumer perceptions and preferences. SusAlgaeFuel is also committed to championing women in STEM and STEM research. To this end, we will conduct a **Women-in-Science Campaign** as part of our communication actions (Section 2.2). In this campaign, ESCI will interview the women in our consortium and provide them with a platform to tell the journey of their scientific careers and how they overcome societal challenges. The interview will then be featured in SusAlgaeFuel dedicated social-media channels to inspire young adults to pursue a career in science.

1.2.7. Open Science

SusAlgaeFuel will develop a proactive Open-Science approach to promote the accessibility and transparency.
Databases and publications will be preserved in open-access repositories: After appropriate IP protection is in place, results will be published in international open access journals via the Green Open-Access route. They will also be archived and made available in trusted open-access repositories (e.g., Open Research Europe and DRYAD Digital Repository). This practice will ensure the latest discoveries can be searched by other actors in relevant fields. Registered pre-prints will be employed to ensure the early and open sharing of the research. Data sets, supporting data, experimental materials, methods, and design will also be archived together with written reports, allowing information to be re-analysed, replicated, and reused. **Research output management** Project outputs will be made accessible to the public through a range of activities: the project's website and social-media (Section 2.2). The Data Management Plan (**Section 1.2.8)** provides details on how research outputs will be managed through FAIR principles.

1.2.8. Data Management

Overview: Data management is an integral part of project implementation. All data management practice will strictly adhere to the FAIR principles to ensure that the data are Findable, Accessible, Interoperable, and Reusable. Moreover, a full Data Management Plan (D8.1) will be delivered within the first six months of the project.
Types of data/ research output, their findability, and accessibility: Project data/research outputs are comprised of quantitative and qualitative data, generated through both primary (i.e. experimental results) and secondary (i.e. policy assessment) research. Data will be deposited into open access repositories such as Dryad Digital Repository, which will be findable using keyword selection, versioning, and a standard identification mechanism (i.e., Digital Object Identifiers). Deposition of commercially sensitive data will not occur until the necessary steps to protect the commercial values of the data have been undertaken. Data sharing with institutions/researchers external to the consortium may occur on an ad-hoc basis at the discretion of SusAlgaeFuel's Executive Board.
Interoperability and Reusability of data/research outputs: Data and metadata produced during the project will be made interoperable through the use of a formal, accessible, widely-used, and broadly applicable vocabulary for all research outputs. Data files will be delivered in a non-proprietary open file format dictated by the Dryad Digital Repository (i.e. tabular data stored as Comma-separated values (CSV) and images as .jpeg) accompanied by a README file providing a clear and concise description of all components of the data set and software used to ensure it is reusable. Where necessary, data and metadata will include qualified references to other data/metadata. For datasets shared on public repositories, we will apply a Creative Commons Attribution license to ensure reusability.
Curation and storage/preservation costs: In order to minimise the risk of confidentiality breach: data storage, handling and retention will be carefully planned by each partner in compliance with the legislation on data protection (GDPR) within their member state. Prior to curation in open-access repositories, research outputs from all WPs will first be catalogued and deposited into the secure UCD-run Google-drive servers and held as temporary soft copies by each partner to assure data preservation. After project completion, all research outputs will be securely stored for at least 10 years in secure and dedicated open-access repositories (managed by UCD).

#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

2. IMPACT #@IMP-ACT-IA@#

2.1. Project's Pathways towards Impact

2.1.1. Unique Contributions towards Expected Outcomes and Wider Impacts

SusAlgaeFuel combines a holistic and innovative approach for microalgae Sustainable Aviation Fuels (SAFs) covering: a) direct capture of CO_2 emissions from biogas upgrading and utilisation of waste digestate as a widely available and low-cost nutrient source to support algae growth; b) novel in-line PAT complemented with machine learning and selective UV irradiation to monitor and purify bacterial contamination from algae culture; c) cascading biorefinery that relies on autolysis (self-disruption) and maximises solvent recycling in order to fractionate biomass into lipids, protein serum and cellulose-rich biomass residue at low energy and solvent requirements; and d) algae-specific thermocatalytic pathways for the efficient conversion of algae-lipids to HEFA-SPK and residue to bio-oil, followed by a range of purification methods for fuel refinement to meet aviation standards and certification.

This innovation has been carefully designed address the HORIZON-CL5-2023-D3-02-09 Call comprehensibly. Through the succession of innovative ideas bringing about the transformative step-change in each process unit across the whole algae-to-fuel process train, **SusAlgaeFuel** is positioned to **concretely deliver 3 out of 6 call-specific Expected Outcomes (EO1, EO2, EO4).**

EO1 Availability of disruptive sustainable renewable fuel technologies in order to accelerate the replacement of fossil-based energy technologies in aviation and/or shipping.

Through innovative strategies SusAlgaeFuel will bring to market the **first certified algal SAF** to propel the shift from fossil-based aviation energy to sustainable alternatives, with the potential to **deliver 20% of the EU's projected SAF requirements of 5 Mt in 2030**.

At its core, SusAlgaeFuel champions **circular production**, making use of waste digestate from AD as a nutrient source for microalgae growth (WP1). This not only minimizes waste but also broadens the sources of feedstock for microalgae cultivation, promoting resource efficiency and sustainability. In alignment with the European Green Deal, the initiative actively captures CO_2 emissions from biogas upgrading and uses them to support microalgae growth (WP1 followed by WP6). This dual-purpose approach not only curbs carbon emissions but also transforms waste CO_2 into a valuable resource for fuel production, contributing to climate change mitigation.

Ensuring the **consistent health of microalgae cultures is vital for sustainable fuel production**. SusAlgaeFuel is developing groundbreaking Process Analytical Technology (PAT) combined with Machine Learning, along with selective UV irradiation (WP2) to monitor and purify bacterial contamination, effectively preventing costly culture crashes and ensuring a stable and reliable supply of microalgae for fuel production.

The cascading biorefinery approach (WP3), which leverages energy-saving autolysis and allows for maximum solvent recycling, minimizes energy and solvent requirements while efficiently fractionating biomass into valuable components, **enhancing overall process sustainability**. Microalgae biofuel technology will be significantly advanced by developing low-energy technologies to disrupt microalgae cells and free intracellular lipids and proteins. Indeed, the proposed wet-processing pre-treatment methods are the potential game-changers, offering a solution to a long-standing need for biomass drying.

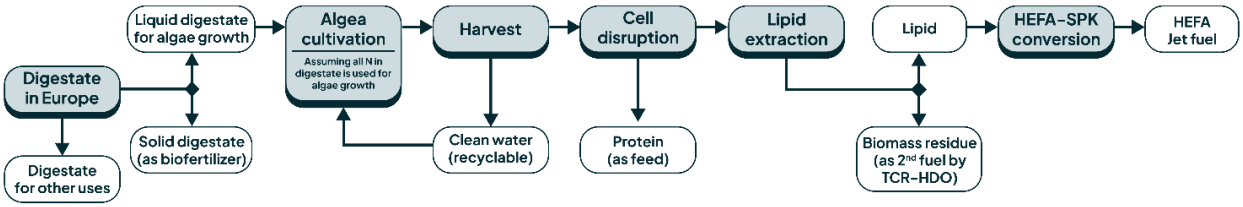
Additional improvements in biofuels production will come from the algal-specific process routes to HEFA-SPK, **improving the attainable yield** of energy-dense aviation fuels by advancing two separate pathways for fuel conversion: lipid hydrotreatment and thermochemical reforming of process waste, i.e. algal residue.(WP4). Quality assurance is paramount, and SusAlgaeFuel will develop purification methods to ensure our microalgal SAF meets international aviation standards and certification requirements (WP4).

SusAlgaeFuel doesn't stop at technical innovations. It conducts **rigorous assessments of the commercial scalability and the social and environmental impacts of its production strategies** (WP5). This holistic evaluation provides valuable insights into the broader feasibility and sustainability of our microalgae biofuel production strategies to enable iterative optimisation. To demonstrate the real-world applicability of the integrated technologies, we will build a pilot-scale system capable of producing 10 kg of algal lipids per year (WP6). This system, located at an AD operator site of partner BIAE in Ireland, allows for the production and quality testing of 1 kg of algal HEFA-SPK within project lifetime. This hands-on demonstration serves as a practical showcase of the scalability and performance of SusAlgaeFuel's disruptive technologies, and ensures we **credibly assess the commercial scalability.** Furthermore, SusAlgaeFuel is harnessing the power of digital technology. Encapsulating all the knowledge, knowhow and data generated in a **digital twin** of the integrated fuel production system will be a major asset in allowing us to project production performance under different scenarios/conditions, serving as a decision toolbox to guide future stakeholders, and facilitating a faster adoption thereby accelerating the replacement of fossil-based aviation energy technologies. **Lastly, SusAlgaeFuel recognizes the pivotal role of policy support.** Actively engaging in policy discussions, it provides concrete policy recommendations at the EU level (WP7). These recommendations aim to foster and promote advanced algal biofuels, aligning with sustainability and climate goals.

Scale and Significance of the contribution to EO1

We have made initial scaling calculations to assess the potential long-term impact of SusAlgaeFuel on AD operation (i.e. digestate treatment) and SAF production in the EU, taking into account the availability of digestate in the EU, the uptake of N by microalgae, the projected growth of AD and projected growth of SAF demand. We have used the innovations outlined in SusAlgaeFuel to calculate potential SAF yield from microalgae production.

Methodology and assumptions: REPowerEU has set a target of 35 BCM of sustainable Biomethane produced from AD in Europe by 2030. This is proposed to be achieved through a combination of 4,000 new smaller/medium AD Plants producing 4M m^3 per annum (@€80MWH) (~5MW) and 1,000 larger AD plants producing 16M m^3 per annum (@€55MWH)(~20MW). At least 56% (19.6 BCM) of the sustainable Biomethane potential is from animal manure and agricultural residues[44] . The European Biogas association (EBA) has estimated 492 Mtonnes of digestate production in Europe in 2030[44] . The use of digestate varies depending on the feedstock from which it is derived, so not all of them will be availed for microalgae valorisation (e.g. digestate from animal by-products or containing heavy metals are not compliant with EU directives for use in food production). Based on the EBA projected digestion production, assuming 50% of digestate is economically available for algal cultivation and 70% of that is remaining as liquid digestate fraction after solids removal, we have assumed that 172 Mt of digestate is available in Europe as nutrient source for algal cultivation. Since nitrogen is likely to be the limiting nutrient for algal growth (considering there is only 0.361% N in the digestate[45]), we have calculated our biomass productivity (and subsequent HEFA yield) based on the amount of N available for algal uptake in the digestate. Other assumptions: lipid content = 30 wt% biomass, protein content = 35 wt% biomass, sugar content = 35 wt% biomass.



Results: As shown in **Table 2.1a**, using SusAlgaeFuel process, we have calculated that each medium-scale and large-scale AD plant has the capacity to produce up to 955 and 3,910 tonnes of SAF per annum respectively (as HEFA). When scaled across all AD plants in Europe, SusAlgaeFuel process has the potential to treat **172 Mt digestate per annum** and generate **6.2 Mt of algal biomass per annum,** leading to the production of **1 Mt of HEFA-SPK per annum (or 20% of Europe's projected SAF requirements of 5 Mt in 2030).** The process will also lead to the **co-production of 2.2 Mt of protein** for feed application and **2.2 Mt of biomass residue that can be further converted to SAF via TCR-HDO pathway**.

Table 2.1a: Potential impact of SusAlgaeFuel to the European AD and SAF industries

		Medium AD ~5MW	Large AD ~20MW	Europe
Digestate produced	Tonnes p.a	217,308	889,315	492,000,000
Liquid digestate available for algae growth	Tonnes p.a	152,115	622,521	172,200,000
N content of liquid digestate	%	0.361	0.361	0.361%
N fraction of algal biomass	%	10	10	10%
Algal Biomass produced (dry weight)	Tonnes p.a	5,491	22,473	6,216,420
HEFA-convertible lipid fraction of algal biomass (SusAlgaeFuel process)	%	30	30	30
HEFA to SAF conversion (SusAlgaeFuel)	%	58	58	58
HEFA-SPK yield (as primary SAF)	Tonnes p.a	955	3,910	1,081,657
Protein yield (for feed)	Tonnes p.a	1,922	7,866	2,175,747
Biomass residue yield (for TCR-HDO conversion as secondary SAF)	Tonnes p.a	1,922	7,866	2,175,747

The impacts on the associated value chain of disruptive SusAlgaeFuel SAF will also be a boost for industrial competitiveness and leadership as outlined below.

Impact on the European SAF industry: The SAF industry in Europe is in its infancy. Aviation as an industry faces a mounting challenge to decarbonise in line with, or even above, set targets. In the foreseeable future there are three energy carriers capable of powering aircraft sustainably: hydrogen, electricity and liquid hydrocarbons (i.e., SAF). Without disruptive technology development, liquid hydrocarbons are the only solution for medium/long-range aircraft, SAF is, therefore, crucial to the EU's future socio-economic development. EU SAF demand in 2030 will be 5 Mt, increasing to 41 Mt by 2050. SAF production analysis by SkyNRG indicates that approx. 1 Mt of SAF capacity

[44] EBA, Tracking biogas and biomethane deployment across Europe. 2022
[45] AFBI, 27 MONTHS PERFORMANCE SUMMARY FOR ANAEROBIC DIGESTION OF DAIRY COW SLURRY AT AFBI HILLSBOROUGH. 2011.

will be operational in the EU by 2027[1], representing significant shortcoming to meet demands. The majority of SAF in Europe is manufactured from feedstocks such as waste fats, oils and greases utilising HEFA conversion pathways. The need to find sustainable alternatives for aviation is clear, but challenges remain in its production, with significant price difference to conventional jet fuel as key barrier to large-scale SAF production in Europe. SusAlgaeFuel is a gamechanger for the EU's SAF industry as it will enable the production of 1 Mt of SAF (or 20% of Europe's projected SAF requirements in 2030), and already this potential has sparked commitment of aviation giant **Boeing** and **CityJet**, who have both agreed to act as advisory panel to the project, and position themselves as early adopters.

Impact on the European microalgae industry: The EU algae sector has historically been dwarfed by larger markets in Asia, accounting for <1% of global algae production, estimated at 32.67 Mt in 2016. Regardless, the EC has steadily introduced policies to promote long-term growth of algal industry (EU Bioeconomy Strategy 2018, EU Blue Growth Strategy 2012) and recently announced 23 actions plans to unlock the potential of algae in healthier diets, lowering CO_2 emissions, and addressing water pollution (11/2022). This is expected to be a driver for the growth of the industry at a compound annual growth rate (CAGR) of 5.8% from 2023 to 2030, reaching US$ 355.48 million (M) by 2030, up from USD 228.66 M in 2022[46],[47],[48]. SusAlgaeFuel will create a more sustainable and cost-competitive EU algae industry through the development of transferrable technologies that can lower the cost of microalgae cultivation (WP1-2) and biorefinery (WP3), while increasing feedstock availability through synergy with AD. The SusAlgaeFuel technologies can add to the arsenal of innovative technologies for microalgae sectors beyond the fuel remit of the current project. Insights into bacteria-microalgae interaction (WP2) can be applied to the growth of other algae species on various agri-food effluents (e.g., brewery waste) for wastewater application. Wet biorefinery using autolysis in WP3 can provide gentle, non-degradative and low-cost extraction methods for higher-value products in nutraceutical applications (e.g., ω-3 lipids). SusAlgaeFuel will also **confer direct benefits to A4F** (microalgae technology provider) by diversifying their service using waste rather than commercial nutrients.

Impact on the European AD industry: The total AD market size was valued at ca. USD 47 billion in 2020, growing at a CAGR of 7.55%, and projected to reach about USD 84 billion by 2028. Currently, **the EU is the global leader in AD** with >17,400 plants[9],[49],[50]. AD is expected to play a critical role in EU's long-term pursuit for energy independence and decarbonisation, being one of the few processes able to supply **drop-in replacement for renewable gas** without the need for infrastructure upgrade. EU-wide Directives such as RED II and EU 2030 Climate and Energy Framework actively promote AD. Yet, one of the primary bottlenecks associated with AD adoption is the high initial investment required to set up the plant. As such, it is critical that any potential revenue stream in the process is exploited. While the full digestate has found valuable use as a biofertilizer, there is significant distribution cost to deliver the product to end users, and excessive land spreading of digestate can lead to nutrient leaching into groundwater (resulting in soil acidification and eutrophication) and NH_3 volatilisation[9,Error! Bookmark not defined.,Error! Bookmark not defined.]. A potential solution to this problem is to fractionate the digestate into liquid and solid fractions and develop novel applications for the liquid fractions, while the nutrient-dense solid fraction (now smaller in volume and suitable for long-distance transport) can continue to be applied as a biofertiliser. SusAlgaeFuel will cement EU leadership in the AD industry through the development of novel strategies/technologies for microalgae treatment of digestate nutrients and direct capture of biogenic CO_2 emission from point source, enabling the valorisation of their waste products into nutrient-dense biofertilisers and valuable microalgae biomass. This creates new commercial circular bioeconomy avenues to diversify income for AD operators and de-risk investment, which in turn can boost AD deployment in the EU.

EO2 Reduced cost and improved efficiency of sustainable microalgae-based and/or direct solar renewable fuel technologies and their value chains.

Aligned with the important relevant key priority in SET Action 8 to reduce the cost of algae advanced fuel to <35 €/MWh (or 0.49 $/kg HEFA) by 2030, SusAlgaeFuel aims to reduce the cost of microalgae SAF from the anticipated **12.3 $/kg HEFA to 6.3 $/kg HEFA (equivalent to a decrease of 49%)**, thus directly addressing the economic barrier that has hindered the widespread adoption of renewable aviation fuels.

Our innovation will achieve **improvements in the overall cost of algae fuels** through the circular use of cheap wastes digestate from AD as a low-cost nutrient source for microalgae growth which in turn reduces/eliminates the need for expensive commercial nutrients and direct capture of CO_2 emissions from biogas upgrading to support microalgae growth providing a low-cost source of CO_2 (WP1). The novel in-line PAT combined with machine learning and selective UV irradiation, by effectively monitoring and purifying bacterial contamination, will **prevent costly culture crashes.** The cascading biorefinery design, which relies on autolysis in wet biomass slurry (thus bypassing the need for drying while minimising the energy requirements for cell disruption) coupled with direct solvent extraction which allows for facile solvent separation and thus maximizes solvent recycling, will directly reduce energy and solvent requirements, and results in **lower operational costs** while efficiently producing multiple valuable components from biomass (i.e. protein serum for feed). The development of optimised algae-specific

[46] DOI: 10.3389/fnut.2018.00058.
[47] Commission, E. Commission proposes action to fully harness the potential of algae in Europe for healthier diets, lower CO2 emissions, and addressing water pollution. 2022
[48] DataBridge. Europe Microalgae Market – Industry Trends and Forecast to 2030. 2023
[49] DOI: 10.1016/j.rcrx.2019.100029.
[50] DOI: 10.1016/j.biortech.2018.07.100.

thermocatalytic pathways for converting algae lipids into HEFA-SPK and residue into TCR-HDO oil **ensures efficient fuel production with higher yields, contributing to cost reduction and improved overall efficiency** (WP4). SusAlgaeFuel will also assess the efficiency and commercial scalability of the entire process train as well as individual sub-process (WP5). This holisti approach addressing multiple stages of the value chain contributes to streamlining the entire microalgae-based renewable fuel production process, enhancing overall efficiency and cost-effectiveness. Importantly, the pilot production of SPKs in WP6 brings the chance to reassess the cost estimates and/or validate approaches, offering more confidence in the cost predictions for the 10,000,000 kg p.a. biorefinery.

The **Energy Return on Investment** (EROI) measures the efficiency of fuel production and is defined the ratio of final energy produced by the biofuel to energy expended in its production along the entire value chain, The EROI of microalgal biofuels to-date remains unfavourable, hindering their swift development. This is contributed by several factors, including low biomass productivity during cultivation, high energy demand associated with cell disruption, high solvent requirements leading to high energy demand for lipid extraction, and low conversion efficiency of HEFA reactions (<50%). SusAlgaeFuel will improve EROI through a) the development of cultivation strategies using nutrient-rich liquid digestate and direct CO_2 capture capable achieving high lipid accumulation using oleaginous species (WP1), b) the development of a biorefinery using autolysis with the capacity to reduce the energy demand for cell disruption by 50%, (WP3), c) the selection of low-polar solvents for facile separation from water (facilitating effective solvent recycling which further reduces solvent usage) (WP3), d) development of HEFA-SPK catalytic pathways tailored for microalgal lipids enabling high conversion efficiency (WP4).

Scale and Significance of the contribution to EO2

We have carried out preliminary calculations to assess the potential impact of SusAlgaeFuel on the cost and EROI of microalgae SAF production. A baseline process and the SusAlgaeFuel process were compared for the production of 10,000,000 kg HEFA-SPK/year. The baseline process has been assumed based on conventional experimental process as there is currently no commercially operating microalgal SAF production process.

Methodology and assumptions: *Baseline process* consists of cultivation in photobioreactors using artificial nutrients (NH_3 for nitrogen, DAP for phosphorus) and commercial-grade CO_2, harvesting via flocculation followed by filtration, cell disruption with HPH, lipid extraction with hexane, fuel conversion and purification to HEFA-SPK. *SusAlgaeFuel process* is comprised of cultivation in photobioreactors using digestate and CO_2 capture from AD, harvesting via flocculation followed by filtration, cell disruption with autolysis + HPH, lipid extraction with hexane at maximum recovery and with emulsion mitigation, fuel conversion and purification to HEFA-SPK via flagship method developed at UCAM. To estimate HEFA-SPK cost, we have calculated **a full mass balance and energy balance for both baseline and SusAlgaeFuel processes** on the basis of 10,000,000 kg HEFA-SPK/year with the following calculation assumptions: Equipment lifetime = 20 years. Cultivation: final biomass concentration = 1 g biomass/L. Cell disruption: biomass concentration in slurry = 250 g biomass/kg slurry, 90% rupture, 6 pass, 1500 bar. Lipid extraction consists of an extraction unit, a centrifugation unit and an evaporation unit. Lipid yield = 80%. Energy content of HEFA-SPK = 47 MJ/kg. The mass balances are shown in **Figure 2.1a** and **2.1b**, while the results for cost estimates and energy demand are shown in **Table 2.1b** and **2.1c**.

Results: Cost calculations in **Table 2.1b** show that the SusAlgaeFuel process can lead to a step-change reduction in the production cost of algae SAFs, **reducing the potential cost per kg of HEFA-SPK from 12.3$ to 6.3$ (equivalent to a decrease of 49%)**. The reduction is primarily driven by a decrease in operating cost (a 56.8% decrease in OPEX) brought about by a series of enabling innovations investigated in each process step along the value chain. The main contributors to the reduced OPEX are: the use of liquid digestate during cultivation as N and P source to replace (as much as possible) costly artificial nutrients, the effective capture of CO_2 from AD during cultivation to replace (as much as possible) commercial CO_2, the reduction in energy cost of cell disruption due to co-development with autolysis, the dramatic decrease in solvent usage during lipid extraction due to effective solvent recycling and emulsion mitigation strategies, and increased lipid-to-HEFA conversion yield coupled with reduced H_2 consumption during fuel conversion due to the use of multifunctional catalyst and looping process. Even though the anticipated cost of HEFA at project completion ($6.3/kg HEFA) is still significantly higher than the cost of conventional jet fuel ($0.7 – 1.4/kg), it compares relatively well with SAFs derived from other feedstocks ($1.9 – 7.7/kg SAF)[1], **towards the attainment of cost competitiveness beyond the project lifetime through process improvement, scale-up, and government subsidies.** EROI calculations in **Table 2.1c** show that the innovations investigated in SusAlgaeFuel can lead to **a significant reduction in the energy requirements for microalgal HEFA production** (up to 66% decrease), consequentially improving the energy balance of the value chain and dramatically altering the EROI of microalgae SAFs to make the process energetically favourable.

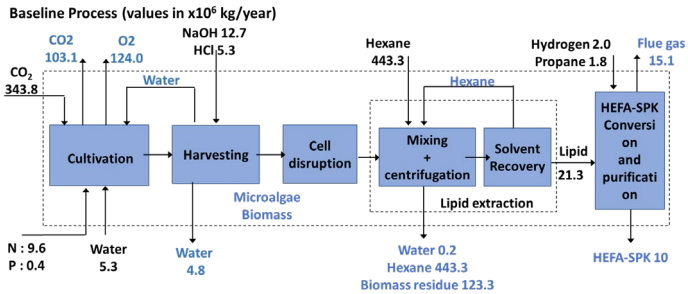


Figure 2.1a: Mass balance of baseline process (10M kg HEFA-SPK/year)

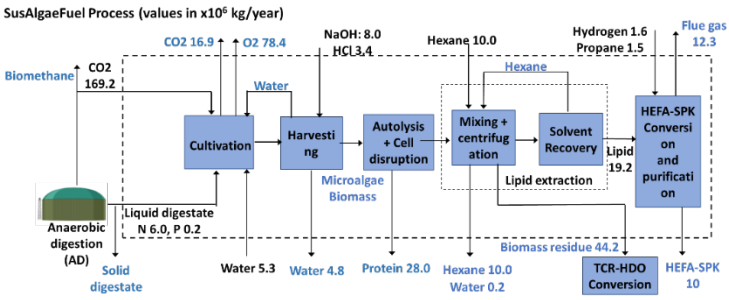


Figure 2.1b: Mass balance of SusAlgaeFuel process (10M kg HEFA-SPK/year).

Table 2.1b: Cost comparison between baseline and SusAlgaeFuel process (Production basis: 10M kg HEFA-SPK/year).

Process Step		**Baseline Process** ($/kg HEFA-SPK)	**SusAlgaeFuel Process** ($/kg HEFA-SPK)	% Change	**Enabling innovation driving cost reduction in SusAlgaeFuel**
Cultivation	CAPEX	0.35	0.22	-36.3	N/A^
	OPEX	2.32	0.47	-79.6	WP1, WP5, WP6: N and P cost reduced to 20% due to use of digestate. CO_2 supply cost reduced to 20% due to direct capture from AD. Lipid content increased from 21 to 30 wt%.
Harvesting	CAPEX	0.64	0.64	0.0	N/A^
	OPEX	0.93	0.65	-30.2	N/A^
Cell disruption	CAPEX	0.25	0.25	0.0	N/A^
	OPEX	1.61	0.56	-65.2	WP3, WP5, WP6: energy cost of HPH reduced from 0.6 to 0.3 kWh/kg biomass due to autolysis
Lipid extraction	CAPEX	0.12	0.08	-35.8	N/A^
	OPEX	2.28	0.49	-78.4	WP3, WP5, WP6: solvent usage reduced from 15 L/kg biomass to 3.5 L/kg biomass. Solvent recycling increased from 65% to 95% due to emulsion mitigation.
Fuel conversion (HEFA) & purification	CAPEX	0.84	0.76	-9.1	N/A^
	OPEX	2.92	2.17	-25.6	WP4, WP5, WP6: HEFA conversion yield increased from 47% to 58%. H2 consumption decreased from 0.34 to 0.25 kg H_2/kg HEFA-SPK
CAPEX		2.19	1.94	-11.3	
OPEX		10.10	4.40	-56.8	
Sum		**12.30**	**6.30**	**-48.6**	

CAPEX includes equipment cost, land cost, and indirect cost. OPEX includes the cost of energy, nutrients, water and labour. Formulas in[Error! Bookmark not defined.],[51],[52] were used for calculations.

^No specific innovation on the cost category. Reduction incurred indirectly due to improvement in other categories (e.g. less usage of solvent during lipid extraction reduces OPEX but also allows for a smaller extraction unit that reduces CAPEX).

Table 2.1c: Energy-demand comparison between baseline process and SusAlgaeFuel process for HEFA production.

Process Step	**Baseline Process** (kWh/year)	**SusAlgaeFuel Process** (kWh/year)	**% Reduction**	**Enabling innovation driving energy-demand reduction in SusAlgaeFuel**
Cultivation	11,357,522	7,185,817	-36.7	WP1: CO_2 delivery using membrane
Harvesting	7,471,169	4,726,951	-36.7	N/A
Cell disruption	75,987,842	24,038,462	-68.4	WP3: Autolysis
Lipid extraction	205,844,732	65,132,212	-68.4	WP3: Less solvent usage leading to less energy spent in solvent evaporation
Fuel conversion	2,063,830	1,705,644	-17.4	WP4: Heat integration
Sum of energy demand	302,725,094	102,789,085	-66.0	
Energy output*	130,555,556	130,555,556	0	
EROI	**0.43**	**1.27**		

[51] DOI: 10.1021/acs.iecr.8b03999
[52] DOI: 10.1016/j.biortech.2013.02.091.

EO4 Enhanced sustainability of aviation and/or shipping fuels, taking fully into account circular economy, social, economic and environmental aspects in line with the European Green Deal priorities.

SusAlgaeFuel directly delivers this expected outcome on multiple levels. Waste digestate from AD as a nutrient source for microalgae growth **exemplifies a circular economy approach**, thereby **minimizing waste, reducing the environmental footprint, and promoting resource efficiency** and direct capture of biogenic CO_2 emissions from biogas upgrading to support microalgae growth, **reducing GHG emissions and contributing to carbon neutrality** (WP1, WP6). Overall, the proposed system will enhance the commercial profitability of AD operation, enabling the valorisation of both of their waste streams (CO_2 effluent and digestate) into valuable microalgae biomass and biofertilizer, which in turn have multiplier impact in promoting AD deployment in the EU, sustainable waste management and displacement of GHG-intensive chemical fertilisers. Production of microalgae-based SAFs via our innovative process leads to a **direct reduction in carbon emissions compared to traditional fossil-based fuels**.

The cascading biorefinery approach, which involves fractionating biomass into lipids, protein serum, and cellulose-rich biomass residue, demonstrates the utilization of multiple components of the biomass, contributing to maximize the value extracted from the feedstock and promoting circularity (WP3, WP4). Indeed, the production of protein-rich serum from biorefinery (WP3) can contribute to food security in Europe through the **development of locally produced feed ingredients, partially alleviating the EU's dependence on imported soya meal feed** and reducing the carbon footprint associated with soy plantation (estimated at 20% of anthropogenic global CO_2 emissions).

Our entire approach is underpinned by a comprehensive Whole Life Cycle Analysis to assess sustainability (WP5), and our emphasis on reducing carbon emissions, resource consumption, and waste generation **directly aligns with the priorities of the European Green Deal** and contributes to Europe's commitment to becoming climate-neutral and fostering a sustainable and circular economy, offering tremendous potential in driving a sustainable aviation sector. Once the certified SusAlgaeFuel SAF is market available, adoption by the aviation industry will have radical emissions reduction impacts, which have already attracted the attention of Boeing and CityJet. The stakes are high in that the CO_2 emissions of all flights departing from EU27+EFTA airports reached 147 MT in 2019, which was 34% more than in 2005[53]. In 2050, it is predicted that in-sector measures **could reduce CO_2 emissions by 69% to 59 million tonnes** compared to a business-as-usual "technology freeze" scenario[53]. **37% of this reduction could come from SAF[53]**, and with SusAlgaeFuel gearing up with potential to occupy 20% of the market (discussed in EO1), it can play a pivotal role in realising this.

Scale and Significance of the contribution to EO4

We have carried out an initial life cycle analysis (SimaPro) on the **proposed SusAlgaeFuel HEFA-SPK pathway** to establish a 'well to wake' gCO_2-e/MJ of neat SAF produced to ASTM D1655 standard.

Methodology and assumptions: The boundary conditions were defined to ensure compliance with the core boundary conditions that ICAO CORSIA approved SAFs use: Feedstock Cultivation – Feedstock Harvesting – Feedstock Processing & Extraction – Feedstock Transportation – HEFA Jet Fuel Conversion – Jet Fuel Transportation – Jet Fuel Combustion. Allocation to co-products is included on a conservative mass balance basis. LCA model was built on the mass and energy balances obtained in **Figure 2.1b**. The following LCA assumptions were made**:** Power supply = 80% AD powered via biogas consumption, 20% Grids Main powered. CO_2 intensity Irelands Grid at 345gCO_2/kWh. Heat supply = 80% biogas, 20% Mains electricity. Allocation: 50% to lipid, 50% to protein co-product. Actual mass ratios are 1:1.458. (LIPID: Protein). Conservative 1:1 allocation is used until RSB RED Certification is reviewed. HEFA conversion step = 10g CO2/MJ for the HEFA Process. Estimated based on ICAO standards. Lipid to SAF conversion assumed at 1.47:1. Shipping and road transport = Dublin-Rotterdam-Sky NRG DSL 01 Plant-EHAM Schipol Fuel Tank Farm. The conversion of biomass residue to TCR-HDO fuel has not been considered in this LCA analysis due to lack of literature data. The process will likely improve the gCO_2-eq/MJ value for SusAlgaeFuel.

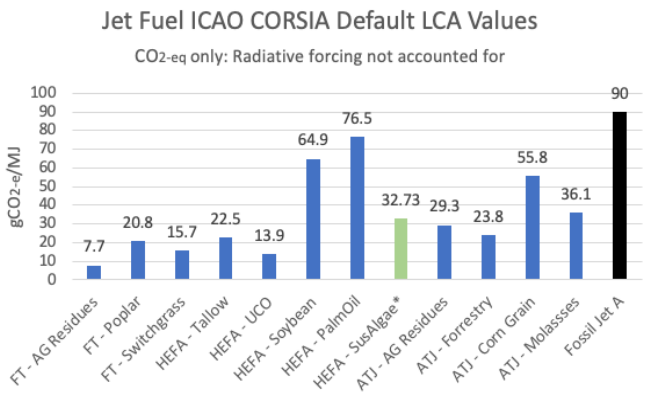


Figure 2.1c: Initial LCA results for SusAlgaeFuel HEFA

Results: The initial results predict an LCA figure of **32.73 gCO_2-eq/MJ of SAF produced on a 'well to wake' basis;** it should be noted that further reductions are expected when a detailed LCA is carried out. This analysis indicates that algal HEFA-SPK produced using SusAlgaeFuel innovations can lead to a **63.6% reduction in g CO_2-eq/MJ compared with conventional Jet Fuel (90 gCO_2-eq/MJ).** The algal SAF also compares favourably with others SAFs as can be seen in **Figure 2.1c**, performing better compared to HEFAs derived from high ILUC feedstock such as soybean and palm oil.

[53] European Aviation Environmental Report 2022.

Contributions to EO3, EO5 and EO6

By concretely delivering EO1, EO2 and EO4, SusAlgaeFuel will consequently play a contributory role in EO6, EO3:

■ **Increase the European energy security and reliability by enlarging the renewable feedstock basis for aviation fuels as well as maintaining and fostering the European global leadership in affordable, secure and sustainable microalgae-based fuel renewable energy technologies (EO6).** As can be seen, SusAlgaeFuel's circular production model has the per annum potential to treat 172 Mt of digestate and generate 6.2 Mt of algal biomass, leading to the production of 1 Mt of HEFA-SPK (or 20% of EU's projected SAF requirements of 5 Mt in 2030) at a reduced cost per kg of algal HEFA-SPK of 6.3$ (a decrease of 49% from estimated 12.3$). The approach of utilizing waste digestate and capturing CO2 emissions for microalgae cultivation significantly expands Europe's renewable feedstock basis, reducing reliance on fossil fuels and thereby enhancing energy reliability even in times of supply disruptions. Attaining affordability as we scale the technologies, building from the TRL4-5 platform of this project, ensures energy accessibility while reducing the vulnerability of European industries to energy price fluctuations.

■ **Increase technology leadership, competitiveness and technology export potential of European industry in possibly game-changing microalgae and/or direct solar renewable fuel technologies (EO3).** SusAlgaeFuel's pioneering technologies will position Europe at the forefront of SAF innovation. This leadership not only enhances EU's competitiveness but also strengthens the region's role, credibility, and influence in shaping global efforts to combat climate change. Underpinned by an IP strategy (Task 7.4) to protect our novel innovations, there will be real opportunity for licensing of these game changing technologies worldwide as global demand for SAF grows.

Finally, inspired by **EO5 Reinforced European scientific basis and European export potential for renewable energy technologies through international collaborations**, through Task 7.3 Stakeholder engagement and networking, we will outreach to and explore collaborations for sharing knowledge and expertise with relevant stakeholders, including other EU projects funded in the area of sustainable fuel and aviation.

Contributions to the wider impacts of the 'Sustainable, secure and competitive energy supply' destination:

SusAlgaeFuel will contribute to diversifying the portfolio of sustainable solutions for difficult-to-decarbonise/electrify transportation sectors, fostering **European global leadership in affordable, secure and sustainable renewable energy technologies.** Developing a whole algae value chain is crucial to achieving climate-neutral energy carriers that are cost-competitive with incumbent solutions. The innovation-driven and diverse approach adopted by SusAlgaeFuel sets the stage for European industry leadership in game-changing renewable fuel technologies, positioning Europe as a technological leader in microalgae-based renewable fuel production. This leadership has the potential to catalyse future partnerships and investments, further driving efficiency agenda.

SusAlgaeFuel objectives overlap with **Destination-specific Impacts**: **(1) Improving the availability of disruptive renewable fuel technologies**. SusAlgaeFuel will help make algal fuels a reality, bringing innovative ideas to improve cost-effectiveness, sustainability and range of product portfolio (two jet fuels, a protein serum as feed, a biofertiliser) that is resilient to future market changes. **(2) Reducing costs and improving the efficiency of the whole technology.** SusAlgaeFuels will tackle the hardest problems in algae technologies, namely the costs of biomass processing and low product yields. One of the most significant cost reductions will come from wet algae biorefinery (employing autolysis) and choice of extraction solvents enabling solvent recycling. **(3) Derisking of renewable fuel technologies**. This overlaps with the disruptiveness of the SusAlgaeFuel solutions, where the aim is to develop a multi-product approach (four different products), with production adapting to the market needs, thus, de-risking the whole biorefinery chain, the same way petrochemical refineries work today. **(4) Enhanced sustainability of renewable fuels** through dedicated work in relation to life-cycle assessments (WP5) and business-model development (WP7), that aligns development with the European Green Deal priorities. **(5) More effective market uptake.** The SusAlgaeFuel team brings together academics, SMEs, industry partners and non-profit organisations all interested in the early commercialisation of the proposed solutions. SusAlgaeFuel will work towards **effective technology rollout from the start of the project**, increasing the chance for success by implementing feedback from discussions with industry stakeholders within the consortium, as well as those reached by C&D activities, to develop a dedicated business canvas/commercialisation model (WP7). (6) Accelerating **Carbon Capture, Use and Storage** through the development of new enabling technology for effective and low-energy delivery of CO_2 to microalgae culture, thus **facilitating direct capture of CO_2 from CO_2 emission point source.**

2.1.2. Target Groups

Table 2.1d: Target stakeholders for SusAlgaeFuel

No.	Target stakeholders (Examples)	Why SusAlgaeFuel is important to stakeholders	How adoption of results contributes to impact achievement
1	AD operators as technology adopters (e.g. BIAE*)	Microalgae as a sustainable strategy for AD waste valorisation	- Adoption by AD operators will increase algae fuel feedstock availability which in turn 'improves the availability of disruptive sustainable fuel technologies'
2	Algae producers, technology providers	Novel technologies for reducing cost of algae production and processing	

	(e.g. A4F*)		- Direct CO2 capture by microalgae will accelerate 'Carbon Capture, Use and Storage' & lead to 'reduced cost and improved efficiency of sustainable microalgae-based fuel technologies & their value chains'. - Adoption by stakeholders will create a full value chain which 'derisks renewable fuel technologies' Adoption will ready the aviation industry in the future for algae fuel and 'accelerates the replacement of fossil-based energy technologies in aviation'
3	Fuel/SAF producers (e.g. Neste)	Technologies to produce algal HEFA-SPK and TCR oil **at low cost and at scale**	
4	Aviation sector: Aircraft manufacturers (e.g. Boeing^), Leasers (e.g. Avolon^), Airports & airlines (e.g. Aer Lingus^)	Possibility of cost competitive algal SAF in the near future and preparation for adoption in the aviation industry (e.g. engine modification). Opportunity to build their strategies and brand around sustainability and attract a growing market share.	
5	By-product users: Biofertiliser producers, Animal-feed producers	Digestate solid fraction and protein serum as novel ingredients for biofertilisers and animal feed	Side products enable income diversification which 'increases resilience of energy system' and 'derisks renewable fuel technologies'.
6	Sensor manufacturers (e.g. IRIS*)	Manufacture of in-line sensor for monitoring algae culture	Adoption will 'speed up new digital technologies in the energy sector'.
7	Fuel certification authorities (e.g. RSB^)	A new SAF feedstock based on algae for certification	Certification enables mass production and 'improves the availability of disruptive fuel technologies'
8	Industry associations: microalgae (EABA), AD (EBA), aviation & SAF (EASA)	Novel technologies in AD waste valorisation, algae biotechnology, SAF conversion and purification for adoption in wider sector	Expand the reach of the project which in turn catalyses wider adoption in the respective industries and 'improves the availability of disruptive sustainable fuel technologies'
9	Academic community	Novel sciences in algae SAF value chain	Results to catalyse further research that can further 'reduce the cost of renewable fuel technologies'. Joint activities with EU projects can promote adoption by Stakeholders
10	Related EU projects (REFOLUTION, FLITE)	Synergies in project scope and results for collaboration and joint activities	
11	Policymakers and MPs (national & EU^)	New algae SAF value chain contributing to EU directives/policy: **Fit for 55, ReFuelEU, REDIII**	Evidence-based algae SAF technologies lead to enactment of policies and carbon trade-offs that accelerate 'the replacement of fossil-based energy technologies'.
12	Investors in green technology	Potentially profitable undertaking in disruptive algal SAF validated in a pilot with credible upscale potential	Adoption can lead to investment to further 'reduce the cost of renewable fuel technologies'
13	University students	Understanding of algae SAFs	Training of future generation of scientists
14	General public/air passengers	Sustainable and eco-friendly fuels for air travel.	Better understanding of sustainable fuels a allows more informed travel choice.

*Consortium members in SusAlgaeFuel, ^ through SFS network and SusAlgaeFuel advisory panel

2.1.3. Barriers and Mitigation

A key driver for adoption of SusAlgaeFuel is the global aviation industry targets to be carbon neutral by 2050. This will be further fuelled by impending legislation, whereby for example, the European Commission has proposed a SAF blending mandate for fuel supplied to EU airports, with minimum shares of SAF gradually increasing from 2% in 2025 to 63% in 2050[Error! Bookmark not defined.]. Despite the tremendous opportunities this opens for SAF, a series of barriers exist:

■ **Availability of abundant and inexpensive nutrients**: This is a major barrier in algal cultivation. In SusAlgaeFuel it will be solved by marrying microalgae farming under different autotrophic regimes with critical AD operation (WP1), thus leading to an increase in the availability of algal feedstock while decreasing cost of cultivation.

■ **Cost of SAFs**: SusAlgaeFuel aims to make a more cost-competitive microalgae biofuel. If successful in delivering our technical innovations, as calculated in Section 2.1.1, we can reduce the cost of algal SAFs to 6.3$/kg, in line with SAFs from other feedstocks ($1.9 – 7.7/kg SAF). Despite this improvement, however, algal will remain significantly more expensive than petroleum diesel (ca. $0.7 – 1.4/kg), meaning that multiple step-change innovations in algae fuel production beyond the project will still be needed to achieve technology rollout. This step change can be brought about by continuing R&Ds which build on SusAlgaeFuel technologies **through larger-scale demonstration and subsequent process improvement or complement them with new paradigms** (e.g. genetic engineering).

■ **Scale:** Another major hurdle faced by algal biofuels, such as the proposed SusAlgaeFuel, is that of **scale**[54]. Current SAF production facilities are tiny compared to the economies of scale available to fossil fuels. Unless we can achieve

[54] Digest, B. The Next Big Step for Algae and Sustainable Aviation Fuels. 2022

meaningful production, the goals so many have set for a low carbon aviation industry will be impossible to meet. To the end, the current call topic rightly calls for proposals to 'also address systemic constraints and opportunities for scaling-up algae-based and/or solar fuel technologies'. To address the fundamental limitations of scale, in addition to investigating improved cultivation techniques, advanced technology, and effective process integration as a blend of approaches, we have also introduced **an Industry 4.0 grade Process Analytical Technology (PAT) driven Digital Twin** (WP6) to assist in understanding the developed processes and simulating their behaviour towards optimisation and predicting capacities at industrial scales. Indeed, a PAT-enabled Quality by Design (QbD) approach can help improve scale-up by combining real-time monitoring, advanced analytics, and systematic optimization to ensure consistent product quality, process efficiency, and successful scale-up. Developing a comprehensive understanding of the relationships between these parameters will lead to the identification of critical process parameters (CPPs) and critical quality attributes (CQAs). The ability to apply Model Predictive Control (MPC) strategies that use real-time data and predictive models to adjust process parameters during scale up would help to maintain optimal conditions and mitigate disturbances.

■ **Lack of financing:** A major barrier to the adoption of our results will be the continuing need for investment beyond project lifetime through either industry/private or government mechanisms at global/EU/national levels. Outreach to investors and venture capital firms to secure a future funding pathway for technology commercialization will be an important endeavour. To facilitate this, a series of strategies will be pursued:

- By combining PAT with Digital Twin technology, we have the ability to simulate process performance and yields in scaled up scenarios, to provide data-driven insights and information for investors. Also, bringing the technology successfully to TRL5 with a physical pilot demonstrator at an AD facility in Ireland thanks to a high-profile Horizon Europe research-industry partnership that can validate pilot scale production, backed by PAT-enabled digital twin technology, can prove to potential investors that the scaled up technology will work.
- Moreover, analysis of resulting fuel products in compliance with ASTM fuel property specifications (WP6), will be another great asset. RSB certification will underpin the case for production, up-scaling and trading the final SAF product within the EU, **advancing the business case for future investors**.
- Wide dissemination of the SusAlgaeFuel results amongst relevant stakeholders, coupled with verified business models to ensure post-project scale up and route to market so the results can be exploited by potential end users (WP7), underpinned by multi-faceted dissemination and exploitation strategies to maximise impact (**Section 2.2**).
- We have also added policy analysis as an integral component of WP7, aiming to engage policy decision makers (through SFS expanded network) early in the project to **further promote the inclusion of algae fuels and carbon/emission trading schemes in future EU and national directives**. This will facilitate a more conducive funding environment for the commercialisation of algae SAFs beyond SusAlgaeFuel.

■ **Cooperation challenges in building a new value chain:** A further barrier to our results uptake is the requirement for **multiple stakeholders to come together in a new value chain to commercialise the results** (e.g. AD operator, algae farmers, fuel/SAF producers, sensor manufacturers), each having different market needs and business models. To mitigate this risk, our consortium includes **representatives from the industry value chain** (A4F as an algae technology provider, BIAE as an AD operator, and IRIS as a sensor developer and real-time monitoring and digital expert), who are each motivated to secure a competitive advantage in these businesses through participation in this project. Moreover, we have integrated **business-model development** into our core activities in WP7, the aim of which is to establish multiparty dialogue for the development of mutually beneficial commercialisation plan. Excellent and validated technologies that can deliver a certified SAF at the necessary future scales and cost, and the interest of flagship early adopters (e.g. Boeing, CityJet) who sit at the demand end of the value chain can have a positive impact for mobilising a new microalgae SAF value chain in Europe

2.2 Measures to Maximise Impact – Dissemination, Exploitation, Communication #@COM-DIS-VIS-CDV@#

2.2.1 Impact Strategy

SusAlgaeFuel will develop an integrated dissemination, exploitation, and communication (DEC) strategy (**Figure 2.2a**) designed to maximise impact and to be sufficiently flexible so as to effectively address the different target groups for the solutions proposed. The impact strategy aims to maximise the legacy of the project and the concrete actions will be outlined in the **Dissemination, Exploitation and Communication Master Plan DECMP (D7.1)** This strategy paper will target stakeholders of the proposed value chain, such as AD operators, microalgae producers, biorefinery technology manufacturers, sensor system developers, aviation fuel/SAF producers, aviation industry stakeholders, potential by-product users (e.g. farmers, fertiliser manufacturers), academic community, fuel certification authorities, investors, policy makers, journalists and media companies, and consumers (i.e. people who fly). **SusAlgaeFuel** will elaborate the following **key DEC goals:**

Goal 1: raising awareness and informing target groups of importance and potential of microalgae alternative fuels
Goal 2: engaging in a dialogue with stakeholders along the algae value chain
Goal 3: enabling knowledge transfer amongst the partners on specific designs, data sets, and methodologies;
Goal 4: fostering exploitation opportunities for further research, policy and commercial adaptation.

Dissemination and training strategy (Figure 2.2a): SusAlgaeFuel will select (1) the most suited journals, topics and events to publish the scientific results and reach relevant audience. They will be presented in (2) selected Open-Access outlets and conferences, while fostering collaboration and engagement with relevant target groups. A mapping exercise will be carried out to identify key EU projects relevant to the area (Task 7.3), such as those funded by HORIZON-CL5-2023-D3-02-08, HORIZON-CL5-2022-D3-03-07, HORIZON-CL4-2021-RESILIENCE-01-16, and REFOLUTION, FLITE, and ALIGHT. SusAlgaeFuel will make a proactive approach to connect with these projects to promote mutual project workshops and seminar, with the view of organising joint seminars/workshops to expand the network and reach of dissemination activities. In addition, students and young researchers will benefit from a Research Summer School in microalgae fuel dedicated lecture series at UCD, UCAM and KIT. Different stakeholders will also get the opportunity to engage with the project through regular demonstration workshop at the pilot-scale algae fuel production facility that will be built at BIAE during the project lifetime The evaluation of these actions (3) will be done through the review of citations, impact factor of journals, altimetric and feedback forms.

Exploitation strategy (Figure 2.2a): The project will (4) identify Key Exploitable Results (KERs), target markets, potential users or clients and define the IP Management Plan in the Consortium Agreement. This forms the basis for developing potential (5) exploitation pathways and associated IPR strategy. By (6) monitoring the market, SusAlgaeFuel will review the exploitation strategy and refine it throughout the project. SusAlgaeFuel will also deliver a commercialisation model (D7.4) and policy white paper (D7.5) in order to harness project results, facilitate commercialisation and legislative adoption by relevant stakeholders beyond project lifetime.

Communication strategy (Figure 2.2a): A multi-layer communication strategy will support and compliment the dissemination and exploitation activities, and will include (7) identification of appropriate communication channels and formulation of key messages for the various target groups. This plan will then be implemented to produce (8) original content including journalistic articles, press releases, videos, infographics and social media posts. Monitoring tools (9) will measure the outreach of these actions and, while some actions may achieve a better outreach than expected (and can be repeated), other actions that are less successful can be adjusted.

Figure 2.2a - Impact strategy of SusAlgaeFuel

2.2.2 Dissemination + Training Actions

Table 2.2a – Planned dissemination actions for SusAlgaeFuel

Industry workshops and networking events, webinars, and seminars (Some will be organised as joint events with other EU projects in the area)	Goals 1-4	M13 – M48
Key motivation: run industry/end-user workshops, webinars, seminars to engage stakeholders. Some of these events will be jointly organised with EU projects in similar microalgae, fuel, aviation areas. **Target audience**: AD operators, algae producers, fuel/SAF producers, aviation sector, by-product users, sensor manufacturers, industry associations (e.g. EABA), investors, university students, air passengers. **KPIs:** 20+ events, reaching 800+ stakeholders		
Workshops at pilot-scale algae fuel facility on AD site (Kilshane, Ireland)	Goals 1-4	M37 – M48
Key motivation: run demonstration workshops to engage stakeholders. **Target audience**: AD operators, algae producers, fuel/SAF producers, aviation sector, by-product users, sensor manufacturers, fuel certification authorities, industry associations (e.g. EABA), policymakers, investors, university students. **KPIs:** 4 workshops, reaching 800+ stakeholders		
Scientific publications (open access)	Goals 1,3,4	M13 – M48
Key motivation: exchange scientific knowledge, provide technical details and open data **Target audience**: Academic community; **Potential Journals (Impact Factor):** Algal Research (IF = 5.276), Bioresource Technology (IF = 11.889), Chemical Engineering Journal (IF = 16.744), Frontiers in Energy Research (IF = 3.858). **KPIs:** >10+ peer-reviewed journal publications, reaching > 500 academics & leading to > 150 citations		
Conference presentations and fairs	Goals 1-4	M13 – M48
Key motivation: present scientific methodologies, results, and their impact; **Target audience**: AD operators, algae producers, fuel/SAF producers, aviation sector, academic community; **Potential Conferences:** AlgaEurope, European Algae Industry Summit (ACI); **KPIs:** >12 presentations and >2 conference workshops reaching > 500 experts		
Microalgae Fuel Summer Research School	Goals 1-4	M25 – M48
Key motivation: provide summer research internships in microalgae fuel production with dedicated lectures on the technologies and methodologies developed in SusAlgaeFuel; **Target audience:** University students; **KPIs**: 2-8 weeks combined research internship and lecture sessions for 5-15 students, to be jointly coordinated/implemented by UCD, UCAM and KIT.		
Technical Brochure	Goals 1-4	M45– M48

Key motivation: produce a 30+ page technical brochure providing an overview of the different results of SusAlgaeFuel featuring photographs, infographics, and an easy-to-understand language; **Target audience**: fuel/SAF producers, aviation sector, policymakers, investors, air passengers; **KPIs:** 200 copies distributed digitally & as printed versions		
White Paper on Policy Recommendation	Goals 1-4	M13 – M48
Key motivation: Dissemination of results to policymakers to enhance exploitation of algae SAFs and to provide advice to EU policy. **Target audience:** Fuel certification authorities, EU projects in related area, and policymakers. **KPIs**: 50 copies distributed digitally & as printed versions		
Engagement with policymakers	Goals 1-4	M1 – M48
Key motivation: consult policymakers on legislative actions to facilitate the adoption of algae fuel technologies **Target audience:** Fuel certification authorities, policymakers. **KPIs**: > 5 joint events leading to >100 new contacts in policy making relevant to energy and environment		
SusAlgaeFuel final conference	Goals 1-4	M46 – M48
Key motivation: present key project results and their impacts; **Target audience**: all, academic community, university students. **KPIs:** engage with >100 stakeholders and discuss project results		

2.2.3 Exploitation Action and IP Management

The consortium (led by UCD) will implement exploitation plans (WP7) to optimise the impact of project's results after project completion. Exploitation will focus on creating new academic and commercial opportunities for algae fuel technologies and delivering fact-based information on renewable fuel technologies for policymakers. The project is set-up in such a way that the future opportunities for the exploitation of project´s results can be maximised, as the technology feasibility (demonstrated at TRL4/5) will be complemented with techno-economic and life-cycle analysis to assess scalability. The 11 Key Exploitable Results (KERs), comprised of novel methodologies and technologies, are listed in **Table 2.2b**. SusAlgaeFuel will identify exploitation means for each KER (e.g. in terms of licensing, spin-outs, or direct use) with the long-term ambition to contribute to the route to TRL9.

Table 2.2b –SusAlgaeFuel's KERs and summary of IP strategies by each partner

KER	Owner	Means of exploitation and/or protection
KER1: Autotrophic growth strategy on digestate	A4F	Utilisation of KER1 for the development of a new service for A4F's clients based on the design, build operation and transfer (DBOT) of microalgae cultivation units using digestate.
KER2: Membrane system for CO_2 delivery to microalgae culture	UCD	Secure IP protection & pursue commercialisation activities via Technology Transfer Office & NovaUCD (innovation hub with experience in supporting 360 start-ups-€760M turnover- from 2004-2019). KER2 will consolidate UCD's growing knowledge basis in microalgae biotechnology to support formation of Irish microalgal spin-out specialising in wastewater treatment.
KER3: In-line sensor for bacterial detection/enumeration	IRIS	Further develop, secure IP and commercialise the pioneering Multirange Spectroscopic Platform ('Fusion') for in-line monitoring of complex samples in Industry 4.0. KER3 will demonstrate the platform's capacity for microalgae and open new market avenues.
KER4: UV irradiation system for bacterial control KER5: Low-energy cell disruption based on autolysis	UCD	KER4 and KER 5 will consolidate UCD's knowledge basis in microalgae for wastewater treatment and biorefinery production of valuable products. Same exploitation pathways as KER2.
KER6: Lipid extraction strategy to minimise solvent	KIT	Secure IP protection through technology transfer division responsible for commercialization. Gained knowledge on solvent system and overall process design (KER 6) will be made available to equipment manufacturers.
KER7: Multifunctional catalysts with H_2 looping route for conversion of algal lipids to HEFA-SPK KER8: Algae-specific HEFA-SPK purification method	UCAM	UCAM will engage with Cambridge Enterprise to identify and protect potential IPs from SusAlgaeFuel, further engaging in i-Teams and Cambridge Venture Project Opportunity (CVPO) projects to work on potential market pathways. The UCAM team already has 4 patent applications on catalytic material development and new process concepts, putting UCAM in a strong position to quickly translate future SusAlgaeFuel research outputs.
KER9: Route for conversion of microalgae biomass residue to TCR-HDO oil KER10: Algae-specific TCR-HDO purification method	UM-SICHT	For Fraunhofer UMSICHT, the new TCR route for the conversion of algae biomasswill offer a new market for fuel conversion and upgrading. By opening the TCR-technology for a new, promising biomass, they expect to be able to generate income in terms of royalties for Fraunhofer UMSICHT based on their worldwide patent **(WO2015158732A1)** and further research offers/investment from industry for the investigation of other algal biomass.
KER 11: Pilot plant for full-system process of converting digestate to microalgae-based aviation fuel	BIAE	Pending outcome from the project, BIAE will consider further scale-up of pilot-scale facility (KER11) from 1,000L to 10,000L of microalgae culture for the treatment their effluents.

We will exploit results from SusAlgaeFuel to create new **commercial opportunities** for algae fuel technologies. A key asset of our consortium is the participation of three industry partners from the value chain: A4F as an **algae technology provider**, BIAE as an AD operator **(technology adopter)**, and IRIS as sensor developer and real-time

monitoring/digital expert for **Industry 4.0**. This cohesive value chain of industry actors complemented with university and academic partners has a clear strategy (Table 2.2c and 2.2d) for how they will use the results of this project for their commercial activities and will ensure **a credible pathway for further scale up and commercial exploitation of project results.** In Task 7.5, the industry partners will work using agile tools such as the Business Model Canvas (BMC) approach to design the business models for the post-project commercialisation of results towards more elaborated Commercialisation Plan (D7.4) that will be delivered at the end of the project.

Management of Intellectual Property Rights (IPR): IP management will lie primarily with the project coordinator (UCD) and with individual partners generating the specific results. For the success of SusAlgaeFuel, it is essential that all project partners agree on explicit rules concerning IP ownership, access rights to any Background and Results and the protection of IP and confidential information before the project starts. SusAlgaeFuel proposes an encompassing **IPR management strategy** from the start of the project for the expected KERs The knowledge required for the project development (Background), the knowledge generated (Foreground) as well as the Access Rights to foreground and background will be managed according to the principles and articles of the **Consortium Agreement** (CA), which will define the rules regarding confidentiality, privacy and patenting, publication and dissemination, and IP ownership. During the project's lifetime, partners will be allowed to submit their own patent applications and exploit their IPs through creation of spin-out companies or licensing. Furthermore, in every project meeting, the consortium will have a standing action to report on commercially sensitive results and discuss specific IP options and plans. After the necessary IP protection has been successfully sought and attained, the results will be deposited in open-science repository and published reports/journal papers. Finally, Deliverable D7.4 (commercialisation plan) in M48, will report on the different steps taken to properly manage IP within the consortium/project, as well as a final IPR strategy for optimal exploitation of SusAlgaeFuel 's results.

Exploitation of the project results to contribute to policy: SusAlgaeFuel will also deliver fact-based information on renewable biofuels for policy makers thanks to SFS partner's extensive experience in aviation sustainability and policy. Since the latest targets set out by the EC under the revision to the Renewable Energy Directive (RED) have seen an increase in the share of renewable energy in the transport sector to at least 29%of final energy consumption, political agreement has been reached by national governments to reach these new targets, including those on advanced fuel and SAFs. SusAlgaeFuel will contribute to national government consultations to promote the introduction of advanced biofuels. This will be facilitated by the membership of the Roundtable of Sustainable Biomaterials (RSB) through SFS and the release of a **White Paper** providing policy recommendations (WP7) that will be aligned with 2018/2001 Carbon Offsetting and the Reduction Scheme for International Aviation (CORSIA).

Certification as an enabler for the exploitation of project results: The project aims to achieve RSB certification, recognised by the EC to meet stringent RED requirements. This RSB certification will underpin the case for production, up-scaling and trading of the final SAF product within the EU, advancing business case for investors.

2.2.4 Communication Actions

Table 2.2c - Planned communication actions for SusAlgaeFuel

Online presence: SusAlgaeFuel website and social media (e.g. LinkedIn)	Goal 1-4	M4 – M48
Key motivation: Communicate project objectives and partners activities. Build a community of interested stakeholders and early adopters. Engage with stakeholders and develop a dialogue. **Target audience**: AD operators, algae producers, fuel/SAF producers, aviation sector, by-product users, fuel certification authorities, academic community, policymakers, university students, air passengers. **KPIs**: >10k web visits/year, 200 social media posts reaching 25k people.		
Internal Communication Training (provided by ESCI)	Goal 3	M6 + M24
Key motivation: improve the consortium's ability to communicate on social media and to the press about the project and its results. **Target audience**: consortium partners. **KPIs:** 2 trainings for >20 persons to increase outreach success by partners by 50-100% (social media impressions)		
Promotional content: roll-ups, posters, infographics, videos	Goal 1,3,4	M1 – M48
Key motivation: provide relevant information to potential adopters. Promote the project and its results. **Target audience**:, AD operators, algae producers, fuel/SAF producers, aviation sector, by-product users, industry associations, academic community, policymakers, investors. **KPIs:** 1 leaflet, 6 infographics, roll-ups, posters, 2 project videos >10,000 stakeholders reached through promotional actions		
Journalistic content & mobilisation of stakeholders	Goal 1-4	M1 – M48
Key motivation: inform about best practices and tell inspirational stories on the research/production of microalgal aviation fuels, report both pros and cons of the proposed solutions and raise awareness of the issue to publishing houses and media outlets, create a dedicated communication flow and launch EU-wide awareness campaigns. **Target audience**: industry associations, general public/air passengers, journalists/social media influencers. **KPIs:** 8+ interviews, 6+ journalistic articles, 1 video news release, press releases >500k citizens reached on various channels (online, print, TV)		
Women-in-Science Campaign	Goal 1,2	M18 – M36
Key motivation: promoting female role models in science by featuring women within the SusAlgaeFuel project, telling their personal stories how they pursued scientific careers and overcome societal challenges. **Target audience**: university students and general public/air passengers. **KPIs:** >20,000 citizens (online, print)		

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2.3 Summary

Specific Needs	Expected Results	D & E & C Measures
• Current SAF supply is low at <0.05% of total EU aviation fuel use. To meet sustainability targets, SAF supply must be increased (~2.3 Million Tonnes by 2030, 28.6 MT by 2050). • SAFs are currently more expensive than fossil-based jet fuel. • Algae biofuels are currently not sustainable and cost-competitive due to a range of factors: - High biomass production cost (& limited feedstock availability) due to lack of access to cheap and abundant supply of nutrients. - Culture crash due to bacterial contamination - High biorefinery cost due to high energy requirements associated with cell disruption and high solvent requirements. - Low fuel conversion efficiency due to limited understanding of the thermochemical routes for conversion of biomass components to SAFs and the required purification steps.	A more **sustainable, cost-competitive and efficient** process for the production of algae SAFs via technologies that enable production of microalgae on cheap and abundant CO2 and waste nutrients from the AD sector: - Autotrophic growth strategy on digestate - Membrane system for CO2 delivery to microalgae culture - In-line sensor for bacterial detection/enumeration - UV irradiation system for bacterial control - Low-energy cell disruption based on autolysis - Lipid extraction strategy to minimise solvent usage - Multifunctional catalysts with H_2 looping route for conversion of algal lipids to HEFA-SPK - Algae-specific HEFA-SPK purification method - Route for the conversion of microalgae biomass residue to TCR-HDO oil (kerosene mixture) - Algae-specific TCR-HDO purification method - Pilot plant for full-system process (TRL4 /5) of converting digestate to microalgae-based aviation fuel - Process simulations, business-modelling, life-cycle analysis and policy assessment/recommendation.	Dissemination measures: 20+ Industry Workshops, networking events, webinars, and seminars; **Workshops at pilot-scale algae fuel facility on AD site;** 10+ open-access publications; >12 presentations and >2 conference workshops; Microalgae Fuel Summer Research School; Technical brochures; Policy white paper; SusAlgaeFuel Final Conference. Exploitation measures: ■ IP protection for new technologies followed by open-access publications and open-science repository according to Data Management Plan; ■ BMCs and Business/commercialisation plans; ■ Policy white paper; ■ RSB certification Communication measures: SusAlgaeFuel public website and social-media channels; Communication training programme; Promotional contents (infographics, leaftets, posters, project videos), Journalistic content (e.g. interviews, press release); Women-in-Science campaign.
Target Groups	**Outcomes**	**Impacts**
- AD operators as technology adopters - Algae producers as technology providers - Fuel/SAF producers - Aviation sector: Aircraft manufacturers; Leasing facilities; Airports & airlines - By-product users: Biofertiliser & feed producers - Sensor manufacturers - Fuel certification authorities - Industry associations - Academic community / researchers - EU projects in related area - Policymakers and MPs (national & EU^) - Investors in green technology - University students (undergraduate, master, PhD) - General public/air passengers	- Further **innovative/disruptive research** in microalgae and SAF production on the basis of our findings, leading to IP generation, licenses, and spin-outs. - **Commercialisation plan for post-project scale up** of SusAlgaeFuel process & adoption by SAF producers, AD operators & businesses along aviation fuel supply chain - Robust mechanisms and research to increase the scale and reduce the cost of algal biofuels to <35 €/MWh by 2030 according to SET plan 8 mandate, increasing the efficiency, and decreasing GHG emissions of algae SAFs. - **Positive policy changes** at national and EU levels in relation to SAF adoption, phasing out of fossil fuels and creating subsidies/incentives for SAF commercialization - Greater awareness of the importance of SAF amongst the general public, with an effective focus on air passengers	- Scientific and technological: Groundbreaking innovative approaches that work in tandem to enable a more cost-competitive, efficient, and circular production of microalgae SAFs, complemented with process simulations, techno-economic and LCA to assess scalability and identify process improvements. - Energy security in the EU and economic: Delivering **world's 1st microalgae HEFA-SPK and TCR-HDO fuels** tested for compliance to ASTM jet-fuel specifications as microalgae substitute aviation fuel with potential to annually treat 172 Mt of digestate for the production of 1 Mt of HEFA-SPK (equivalent to 20% of Europe's projected SAF requirements of 5 Mt in 2030).

3 QUALITY AND EFFICIENCY OF IMPLEMENTATION #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1. Work Plan and Resources

SusAlgaeFuel is divided into eight interconnected WPs which work to complement each other in delivering an integrated technology, business model, sustainability and policy solution for algal aviation fuel development in the EU. The interconnectivity between the WPs is illustrated in **Figure 3.1a**. WPs 1-4 will provide core technology development at bench scale, while WP5 will use the information obtained in these WPs to assess commercial and socio-environmental scalability of the system. Each WP is further divided into **several Tasks** to address specific objectives. In WP6, the processes developed in WPs 1-4, complemented with process optimisation based on WP5, will be transferred to pilot scale to demonstrate scalability, culminating in the production of sufficient SAF for ASTM compliance test at the end project of lifetime. The project will adopt a staggered approach, with WP6 starting only in Month 25 (2nd phase of the project), allowing investigation in WPs 1-4 to reach sufficient maturity prior to scale-up. WP7 provides blueprint for results dissemination, commercialisation and policy guidance.

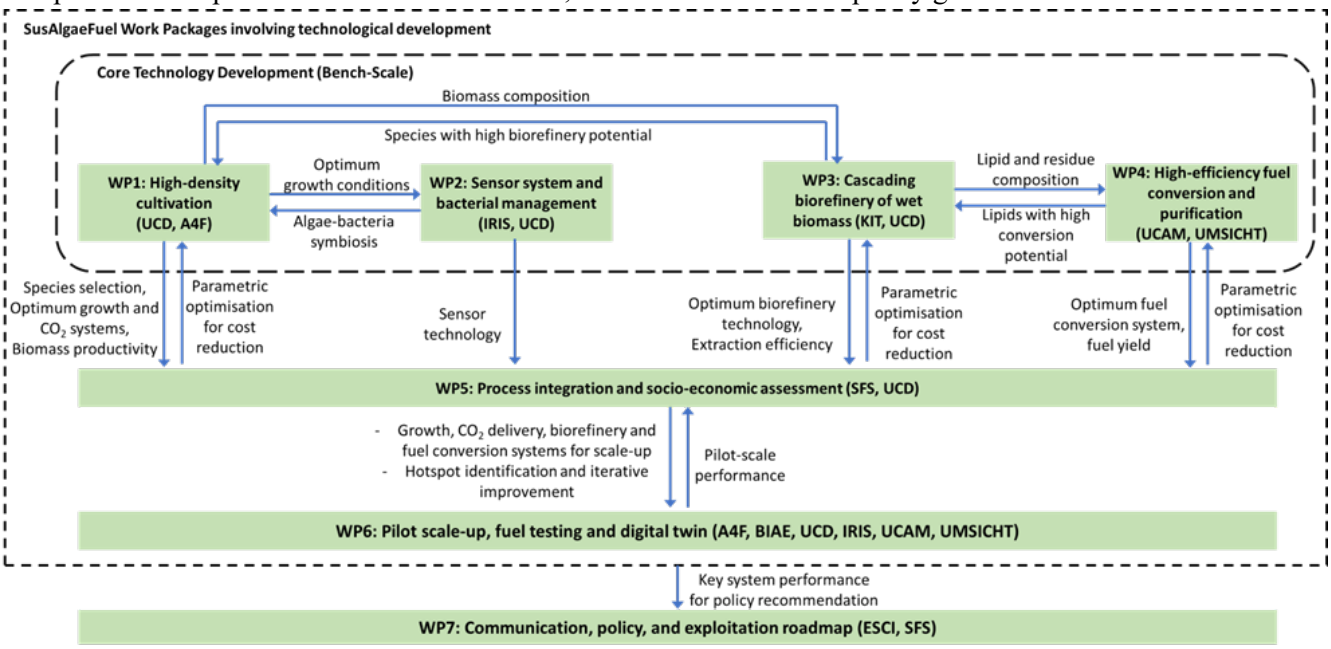

Figure 3.1a: SusAlgaeFuel's approach and interconnectivity between WPs

Management Structure: The **Executive Board (EB)** assumes the overarching leadership for the project. It decides on the general policy and strategic orientations of the project to ensure that milestones and deliverables are achieved according to the timeline. EB is responsible for monitoring the overall progress of the project and designing corrective actions or authorising the execution of risk mitigation strategies when significant project delay/deviation is observed. The EB comprises all WP leaders and the project coordinator. The experienced **Project Coordinator** (UCD,) has the overall responsibility to manage and execute the project. They also act as an intermediary between European Commission, Advisory Board (which currently consists of Boeing and CityJet), EB and rest of the consortium. **WP Leaders** are in charge of: (a) implementing decisions taken by the EB for their WPs, (b) coordinating the activities within their WPs and monitoring progress, (c) reporting to Project Coordinator, and (d) organizing and chairing online WP meetings. The **Project Office** (**PO**) at UCD will manage the daily logistics of coordination activities on behalf of Project Coordinator, including the organisation of on-site consortium meetings at M1, M18, M36 and M48

Table 3.1a: List of Work Packages

WP No.	WP Title	Lead Part.No	Lead Partner	Person-Months	Start Month	End Month
1	High-density cultivation through biogenic CO2 capture and waste utilisation	1	UCD	39.5	1	36
2	Sensor system for bacterial detection and culture purification	3	IRIS	42.0	1	36
3	Cascading biorefinery for biomass purification	5	KIT	63.0	1	36
4	High-efficiency fuel conversion and purification	6	UCAM	70.7	1	36
5	Process integration and whole life-cycle analysis	8	SFS	38.0	13	48
6	Pilot scale-up, fuel testing and digital twin	3	A4F	154.0	25	48
7	Communication, policy and exploitation roadmap	9	ESCI	46.7	1	48
8	Project coordination	1	UCD	29.7	1	48

		Year 1						Year 2						Year 3						Year 4					
	Milestones (ML) and Deliverables (D)	M2	M4	M6	M8	M10	M12	M14	M16	M18	M20	M22	M24	M26	M28	M30	M32	M34	M36	M38	M40	M42	M44	M46	M48
WP1: High-density cultivation through biogenic CO2 capture and digestate utilisation																									
T1.1: Digestate characterisation and pre-treatment	**ML1.1, D1.1**			ML	D																				
T1.2: Development of cultivation procedures on digestate	**D1.2**												D												
T1.3: Novel membrane system for the delivery of biogenic CO2 to aqueous solution	**ML1.2**						ML																		
T1.4: Hybrid cultivation integrating direct CO2 capture and digestate bioremediation	**D1.3**																		D						
WP2: Sensor system for bacterial detection and culture purification																									
T2.1: In-line sensor technology to monitor bacterial level	**D.2.1**												D												
T2.2: UV irradiation for bacterial management	**D2.2**												D												
T2.3: Symbiotic algae/bacteria interaction and adaptive growth	**D2.3**																		D						
WP3: Cascading biorefinery for biomass purification																									
T3.1: Autolytic cell-wall disruption for direct processing of wet microalgae	**ML3.1, D3.1**						ML			D															
T3.2: Solvent screening, recycling and emulsion management	**ML3.2, D3.2**												ML						D						
WP4: High-efficiency fuel conversion and purification																									
T4.1: Multifunctional catalyst for hydrotreatment conversion of microalgae lipids into HEFA-SPK	**ML4.1, D4.1**						ML						D												
T4.2: Chemical looping to reduce hydrogen and energy during HEFA-SPK synthesis	**ML4.2, D4.2**												ML						D						
T4.3: Algal HEFA-SPK fuel purification	**ML4.3, D4.3**															ML			D						
T4.4: TCR-HDO treatment for the conversion of microalgae residue to kerosene mixture	**ML4.4, D4.4**											ML							D						
T4.5: Algal TCR-HDO fuel purification	**ML4.5, D4.5**															ML			D						
WP5: Process integration and whole life-cycle analysis																									
T5.1: Techno-economic assessment and identification of major cost drivers	**ML5.1, D5.1**											ML							D						
T5.2: Life-cycle analysis and identification of socio-environmental hotspots	**D5.2**															ML									D
WP6: Pilot scale-up, fuel testing and digital twin																									
T6.1: Design, build, and commission of a pilot-scale system comprised of cultivation and biorefinery units	**ML6.1, ML6.2**														ML				ML						
T6.2: Operation of pilot system for production of lipids and biomass residue	**ML6.3, ML6.4, D6.1, D6.2**																					ML D	ML D		
T6.3:Pilot fuel production, purification and testing for compliance with ASTM standards	**ML6.5, D6.3**																							ML	D
T6.4: Digital twin of the pilot system	**D6.4**																								D
WP7: Communication, policy and exploitation roadmap																									
T7.1: Communication, dissemination and exploitation strategy	**D7.1, D7.2, D7.3**			D						D	D														D
T7.2: Communication activities																									
T7.3: Stakeholder engagement and networking																									
T7.4: Exploitation pathways and business-model development	**D7.4**																								D
T7.5: Policy analysis	**D7.5, D7.6**																								D, D
WP8: Project coordination																									
T8.1: Scientific, administrative and financial coordination																									
T8.2: Risk management and contingency planning																									
T8.3: Data management and open science	**D8.1**			D																					

Figure 3.1b: Gantt chart for SusAlgaeFuel

Table 3.1b: Work package description

Work Package No.		**1**	**Lead: UCD**
Work Package Title		**High-density cultivation through biogenic CO2 capture and digestate utilisation**	
Start Month	**1**	**End Month**	**36**

Objectives: To develop high-density microalgal cultivation strategies by capturing biogenic CO_2 and recovering waste nutrients from digestate.

We will **a)** formulate a cultivation medium based on liquid fraction of digestate that minimises or overrides the need for artificial nutrients (Task 1.1), **b)** develop autotrophic cultivation strategies for the investigated oleaginous species (Task 1.2), **c)** develop a novel membrane system for the efficient delivery of CO_2 into microalgae culture, and **d)** integrate CO_2 delivery system with cultivation strategies on digestate in a single system and optimise this hybrid system to attain biomass concentration > 3 g biomass/L and lipid content >30 wt% of biomass (Task 1.4).

Description of work:

In this WP, A4F and UCD will work together to establish optimum cultivation strategies for each investigated species (*C. pyrenoidosa, A. protothecoides, N. salina*) at bench scale (**2 – 10 L culture**). Cultivation will utilise the liquid fraction of anaerobic digestate (supplied by partner BIAE) as N and P source and CO_2 as carbon source. The characteristics and consistency of the supplied digestate in terms of physical, biochemical and microbiological properties will first be analysed. The digestate will then be subjected to a range of pre-treatment steps (e.g. dilution, filtration) to generate liquid medium with dissolved nutrients configurations suitable for algal growth. Afterwards, the three species will be cultivated on the digestate-based medium under different growth parameters and the ability of each species in recovering nutrients will be analysed. For the CO_2 delivery system, UCD will develop a non-porous membrane CO_2 delivery (or carbonation) system and compare their performance to standard methods for delivering CO_2 into microalgae culture (e.g. sparging). The membrane carbonation system will then be integrated with cultivation strategies on digestate medium and optimised for maximum CO_2 utilisation, biomass productivity and lipid accumulation. The results and technologies obtained in this WP will be used for techno-economic and LCA in WP5 and transferred to the pilot scale-up in WP6.

Task 1.1: Digestate characterisation and pre-treatment (M1 – 8). Lead: UCD. Participants: A4F **.**

An initial characterisation of digestate supplied from BIAE will be carried out in terms of total N (by colorimetric methods), NH_4-N (by colorimetric methods), total P (by colorimetric methods), volatile fatty acid (VFA) concentrations (by GC), turbidity, micronutrient concentrations (by ICP-AES), and total solid content. Microbiological contaminants (by plate count) and potential growth inhibitors, such as heavy metals (by ICP-AES), will be identified. The digestate will be partitioned into solid and liquid fractions via a series of pre-treatment steps that include settling and centrifugation. The liquid fraction isolated from centrifugation will be analysed through the same set of characterisations as above before being subjected to further pre-treatments using a combination of microfiltration, ultrafiltration ,or two-step filtration, in tangential flow configurations for removing solid particulates and bacteria and producing filtrate with accessible nutrients for microalgal growth. The filtrate will then undergo the same set of characterizations as raw digestate and isolated liquid digestate. The sequences of pre-treatment steps and parameters (e.g. transmembrane pressure for filtration) will be optimised to achieve maximum nutrient extraction. This task will guide A4F in the formulation of cultivation medium from digestate, minimizing the use of artificial nutrients in Task 1.2. Different batches of digestate received from BIAE will be analysed for their consistency in physical properties and chemical compositions and the effect of inter-batch variation on pre-treatment efficiency and filtrate composition.

Task 1.2: Development of cultivation procedures on digestate (M9 – 24). Lead: A4F. Participants: UCD.

In this task, cultivation procedures will be implemented at lab scale for achieving high biomass concentration and lipid content on pre-treated liquid digestate obtained Task 1.1. A microalgae cultivation medium with the pre-treated liquid digestate will be formulated by including the digestate in standard medium at different dilution ratios (ranging from no dilutions to digestate:standard medium ratio = 1:8 v/v), testing the effects on the three microalgae species selected, using lab-scale controlled conditions. The effect of digestate concentration on microalgae growth rates and on lipid accumulation will be assessed and the best dilution for each species will be identified. Specifically, the potential influence of carboxylic acids present in the digestate on the lipid profile will be investigated using GC-MS analysis. The need for supplementing the growth medium with crucial microelements will also be assessed. The pre-cultivation and post-harvest medium will be subjected to same set of biochemical characterization to assess the extent of nutrient utilisation by microalgae. The achieved medium formulation will be used for all strains growth and improvements on light, temperature, pH conditions will be performed to achieve maximum biomass concentration and lipid content. Final standard operation protocols (SOPs) will be prepared for the cultivation of the three species using the developed strategy. The best performing strain will be used in Task 1.4 and WP6.

Task 1.3: Novel membrane system for the delivery of biogenic CO_2 to aqueous solution (M1 – 24). Lead: UCD. Participants: A4F.

The membrane carbonation system for CO_2 delivery into aqueous solution will first be designed and operated at bench scale without the presence of microalgae. In order to achieve maximum CO_2 transfer into aqueous solution, the performance of the system will be optimised in terms of membrane types, membrane configurations, CO_2 partial pressure and pH of the solution. In terms of membrane configurations, we will first use a composite hollow-fibre membrane made of a nonporous polyurethane dense core as shown in **Figure 1.2a** but will explore other options as

the Task progresses. The mass transfer of molecular CO_2 across the membrane is dependent on intrinsic permeability of the membrane, the difference in CO_2 concentration between the supplied gas and the solution (as indicated by CO_2 partial pressure), and solution pH[7],[55]. The membrane system will be equipped with an automated pH controller to correct pH levels within optimal range throughout operation. The rate of CO_2 delivery into the solution will be determined by the difference between total carbon and total organic carbon in the solution (with a TOC analyser).

Task 1.4: Hybrid cultivation integrating direct CO_2 capture and digestate bioremediation (M19 – 36). Lead: UCD. Participants: A4F

In this Task., we will combine the membrane carbonation system developed in Task 1.3 with the digestate bioremediation strategies investigated in Task 1.2 at bench scale. The Task is divided into two stages where we will first operate the membrane carbonation system in microalgae culture grown on standard medium before transferring into dual system with digestate.

The membrane system is integrated into a 2L photobioreactor and its effectiveness in delivering CO_2 to microalgae culture grown standard medium is evaluated using the best performing strain found in Task 1.2. The mass transfer of CO_2 in the integrated system is driven by a carbon-concentration gradient maintained through the continuous photosynthetic uptake of CO_2(aq) and HCO_3^- by microalgal cells[7,55,56]. The rate of photosynthesis, in turn, is dependent on cultivation parameters, such as nutrient availability (N and P), light intensity, pH and growth phase[9,57,58]. The performance of the system will be measured in terms of biomass productivity and CUE, calculated as the ratio of carbon assimilated in the biomass (measured with CN analyser) to the amount of carbon delivered into the microalgae suspension (measured with a flow meter). In addition to carbon mass balance, nitrogen and phosphorus mass balances (by colorimetric methods and ICP-AES) will also be developed to understand the limiting factors in the system. Mass-transfer model to describe the kinetics of carbon transport across the membrane and subsequent carbon utilisation by microalgae will be developed and solved using MATLAB.

Once fundamental mass transfer of the system is understood, the dual system will be operated using digestate medium derived from Task 1.1 with dilution ratio found in Task 1.2, bringing together a full system where microalgae growth (optimum strain from Task 1.2) is supported using CO_2 delivered by the membrane system and N, P and micronutrients supplied by the digestate. The performance of the system will be assessed in terms of biomass productivity, and bioremediation (or nutrient utilisation) of the digestate. The system parameters (e.g. light intensity, pH, growth phase and harvest time) will be optimised to maximise CUE (> 90%), final biomass concentration (> 3 g biomass / L) and biofuel-lipid accumulation (>30 wt% of biomass), while achieving complete bioremediation. The neutral lipid content and fatty acid composition of the biomass will be determined using total lipid extraction (Bligh and Dyer) followed by SPE and transesterification/GC-MS analysis respectively.

Work Package No.		**2**	**Lead: IRIS**
Work Package Title		**Sensor system for bacterial detection and culture purification**	
Start Month	**1**	**End Month**	**36**

Objectives: To develop effective strategies (for the in-line monitoring and management of bacterial contamination in algal culture, based on: **a)** developing rapid *in-situ* methods based on advanced NIR and Raman spectroscopy coupled with machine learning algorithm to quantify the level of bacteria in microalgae culture (Task 2.1), and **b)** using the findings from the sensor system to develop a selective UV irradiation system for reducing bacterial level (Task 2.2), and **c)** refine cultivation strategies that favour the growth of beneficial bacteria (Task 2.3).

Description of work:

In this WP, IRIS and UCD will work together to develop effective sensor technologies for in-line monitoring of bacterial growth in algae culture and, using these technologies, to develop an effective digestate pre-treatment and cultivation management strategies to control bacterial growth during algal cultivation on digestate-based medium (thus preventing culture crash). Alternative approaches will be investigated that will aid in overcoming challenges associated with bacterial contamination during algal cultivation in digestate-based medium, such as 2-stage illumination using UV at different wavelengths and directed adaptation with symbiotic bacteria. WP1 and WP2 will be pursued in parallel with each other, enabling findings from WP2 to be directly integrated into work in WP1.

Task 2.1: In-line sensor technology to monitor bacterial level (M1 – 24). Lead: IRIS. Participant: UCD.

IRIS will leverage a novel Multirange Spectroscopic Platform designed and engineered as a test rig to deliver the most suitable from the following sensing techniques from the one comprehensive platform: SWIR spectroscopy, NIR spectroscopy, VIS spectroscopy, Raman spectroscopy, Fluorescence spectroscopy. This self-contained multirange spectroscopic platform will allow the most successful configuration of techniques for tackling the demanding application of the monitoring of bacterial contamination in algal culture (optical techniques for parameters such as pH, biomass concentration, glucose and/or other bacteria metabolites such as butyric acid, fluorescence-based techniques to monitor chlorophyll content, which provides an indication of algal photosynthetic condition), as well as enable seamless data fusion, to obtain dramatic improvements in precision, accuracy, and limit of detection. As such, the platform will be used for measuring the predefined critical parameters and bacterial

[55] DOI: 10.1039/C5EE02005K
[56] DOI: 10.1016/j.biotechadv.2019.107444.
[57] DOI: 10.1111/tpj.12823.
[58] DOI: 10.1016/j.algal.2015.12.020

indicators and verified against the flow cytometry reference methods that have recently been optimised at UCD for the analysis of bacterial levels. Based on the results, the optimum spectral data fusion techniques will be defined for their configuration into a reliable and cost effective in-line monitoring solution based on a smart sensing device with automated data acquisition system and machine learning algorithms to enable real-time data analysis and early warning systems when abnormal bacterial growth or contamination is detected, so that the system can trigger appropriate responses such as adjusting culture conditions, initiating mitigation measures, or isolating affected batches. Data gathered from this task will be employed in Task 2.2 for the development of bacterial management toolkit and WP6 for the building of the Digital Twin of the system (Task 6.4).

Task 2.2: UV irradiation for bacterial management (M1 – 24). Lead: UCD. Participant: IRIS.

Recent studies have demonstrated that bacteria and microalgae can acclimatise differently to UV irradiation, there the same treatment that inactivates bacteria was found to be ineffective for microalgae[5]. Based on these findings, we plan develop a 2-stage illumination system where we first expose the digestate growth medium (before algal inoculation) to UV irradiation to inactivate bacteria in the medium and then subject microalgae culture grown on the sterilised digestate to a range light quality sources (e.g. artificial day light, standard white light) and UV irradiation under different wavelength ranges and regimens (e.g. red/blue light supplementation at 662nm/430 nm, red/blue light mix, continuous and pulsed) in order to maintain low bacterial count, while boosting algal growth and lipid accumulation. The system thus has a dual purpose, enabling the sterilisation of bacteria on the digestate in the first stage while suppressing the growth of native bacteria present in the culture, and enhancing microalgae growth in the second stage. The efficiency of bacterial inactivation will be assessed using standard baseline methods (e.g. CFU count and flow cytometer) as well as in-line sensors developed in Task 2.1. Lipid content and fatty acid composition during growth in the 2nd stage will be assessed using standard gravimetric and GC-MS methods, allowing for assessment of the effect of targeted UV illumination on lipid accumulation. UV Parameters to optimise during the bacteria inactivation: wavelength, duty cycle, mode and energy efficiency. IRIS will contribute as a designer/provider of the engineered solution for UV irradiation with tailor-made UV LED lamps.

Task 2.3: Symbiotic algae-bacteria interaction and adaptive growth (M19-36). Lead: UCD. Part.: IRIS.

To further reduce the chances of bacterial proliferation leading to culture crash, the algae-bacteria interactions during algal growth on digestate will be studied at lab scale. Bacteria present in the algal culture can be divided into contaminant bacteria present in the digestate as well as native bacteria present in algal inoculum. The presence of contaminant bacteria is expected to be significantly reduced if the digestate based medium has been previously subjected to microfiltration (Task 1.1) or UV irradiation (Task 2.2); however, native bacteria in the culture derived from inoculation can also pose risk of culture crash if they proliferate aggressively under nutrient-rich conditions afforded by the digestate medium. Hence, it is important to distinguish different sub-populations of bacteria and their roles in affecting algal growth. The algae species selected in Task 1.2 will be cultivated on standard medium, digestate medium or treated digestate medium and the bacteria community at these different conditions will be plated in universal media for the isolation of colonies. The bacterial species will be identified by 16S rDNA, and the main bacteria found will be inoculated with axenic algal cells to understand the symbiotic/beneficial or harmful relations between the algae and bacteria. Based on these results, culture conditions will then be adapted to favour the growth of beneficial bacteria, while reducing the growth of the harmful ones. An enriched cultivation with the beneficial bacteria will be also carried out to assess a possible improvement in algae growth. Flow cytometry and spectroscopic platform developed in Task 2.1 will be used to map the dynamics between algae and bacteria populations.

Work Package No.		**3**	**Lead: KIT**
Work Package Title		**Cascading biorefinery for biomass purification**	
Start Month	**1**	**End Month**	**36**

Objectives: To develop a cascading biorefinery for low-energy and low-solvent fractionation of wet microalgae slurry at high biomass concentration (100 – 200 g biomass / kg slurry) into aviation-fuel precursors (i.e. biofuel-convertible lipids and cellulosic biomass residue) and food/feed components (i.e. protein-rich serum).

We will **a)** investigate microalgal ability to self weaken (or autolysis) as the basis for efficient cell disruption (Task 3.1) and **b)** develop extraction strategy using solvents (including bio-based solvents) which can be easily separated from wet algae biomass and enable solvent recycling. Overall, we aim to achieve a biorefinery with a target energy demand for cell disruption < 1 MJ/kg biomass and target solvent usage for lipid extraction < 3.5 L/kg biomass.

Description of work: In this WP, KIT and UCD will work together at bench scale (**10 – 100 g slurry**) to first investigate different combinations of autolytic incubation with mechanical and chemical cell disruption (Task 3.1). This will then be followed by the identification of the best wet processing pathway for cost-effective solvent extraction of lipids at low solvent usage. To achieve this objective, we will specifically investigate the use of non-polar or low-polar solvents with high affinity to neutral lipids and capacity to form bi-phase with water. In order to achieve high purity lipid extracts, solvent systems will additionally be selected for low solvability to proteins and carbohydrates or to components from cultivation medium. Once the best extraction solvent has been selected and the process is optimised to achieve the target lipid yield, WP3 will also investigate different methods to mitigate emulsion formation (e.g. mixer type) to facilitate optimum solvent recovery (target solvent recovery > 95%).

Task 3.1: Autolytic cell-wall disruption for direct processing of wet microalgae (M1 – 18). Lead: UCD. Participants: KIT.

Task 3.1 will investigate the development of low-energy cell disruption for wet microalgae biomass. For both saltwater species (*N. salina*) and freshwater species (*C. pyrenoidosa, A. protothecoides*), the coupling of autolytic incubation with a range of mechanical (HPH and HSM) and chemical methods (acid hydrolysis) will be investigated. The parameters associated with autolytic incubation (degree of anoxia, temperature) will be optimised to achieve maximum cell-wall weakening. The use of autolytic incubation as a pre-treatment that weakens microalgae cell walls is intended to reduce the intensity of operating conditions for mechanical or chemical disruption (i.e. lower pressure configuration and less number of passes for HPH, milder pH for acid disruption) and thus the operating cost of biorefinery. The parameters for mechanical and chemical steps (e.g. pressure configuration for HPH) will be initially selected based on literature. The extent of cell disruption after the combinatorial treatment (or hybrid treatment) are to be assessed with a combination of light microscopy (cell counting + staining), flow cytometry and protein leakage in the supernatants. The performance of the hybrid treatments will also be compared against standalone disruption treatments (e.g. HPH only, HSM only and acid hydrolysis only). The hybrid treatments that lead to the highest disruption for each species (as indicated by flow cytometry) will be further parametrically optimised (e.g. pressure configuration, number of passes for HPH; rotation speed, temperature for HSM; pH, acid requirements for acid hydrolysis) to achieve maximum disruption (90 – 100%). Central composite factorial design complemented with response surface methodology (RSM) approach will be used to reduce the number of experiments and develop predictive mathematical models.

For the freshwater species (*C. pyrenoidosa, A. protothecoides*), the combination of autolytic incubation with PEF-treatment will be investigated in parallel. As autolytic incubation is known to increase cell wall permeability, a combination with PEF can be a promising approach to reducing disruption energy demand. In our previous studies on *A. protothecoides*, KIT has already shown that typical energy demands are of max. 1.5 MJ/kg biomass and can reasonably be reduced to 0.5 MJ/kg biomass by optimizing the processing pathways. Similar protocols will be transposed to *C. pyrenoidosa* and efficiency of the treatment will be assessed by flow cytometry. Results from autolytic incubation + PEF hybrid treatment will be compared to those obtained for the other combinations and only the most optimum treatment will be used in T3.2 and, more importantly, transferred to scaling activities in WP6.

The protein released into the aqueous medium of the microalgal slurry as a result of the hybrid treatment (i.e. autolytic incubation + HPH or autolytic incubation + PEF) will be measured via spectrophotometric methods and protein yield quantified in terms of total water-soluble protein in the biomass, demonstrating the cascading biorefinery potential of the approach for the production of both fuels and feed. The application of protein, however, will not be further investigated given the fuel focus of the project.

Task 3.2: Solvent screening, recycling and emulsion management (M13 – 36). Lead: KIT. Participant: UCD.

Solvent selection: screening will be performed on biomass that has been submitted to optimal disruption conditions as determined in Task 3.1. The solvents will be chosen among non-polar solvents and solvents of low-polarity. Solvents which are highly problematic for health, environment and safety issues are excluded. Possible candidates to be considered are: hexane, heptane, cyclohexane, ethyl-acetate, MTBE and 2-MeTHF. Disrupted wet microalgae slurry will be mixed with the selected solvents at increasing biomass concentrations to achieve different solvent usage (2 – 10 L/kg biomass). The mixture will be centrifuged, solvent phase collected and evaporated, and lipid yield measured gravimetrically. The extracted lipids will be further fractionated using solid-phase extraction and quantity of neutral lipids measured via both gravimetry and GC-MS analysis. Lipid extraction efficiency will be calculated as a percentage of neutral lipids available in the biomass, determined via total lipid extraction (Blight and Dyer) followed by solid-phase fractionation. The best solvents for each species are those able to achieve lipid extraction efficiency >80% at low solvent-to-biomass ratio (<5 L / kg biomass). The extraction conditions using these solvents (e.g. extraction duration and agitation level) will be further optimised to reduce solvent usage to < 3.5 L/kg biomass. This target seems reasonable since hexane extraction from PEF-treated wet *A. protoheorids* biomass at KIT and HPH-treated wet *Nannochloropsis* biomass at UCD have been demonstrated to successfully extract up to 80% of the lipids with 2-3 L of solvent per kg biomass.

Under the same development boundary conditions as described above, the use of green or bio-based solvents (D-limonene, ethyl lactate and methyl soyate) will also be tested. First, the technical boundary conditions will be established through comprehensive literature survey. If green solvents appear to be economically promising (based on the volumetric cost of the solvent and yield considerations), they will be included in the solvent screening.

Emulsion mitigation to enable recycling: The issue of solvent/water emulsion formation resulting in solvent losses during centrifugation step of solvent recovery has not been extensively investigated in literature. The phenomenon, however, is difficult to observe and measure at bench scale. For this reason, once extraction conditions have been optimised at bench scale, the best extraction systems will be tested at intermediate scale, i.e. an increase by two orders of magnitude with typically 30 g of biomass (ca. 300 g of wet slurry). The amount of solvent that can be recovered from the extraction process will be monitored and methods that minimize the trapping of solvent in solvent/water emulsions will be investigated and optimised, such as adjustment of pH, temperature or ion strength of the extraction mixture as well as agitating the extraction mixture at different speed and cycles. Methods that do not minimise emulsions and ultimately achieve solvent recovery to >95% will be disregarded.

Work Package No.	**4**	**Lead: UCAM**
Work Package Title	**High-efficiency microalgal fuel conversion and purification**	

Start Month	1	End Month	36

Objectives: To develop and optimise catalytic pathways for high-efficiency conversion of microalgae biomass components (lipids and biomass residue) to aviation fuels and to develop purification methods for the production of high-quality fuels capable to achieving aviation specifications.

WP4 will **a)** develop tailored multifunctional catalytic pathways and chemical looping process for hydrotreatment/hydrocracking of microalgae lipids into HEFA-SPK (Task 4.1, 4.2), **b)** investigate the application of thermo-catalytic reforming + hydro-deoxygenation (TCR-HDO) treatment for the conversion of microalgae biomass residue to a kerosene fuel (Task 4.4), and **c)** investigate purification methods of the fuel molecules for the synthesis of high-quality aviation fuels meeting ASTM specifications (Task 4.3 and Task 4.5).

Description of work:

In this WP, UCAM and Fraunhofer UMSICHT will work together at bench scale (**1-20 g lipids and biomass residue**) to develop advanced fuel conversion and purification technologies for microalgae biomass components.

Task 4.1: Multifunctional catalyst for hydrotreatment conversion of microalgae lipids into HEFA-SPK (M1 – 24). Lead: UCAM.

Previous studies producing SPK from algal oils have indicated problems with yields and catalyst deactivation due to the high content of oleic acid and metals in algal lipids[Error! Bookmark not defined.], which suggests that dedicated catalysts and lipid extraction methods can help optimise the hydroprocessing route with the algal feedstock.

UCAM will start with catalyst development for two-step and one-step hydrotreatment of algae lipids. The two-step process will use transition metal (Ni, Mo, Co, Cu, Fe)-based catalysts for hydrodeoxygenation and Pt-C for isomerisation and hydrocracking, comparing to published results for other feedstocks[59]. The use of transition metals has been demonstrated beneficial for the deoxygenation of oleic acid[60], while the effective removal of other heteroatoms with Pt-based catalysts helps improve the hydrocracking yield from algal-lipids[61].

After learning from the proposed two-step hydrotreatment route, we will then work on developing an algae-specific bi-functional catalyst using transition metals-based catalysts on mild/medium-acidic supports[62] (mesoporous zeolites, silicoaluminophosphates, alumina, graphitic and activated carbon) for one-step hydrotreatment, with and without the addition of sulphidation agents[63] (e.g. dimethylsulphide). Acidic surfaces suppress decarboxylation and affect metal-support interaction; thus, the acidity levels of supports will be controlled by dedicated support pre-treatment steps. The development of bifunctional catalysts will progress systematically, varying Ni-Mo-Co-Cu-Fe/support combinations, with the optimal composition established by employing machine learning (ML) methods. The catalysts will be characterised through temperature-programmed desorption and reduction (TPD-NH3, TPR), gas adsorption analysis (BET), X-ray diffraction, thermo-gravimetric analysis, SEM, TEM, and EDX.

The hydroprocessing steps will be conducted in a bench-top micro-reactor (Autoclave Engineers BTRS, 5 MPa, 650ºC) with a 30 cm-long tubular reactor, operating in a continuous down-flow mode, with two gas-liquid separators (hot and cold) for product fractionation. The composition of the collected fractions will be assessed with GC and LC/MS, with results used to identify main reaction routes (hydrodeoxygenation, decarboxylation) by looking at the product distributions and length of alkane hydrocarbons. Deconvolution of the possible pathways will be conducted using multivariant correlations (principle-component analysis) supported with ML- algorithms for data predictions. Apparent or single-step kinetics will be assessed for selected catalysts in a Carberry Spinning Basket Reactor (Autoclave Engineers, continuous flow, up to 50 bar).

Results from small-scale experiments will reveal (1) the best single-functional catalysts for a two-step process, and (2) the best bifunctional catalysts for a single-step HEFA-SPK process. We will then try the final approach: a combination of the catalytic material in one reactor (i.e. multilayer catalytic bed) for a novel combinatory single-step algae-lipid HEFA-SPK conversion process.

In months 1-36 of the project, in anticipation of a shortage of microalgal lipids generated within the project, HEFA-SPK experiments at the mL-production scale will first be carried out with in-house blended lipids that mimic the algal lipid composition. The blend will be created by mixing different vegetable oils at various ratios to produce fatty acid compositions similar to those extracted with low-polar solvents from the three microalgae species investigated in SusAlgaeFuel. Lipid composition results from WP3 will guide the blending process. The lipids will also be enriched in metals (Mg, Ca, Cu) to assess their effect on catalyst deactivation in the HEFA conversion steps.

Task 4.2: Chemical looping to reduce H2 and energy requirements during HEFA-SPK synthesis (M13 – 36). Lead: UCAM.

Small-scale experiments in Task 4.1 will provide information on catalysts and reaction mechanisms in hydrotreatment. In the next step, we will innovate the hydrotreatment process route by introducing a **Cambridge-developed flagship process** to produce hydrogen via a chemical looping route from propane (a by-product from HEFA hydrotreatment). An iron-containing oxide, exposed to the gas product stream from the HEFA route, will reduce while consuming propane. In a separate step, the reduced oxide will be regenerated with steam, producing H2. This H2 will be recycled to the HEFA-SPK process, making it more self-sufficient (**Figure 1.2b**).

[59] DOI: 10.1016/j.biombioe.2017.07.008
[60] DOI: 10.1016/j.renene.2019.03.030.
[61] DOI: 10.1021/ef301977b.
[62] DOI: 10.3390/catal9121063
[63] DOI: 10.1016/j.apcata.2014.11.007

During Task 4.2, first, we will assess the achievable H_2 yields, dependable on the propane stream parameters and applied temperatures. Then, we will investigate the heat requirements and heat release in both chemical looping steps, searching for possible heat integration opportunities. Finally, we will assess the influence of impurities in the HEFA-propane stream on the longevity of the chemical looping oxide. Task 4.2 will produce a novel HEFA-SPK route with renewable H2 sourced internally. Information on the optimised process parameters in HEFA-only and HEFA-H2 configurations will enable further integration work and the sustainability assessment of algal-dedicated pathways to SPKs in subsequent WPs.

Task 4.3: Algal HEFA-SPK fuel purification (M13 – 36). Lead: UCAM.

SPK samples from Tasks 4.1 will be assessed for their carbon distribution in normal, iso and cyclo-paraffins, and remaining content of other types of hydrocarbons and heteroatoms, as property indicators. The analysis will be based on high-throughput gas and liquid-chromatography, combined with mass spectroscopy and CHNS. Impurities at a lower level (metals, halogens) can still affect the fuel performance, thus, will be investigated using ICP-MS. Physicochemical properties from thermogravimetric experiments (thermal stability, volatility) will be correlated to the obtained results for composition. We will then consider suitable purification techniques, assessing the potential routes depending on the required level of separation. Those will include further hydrothermal treatment (polishing and refining) using the equipment from Task 4.1, caustic treatment in batch experiments, and commonly used separation methods, mainly filtration and distillation. For the latter, we will device a microdistillation, microfluidic unit, suitable to work with the small feedstock of samples produced in Task 4.1. After the purification steps, samples will be again assessed for their composition and physicochemical properties, comparing the obtained results to the literature and state-of-the-art SPKs that fulfil ASTM requirements. If the property indicators proposed here suggest that ASTM requirements cannot be met directly after the purification steps, further discussions will consider potential fuel blending. The large set of data from Task 4.3 will enable us to devise a decision toolbox, proposing appropriate purification steps tailored to algal-derived fuels.

Task 4.4: TCR-HDO treatment for the conversion of microalgae residue to kerosene mixture (M13 – 36). Lead: UMSICHT.

Fraunhofer UMSICHT will purchase algae biomass residue from commercial suppliers to carry out preliminary TCR investigation and develop specific experimental protocols for microalgae biomass (M19 – 36). The specific activities to be carried for this Task are: 1) pre investigations of the pyrolysis condition in microscale using thermogravimetric analysis (TGA) and/or differential scanning colorimetry (DSC), 2) in lab scale, pelletisation of algal biomass to make sure that they are compatible with the TCR reactor (TCR2 plant), 3) screw modification of TCR reactors to algal feedstock and adaptation of the TCR's post-reforming section (fitting the height of the char bed) to ensure the right gas velocity and gas residence time within the bed, 4) optimisation of TCR operation through several trials to produce maximum oil yield from residue, 5) analyses of products: oil, gas and char in terms of their higher heating value (HHV), water content, elemental and mineral contents (ICP), and composition (gas online and gas calorimeter online), 6) hydro-deoxygenation or hydro treatment of oil fraction using different catalysts to produce drop-in biofuel comprising a mixture of petrol, kerosene and diesel with physical and chemical characteristics that, after purification, are compliant with requirements according to EN590 and EN228 standards. Activities 4-6 will be repeated on the biomass residue feedstock obtained in WP6 using the procedure optimised on commercial biomass in Task 4.3.

Task 4.5: Algal TCR-HDO fuel purification (M13 – 36). Lead: UMSICHT.

After the TCR-HDO steps of the algae residues, the purification of the generated hydrocarbons will be carried out. For standard hydrocarbons, a distillation will be the required method, while in certain cases, an extraction with non-polar solvents e.g. pentane might be required. Distillation will generate several fractions that will be isolated according to boiling temperature (petrol fraction from room temperature to 170 °C; kerosene fraction from 170 °C to 230 °C, and diesel fraction from 230 to 350 °C. The residue with boiling point higher than 350 °C can be recycled to the TCR plant for further thermal degradation. Extraction of fuel molecules with non-polar solvents is needed if there are still heterocompounds remaining in the hydrogenated fuel (e.g. sulfonated compounds due to a lack of hydrogenation). In this case, the solvent will be applied to extract the fuels out of the hydrogenated product.

The content of aromatic compounds in aviation fuels is an important indicator of the overall quality of the fuel. As they are required to swell the gaskets in the tank of the aviation, a certain amount aromatic compounds are needed in the (post purified) kerosene fraction. If the concentration of aromatics in the kerosene fraction falls below certification requirements, we can extract additional aromatics from the petrol fraction or diesel fraction and used the extracts to enrich the chemical profile of the kerosene fraction.

Work Package No.		**5**	**Lead: SFS**
Work Package Title		**Process integration and whole life-cycle analysis**	
Start Month	**1**	**End Month**	**48**

Objectives: To assess the commercial scalability, well-to-tank social, environmental and GHG-emission impacts of the investigated microalgae biofuel production strategies in relation to systemic constraints.

WP5 will **a)** use process simulations for the assessment of commercial scalability (Task 5.1), and **b)** carry out a comprehensive Whole Life Cycle Analysis (including social LCA) to assess the sustainability of the process (Task

5.2). This WP will use results obtained from bench-scale development in WPs1-4 for their analysis and provide feedback to the WPs to enable iterative improvement. WP5 results will also guide decisions for scale-up in WP6.

Description of work: In this WP, UCD and SFS will work together to conduct a rigorous evaluation of techno-economic and socio-environmental impacts of the proposed value chain.

Task 5.1: Techno-economic assessment and identification of major cost drivers (M13 – 36). Lead: UCD.

UCD will use various process simulation software (e.g. Aspen Hysis, Pro-II) as well as open-source models to project the scalability of the proposed system if operated to produce 10,000,000 kg of algae HEFA-SPK per annum. The plant design will be based on the methods developed in WPs 1–4 utilising the data generated in the laboratory studies (such as growth rates, yields, water, energy and chemical requirements) to validate the model assumptions. We will utilise generally accepted cost models for CAPEX and OPEX for current biorefineries. Two integrated process plant models will be created, one using a commercial refinery simulation software and the other an open-source approach to enable a comparison with other standard refinery models and afford the flexibility to evaluate uncertainty and improve optimisation. The production of algal biomass, the extraction of lipids from biomass, and the conversion of lipids to fuel will be simulated and the costs per litre of SAF determined. An optimization analysis and uncertainty analysis will also be carried out to allow for the identification of major cost sensitivities and provide early feedback to focus laboratory efforts in other WPs during bench-scale experiments.

When developing the cost model, we will also investigate other means of enhancing the techno-economic and commercial viability of the biorefinery processes, including the recycling of water (e.g. reusing water obtained after biomass harvest for subsequent cultivation cycles) and heat recovery within the biorefinery. We will also carry out a sensitivity analysis to measure the impact of process variations (e.g. growth conditions during cultivation and solvent-to-biomass ratio during lipid extraction) and uncertainty in the supply chain (e.g. availability of digestate, availability of water and transport distance for raw materials/nutrients) on the yield and overall cost of production of fuels. The task will provide a roadmap for cost reduction beyond project lifetime.

Task 5.2: Life-cycle analysis and identification of socio-environmental hotspots (M13 – 48). Lead: SFS.

The Whole Life Cycle Analysis (W-LCA) process encompasses four key components: goal definition and scoping, inventory analysis, impact assessment, and interpretation. Additionally, W-LCA incorporates a Social Life Cycle Analysis (S-LCA) as an integral part of the full system Life Cycle Assessment (LCA). Standardisation is a crucial part of the LCA process for SusAlgaeFuel. This Task ensures cross-party standardisation for the first two LCA components: boundary conditions (goal definition and scoping) and inventory analysis for microalgae-based SAF production. Establishing robust cooperation among various WPs is crucial for attaining accurate and reliable results. In this Task, SFS will conduct a comprehensive assessment of the environmental and sustainability impact of the proposed microalgae value chain. The evaluation will utilise the latest approaches and simulations in life-cycle analysis (LCA) at both individual stages (microalgae cultivation, biorefinery, and fuel conversion) and the full system level, with a strong focus on carbon mass balance. A well-to-wake Life Cycle Analysis will be carried out following EU RED II (Plus upcoming RED III) and ICAO's CORSIA methodology, utilising SimaPro LCA software with EcoInvent's latest databases. The objective is to determine the CO2-eq/MJ value of the neat and blended algal SAF, including core LCA values and separately assess induced land-use effects (ILUC). The LCA will compare the environmental impact of the proposed system to conventional microalgae systems, including cultivation using digestate and direct CO2 capture from AD industry. A wide range of environmental indicators, such as climate-change, air-quality, water-quality, and non-renewable energy indicators, will be considered, providing results on a full range of Characterization Factors using CML 2001 methodology.

To establish specific emission values of the blended SAF produced, the Greenhouse gases, Regulated Emissions, and Energy in Transportation (GREET) software (USA) will be utilised. Empirical verification will be conducted in partnership with an aircraft/engine Original Equipment Manufacturer OEM from the ATM D4054 test facility index. SFS will apply the RSB's 12 Principles of sustainability during WP5 to help align fuel production with international best practice and obtain 'RSB EU RED Fuel' Certification. This certification is recognised by the European Commission, demonstrating compliance with the EU RED and RSB's stringent sustainability principles.

Identification of opportunities for improvements in microalgae fuel production will involve calculating hotspots, calculated using bench-scale results from WPs 1-4 in the initial phase of the project (M13 – M36), and later derived/validated using project results from the scaling activities in WP6, as well as relevant information from existing literature. These findings will be fed back to the WPs to facilitate informed improvement and optimization. The final CO2-eq/MJ value for the investigated microalgal value chain will be compared with ICAO CORSIA default LCA values for similar SAF pathways and microalgae SAF pathways by other pathways. The WP will establish the final LCCA, including the full levelized cost of energy (LCOE).

In parallel, a social LCA (sLCA) will be carried out to explore the social impacts and sustainability of the developed products and technologies, and their actual and potential positive as well as negative impacts along the life cycle. We will conduct a social assessment of the developed process and the pilot scale facility (built in WP6) to identify the main social risks and social impact categories (e.g working conditions, health and safety impact) associated with the process. Social analysis is important for all value chain actors. Key stakeholder groups will be identified, in collaboration with stakeholder engagement strategy activities in WP7, and accordingly social impacts will be assessed through sLCA study parameters. Limitations will be determined by considering that all stakeholders such as employees, raw-material suppliers and end users will be affected in the project value chain.

Work Package No.		6	Lead: A4F
Work Package Title		Pilot scale-up, fuel testing and digital twin	
Start Month	25	End Month	48

Objectives: To demonstrate the technical scalability of the technologies developed in WP1-WP4 to TRL4/5 through the build of a pilot-scale system an AD operator site (BIAE) and the operation of the system for the production and quality test of algal HEFA-SPK and TCR-HDO fuels according to ASTM standards.

WP6 will **a)** build a pilot-scale algal production facility with the capacity of producing 10 kg of algal lipids p.a. on BIAE site (Task 6.1), **b)** operate the pilot system to produce sufficient lipids for the manufacture of 1 kg of algal HEFA-SPK (Task 6.2), **c)** convert the generated algal lipids and biomass residues to 1 kg of HEFA-SPK and TCR-HDO fuels respectively, purify the fuels and test them for compliance with ASTM standards (Task 6.3), and **d)** develop a digital twin of the integrated fuel production system as a decision toolbox (Task 6.4).

Description of work: A4F, BIAE, and UCD will first build/procure, and commission pilot-scale infrastructure comprised of cultivation and biorefinery unit operations on partner BIAE site (Kilshane, Ireland) according to the optimum technologies/specifications found in WP1-WP3. The work will demonstrate a scalable closed-loop system where microalgae growth is supported using waste nutrients directly derived from on-site AD operation. Once built, the pilot-scale system will be operated semi-continuously throughout Year 4 by the UCD team, after a period of training supplied by A4F. The operation will provide the lipid and biomass residue needed for fuel conversion and testing. Once sufficient amounts of lipids have been accumulated for the production of 1 kg of algal HEFA-SPK, both lipid and biomass residue will be dispatched to UCAM and UMSICHT for fuel conversion via the HEFA-SPK and HDO-TCR pathways, respectively. Rather than building a new pilot-scale infrastructure, fuel conversion will utilise large-scale infrastructure that already exists at UCAM and UMSICHT capable of processing the required fuel targets (1 kg of HEFA-SPK from the lipids and a minimum 1 kg of TCR-HDO oils from biomass residue) to reduce project costs. These infrastructures will be modified according to the specific requirements for algae as investigated in WP4. The resulting fuel molecules will then be purified according to the methods developed in WP4 and assessed for compliance to ASTM standard. In parallel, IRIS will develop a digital twin of the integrated system to simulate production performance under different scenarios/conditions, serving as a decision toolbox to guide stakeholders.

Budget lines for building and commissioning of pilot-scale infrastructure have been distributed between BIAE, A4F and UCD, while appropriate budget for modifying fuel-conversion rigs has been allocated to UCAM & UMSICHT.

Task 6.1: Design, build, and commission of a pilot-scale system comprised of cultivation and biorefinery units (M25 – 36). Lead: A4F. Participants: BIAE, UCD, IRIS.

In this task, the detailed engineering design of the microalgae cultivation system will be developed by A4F according to the developments of WP1. This includes the photobioreactor for autotrophic cultivation (**1,000 L**), a pre-treatment system (microfiltration unit and/or UV irradiation system) to treat digestate and remove contaminating bacteria as developed in WP1 and WP2, a membrane-based CO2 delivery system to directly capture flue CO2 from biomethane upgrading facility as investigated in WP1, and a PAT spectroscopic sensor system to monitor bacterial level in the culture as developed in WP2. BIAE will carry out the civil engineering work needed to prepare their site for algae infrastructure, before building and commissioning the system with the help and guidance of A4F. Unit operations needed biorefinery fractionation will be designed and sized by UCD according to the development in WP3. The system will include a cell disruption unit (selected based on performance in WP3), a lipid extraction unit, a phase separation unit, and a solvent recovery system with an integrated heat recovery system. The individual units will be purchased from commercial suppliers and assembled by UCD Workshop on BIAE site, adjacent to cultivation unit.

Task 6.2: Operation of pilot system for production of lipids and biomass residue (M37 – 44). Lead: UCD. Participants: BIAE, A4F.

After receiving training from A4F, UCD will operate the pilot-scale system built on BIAE site semi-continuously for 3 – 8 months in order to generate the lipids and biomass residue needed for fuel production in Task 6.3. Only one species will be selected for the operation; the selection of which will be made based on findings from bench-scale experiments in WP1. Growth and biorefinery conditions identified in WP1 and WP3 will be initially adopted, but will be further optimised in order to achieve maximum final biomass concentration (>3 g biomass/L) and lipid accumulation (> 30wt% biomass) within short harvest cycle (7-14 days) for the growth system and the recovery of >80% lipids at energy demand < 1 MJ/kg biomass for the biorefinery system at pilot scale. The performance of in-line optical sensors in analysing bacteria/microalgae population dynamics will be analysed using real data obtained from the operation of the pilot system and compared against baseline methods (e.g. flow cytometer + staining). System performance data of the pilot system (e.g. energy and water requirements, solvent requirement) will be fed to WP5 (Task 5.2) for iterative life-cycle analysis and projection of environmental impacts of the proposed system, and to Task 6.4 for the development of a digital twin. The extracted lipids and biomass residues will be stored at -20°C until sufficient amounts have been generated for Task 6.3.

In parallel to the biorefinery work carried out in the pilot-scale facility, A4F will investigate bead milling for wet cell disruption as baseline/standard method to provide direct performance comparison to the system in Task 6.1.

Task 6.3: Pilot fuel production, purification and testing for compliance with ASTM standards (M31 – 48). Lead: UCAM. Participant: UMSICHT.

From month 25, the team at UCAM we will start designing a new rig/modifying our existing rig at UCAM to scale up the production of algae-derived HEFA-SPK. Conclusions from Tasks 4.1, 4.2 and 4.4 will guide the design, primarily including the hydrotreatment and purifications steps. If the scale up is successful, H_2 production using our chemical looping approach will be added, connecting the new scaled up HEFA rig to our existing fixed bed reactors for thermochemical water splitting. The main goal of designing and building the scaled up pilot is to achieve 1 kg of HEFA-SPK product from the lipids generated in pilot-scale production in Task 6.2.
The biomass residue generated from pilot-scale production will be processed at UMSICHT for the production of TCR-HDO oil. UMSICHT can operate a TCR 2 plant with a throughput of 2 kg of biomass (currently available at UMSICHT site from previous project TOSYNFUEL). Drying and pelletisation of the algae residue will be required beforehand. The oil will then be hydrotreated and distilled/extracted by solvents.
The resulting HEFA-SPK product (UCAM) and final TCR-HDO biocrude oil (UMSICHT) will be purified using the methods developed in Task 4.3 and 4.5 respectively before being analysed for compliance with ASTM D7566 & DEFSTAN 91-91 standards by accredited service provider. Parameters to measure include acid number, content of aromatics and sulphur, density at 15°C, flash point, freezing point, viscosity at 20°C, heat of combustion, smoke point, naphthalenes, corrosion copper strip, thermal stability, metal content, halogens, N content, water content.

Task 6.4: Digital twin of the pilot system (M37 – 48). Lead: IRIS.

IRIS will develop a digital twin of the pilot-scale facility based on an agent-based approach. The digital team from IRIS will engage with the design team from Task 6.1, to understand the design and operation of the pilot-scale algal SAFs production system and its various components, processes, equipment, and their interactions. The critical variables and parameters that affect the production process and performance will be identified (e.g. algal growth rate, nutrient levels, and any other relevant factors). A detailed model of the pilot-scale system will be built, and each component, process and identified variables and parameters will be represented with appropriate equations, models, or algorithms (agent Interactions): and incorporated into the model. Real-world data from the operation of the pilot-scale production system in Task 6.2 will be implemented into the digital twin for both 'calibrating' the parametrizable features and building reduced-scope (dedicated) models, when required, to define the agents behaviour. The model's accuracy will be validated by comparing its predictions with actual production data from the pilot plant. A series of different scenarios (such as varying nutrient inputs, light conditions, or different operational strategies) will be defined to explore the behaviour of the production system under different conditions,. The simulation outputs, i.e. the result from running the whole digital twin under certain parametrization (such as algal biomass production, SAFs yield), will be analysed to understand the impact of each scenario on production performance and to help in optimising the process and identify potential bottlenecks.

Work Package No.		**7**	**Lead: ESCI**
Work Package Title		**Communication, policy and exploitation roadmap**	
Start Month	**1**	**End Month**	**48**

Objectives: To develop and implement successful strategies for dissemination, communication and engagement with stakeholders in order to maximise the impact of the project outcome and ensure its legacy.
WP will **a)** implement a comprehensive communication and dissemination strategy to reach relevant stakeholders (Task 7.1-7.3), **b)** manage IPs and explore exploitation pathways for results commercialisation (Task 7.4), and **c)** produce policy recommendations on algae SAFs at EU levels (Task 7.5).

Description of work:

Task 7.1: Communication, dissemination and exploitation strategy (M1-48). Lead: ESCI. Participants: All.

The Dissemination; Exploitation and Communication Master Plan (DECMP) (D7.1) as outlined in section 2.2 will outline the multi-layered activities to maximise impact of the project results. Dissemination, Communication and Exploitation actions will be monitored using MATOMO (a GDPR friendly version of Google Analytics) and BRANDWATCH for online activities, and questionnaires for events. Monitoring results will feed into Best Practices of Communication, Dissemination and Dialogue Actions (D7.2; M18). The first version of D7.2 will be used to update the CDMP in M20. ESCI will train partners (2 courses) on maximising impact with social media. The consortium will produce open-access peer-reviewed articles in high-impact scientific journals and present their results in relevant conferences and workshops. Within the last three months of the project, a final event will be held to present the project results to 100+ stakeholders. The Conference will include open sessions, round tables and workshops to enable different groups to get engaged during the event. UCD and ESCI will also produce a comprehensive Technical Brochure in an easy-to-understand language (D7.3). Research training will be offered on microalgae-related research, including an algae fuel summer research school for undergraduate students.

Task 7.2: Communication activities (M1-48). Lead: ESCI. Participants: All.

A variety of editorial, promotional and visual content will be developed to share on the project's channels and media multipliers. The SusAlgaeFuel website and social media channels (e.g. Twitter, Linkedin) will become a meeting place for all stakeholders and establish links with other relevant projects and initiatives. Furthermore, promotional communication materials promoting the project and/or key results will be developed, including leaflet, infographics; journalistic articles and interviews with technology and environment experts, end-users and other stakeholders. In addition, 3 videos will be produced, an introductory project video and one final summarising results, and 1 Video News Releases to be distributed through ESCI's network of 250+ broadcast journalists to ensure broadcasting in 3

national TV stations across Europe. Project visibility will be enhanced with a Women-in-Science campaign promoting female scientists on SusAlgaeFuel.

Task 7.3: Stakeholder engagement and networking (M1-48). Lead: ESCI. Participants: SFS, All.

A mapping exercise will be initially carried out to identify key stakeholders relevant to the project. ESCI will extract this list from publications, media partners, CORDIS database, and websites across Europe. Of particular importance is creating a data set of previous/current EU/H2020/HE projects that have been funded in areas under CSA or RIA call topics related to microalgae biofuels or sustainable aviation. This includes other projects funded under the call topics HORIZON-CL5-2023-D3-02-08, HORIZON-CL5-2022-D3-03-07, HORIZON-CL4-2021-RESILIENCE-01-16. Based on identification and mapping of relevant stakeholders for the project, SusAlgaeFuel will connect with other EU projects in the area to regularly exchange information and results, cooperate at workshops, conferences, and promote mutual project activities, such as joint webinars/seminars and cross-fertilisation activities. Additionally, SusAlgaeFuel will carry out regular demonstration workshop at our pilot-scale production facility (after the build is completed in M36) and invite key stakeholders relevant to the project from both Ireland and the EU to the workshops in order to increase awareness and promote pathways for commercialisation. SFS, UCAM and BIAE will also send communication bulletins to their wider network of fuel-industry and AD stakeholders, including SkyNRG, Avolon, Boeing, Orix Aviation, Marshall Group, Phillips 66.

Task 7.4: Exploitation pathways and business-model development (M1-M48). Lead: UCD. Participants: all.

Building from Section 2.2.3, UCD will draft the Consortium Agreement in M1 (to be signed by all project partners) outlining the project's IP management strategy. A review and analysis of the most appropriate IP tools to protect potential research results will be carried out and catalogued during the preparation of the DECMP plan. The Agreement will also outline the procedure to determine ownership of foreground IPs generated during the project, allowing each partner to patent their own set of results, and exploit IPs. UCD, through its experienced technology transfer office (TTO) and innovation hub NovaUCD, will leverage its expertise in supporting the partners in managing their IP and commercialisation activities. Project results will be analysed by the respective partners as they emerge for Freedom to Operate (FTO), patentability and protectability with up-to-date reviews.

UCD will lead partners in developing a roadmap for commercial-scale deployment of the proposed value chain in the EU according to feedstock availability and distribution, and (anticipated) policy direction in renewable energy. On the one hand, the Business Model Canvas (BMC) will be used as a lean tool for facilitating business model development for each of the industrial partners (A4F, BIAE, IRIS, SFS) to define the core aspects of the commercialisation of their results. These BMC will then be iterated out to a more complete business/commercialisation plan, which will also consider financing route for the post project development lifecycle as well as the funding of the market entry towards TRL9 activities, in order to ensure a competit8ive Time to Market (TTM). On the other hand, because the proposed system will combine several stakeholders (AD operator, microalgae producers, sensor technology developer, fuel producer, various aviation stakeholders) in separate sectors within a single value chain, the financial benefits of the system for the individual stakeholders need to be verified. Of particular importance is understanding the potential market and revenue stream of the systems for each stakeholder and how this can translate to a profitable undertaking/model for all stakeholders in the future. This task will involve carrying out research/market interviews with industry stakeholders and lead to the publication of a commercialisation plan outlining strategies for adoption by stakeholders (D7.4).

Task 7.5: Policy analysis (M25 – M48). Lead: SFS.

SFS will provide up-to-date information on the EU policy landscape over the project lifetime to greatly improve the environmental and commercial viability of the SAF, as policy compliance can be assured from an early stage. Attention will be given to advanced biofuel targets, energy multipliers and certificates awarded, as well as incentives and mandates to de-risk potential investment in microalgae biofuels. Furthermore, SFS will contribute to national government consultations to promote the introduction of advanced biofuels and accelerate the phase-out of fossil fuels. A membership of the Roundtable of Sustainable Biomaterials (RSB) through SFS offers the project access to a real-life sustainability framework. Through networking, webinars and exclusive RSB Platforms for SAF and policy discussion, the project aims to achieve RSB certification (D7.5), recognised by the EC to meet stringent RED requirements. This RSB certification will underpin the case for up-scaling and trading the final SAF product within the EU, advancing the business case for future investors. At the end of the project, SFS will publish a white paper consisting of policy recommendations to promote algae SAFs based on project findings (D7.6).

Work Package No.		**8**	**Lead: UCD**
Work Package Title		**Project coordination**	
Start Month	**1**	**End Month**	**48**

Objectives: To implement effective project management for timely achievement of milestones and deliverables: **a)** To coordinate all scientific, gender, administrative and financial activities (Task 8.1); **b)** To identify risks and implement mitigation (Task 8.2); **c)** To manage data according to FAIR and open-science principles (Task 8.3).

Description of work:

Task 8.1: Scientific, administrative, and financial coordination (M1 – M48). Lead: UCD. Participants: All.

The Executive Board EB (comprised of Project Coordinator and WP leaders) will effectively coordinate the interdisciplinary project activities according to the proven project management structure outlined in Section 3.1 and

monitor the overall project progress to ensure timely achievement of milestones and deliverables. The Project Coordinator (Dr. Halim, UCD) support by Project Manager (hired at UCD) will be responsible for all aspects related to financial management, including payments to the partners and financial reports to the EC, and project administration, which includes the preparation of the Consortium Agreement, the timely delivery of project reports to the EC, preparation/implementation of Data Management Plan (D8.1), and consortium meetings. UCD is experienced in inclusive project coordination and will ensure effective management of gender and ethics aspects.

Task 8.2: Risk management and contingency planning (M1 – M48). Lead: UCD. Participants: All.

The EB will continuously monitor the project progress and, in the event of significant project delay/deviation, design corrective actions or authorise the execution of risk mitigation strategies described in **Table 3.1e.** The EB can also propose an amendment to the project plan and deliverables in cases when risk mitigation strategies were unsuccessful in resolving risks. The EB will meet online every three months to discuss contingency planning.

Task 8.3: Data management and open science (M1 – M48). Lead: UCD. Participants: All.

The project will adopt a sound data management strategy in compliance with FAIR practices as proposed in Section 1.2.8. The project coordinator (UCD) will draft a full DMP by month 6 (D8.1). SusAlgaeFuel will champion an open-science practice as described in Section 1.2.7, including open-access publications for all journal papers resulting from the project via the ´gold´ or 'green' open access route and curation of data in open-access repositories.

Table 3.1c: List of Deliverables

No.	Deliverable Name	Short Description	WP	Lead Part	Type	Delivery
1.1	Digestate pretreatment	Report: effective methods to treat liquid digestate for microalgal medium formulation	1	UCD	R (SEN)	M8
1.2	Microalgae growth strategies	Report: effective cultivation strategies on digestate to achieve biomass concentration >3 g/L	1	A4F	R (SEN)	M24
1.3	CO_2 delivery / digestate bioremediation system	Report: microalgae system integrating membrane for CO_2 delivery and digestate bioremediation,	1	UCD	R (PU)	M36
2.1	In-line sensor for bacterial/microalgae quantification	Report: development and performance of sensor utilising various spectroscopic principles as in-line methods to measure bacterial level in culture.	2	IRIS	R (SEN)	M24
2.2	UV irradiation system	Report: development and performance UV irradiation system to manage bacterial population	2	UCD	R (SEN)	M24
2.3	Algae/bacteria population	Report: algae-bacteria interaction on digestate and how they can be harnessed to improve growth	2	UCD	R (PU)	M36
3.1	Low-energy cell disruption based on autolysis	Report: low-energy biomass treatment for processing/fractionating wet biomass slurry (target energy demand < 1 MJ/kg biomass).	3	UCD	R (SEN)	M18
3.2	Lipid extraction strategy	Report: low-polar solvent screening, solvent consumption, recycling and emulsion mitigation	3	KIT	R (SEN)	M36
4.1	Algae-specific HEFA-SPK conversion route	Report: C-product distribution, mechanisms for algal HEFA-SPK route (>50% conversion)	4	UCAM	R (SEN)	M24
4.2	Chemical looping system to reduce H2	Report: performance of the chemical looping for H_2 production from propane.	4	UCAM	R (SEN)	M36
4.3	Algae-specific HEFA-SPK purification	Report: composition and properties of fuels before and after purification steps.	4	UCAM	R (SEN)	M36
4.4	Algal specific TCR-HDO conversion route	Report: mass and energy balance of the conversion process and yield of hydrogenated product	4	UMSICHT	R (SEN)	M36
4.5	Algal specific TCR-HDO purification	Report: method development for purification of TCR-HDO oil to produce hydrocarbons for drop-in fuels.	4	UMSICHT	R (SEN)	M36
5.1	Commercial scalability assessment	Report: techno-economic assessment of the proposed value chain for production of algal SAF	5	UCD	R(PU)	M36
5.2	Environmental and social impacts	Report: the well-to-tank environmental and social impact assessment of algal SAF (LCA, S-LCA)	5	SFS	R(PU)	M48
6.1	Production of biomass using pilot facility	Report: the performance and resource needs for pilot-scale production of microalgae biomass.	6	BIAE	R(PU)	M42
6.2	Production of protein, lipids and residue using pilot facility	Report: the performance and resource requirements of pilot-scale production of microalgae lipids and by-products	6	UCD	R(PU)	M44
6.3	Production of fuel using pilot facilities	Report: Production of algal HEFA-SPK and TCR-HDO fuels, assessment of quality (ASTM standards).	6	UCAM	R(PU)	M48
6.4	Digital twin simulations	Report: development and performance of digital twin	6	Iris	R(PU)	M48
7.1	DEC Master Plan	Provides overarching strategy on communication, dissemination, and exploitation activities	7	ESCI	R(PU)	M6, M20
7.2	Best Practices of DEC	Report: key DEC activities of Month 0-18.	7	ESCI	R(PU)	M18
7.3	Technical Brochure	Provide overview of project results	7	UCD	R(PU)	M48
7.4	Commercialisation plan	Comprehensive commercialisation / research exploitation plan for project results	7	UCD	R(PU)	M48

7.5	RSB EU RED Fuel Certification	Attain the RSBs EU RED Certification to achieve internationally recognised sustainability standards	7	SFS	R(PU)	M48
7.6	Policy advice	**White paper** promoting algae SAFs in the EU	7	SFS	R(PU)	M48
8.1	Data Management Plan (DMP)	A FAIR-compliant plan on how to collect, store and share data/research outputs	8	UCD	R(PU)	M6

Table 3.1d: List of Milestones

No.	Milestone Name	WP	Due Date	Verification
1.1	Digestate-based culture media formulated for autotrophic growth	1	M6	Method running
1.2	Build and operation of membrane system for CO_2 delivery	1	M12	> 90% CUE
3.1	Demonstration of autolytic cell disruption	3	M12	<1 MJ/kg biomass
3.2	Demonstration of wet processing pathway for recovery of lipids	3	M24	<3.5L/kg biomass
4.1	Efficient HEFA-SPK catalysts and process configurations	4	M12	>50% conversion
4.2	High conversion of propane and H_2 production corresponding to >90% of attainable FeO to Fe_3O_4 conversion.	4	M24	Method running
4.3	Minimal combination of required purification steps to reach properties and proxy indicators for ASTM quality standards.	4	M30	Method running
4.4	Process for conversion of algae to TCR crude oil to hydrogenated oil	4	M22	Method running
4.5	Kerosene fraction from hydrogenated algae oil	4	M30	Method running
5.1	Accurate Life Cycle Inventory	5	M22	Method running
6.1	Design of pilot plant completed	6	M27	Method running
6.2	Build of pilot plant completed on AD site	6	M36	Infrastructure
6.3	Production of wet microalgae slurry using selected species and optimum cultivation strategies	6	M42	ca. 5 kg biomass
6.4	Production of microalgal lipids using optimum biorefinery	6	M44	>1 kg lipid
6.5	Production of 1 kg of microalgal HEFA-SPK and a minimum of 1 kg of microalgal TCR-HDO oil using optimum conversion systems	6	M46	>1kg HEFA and 1kg TCR-HDO

Table 3.1e: Critical Risks for Implementation #@RSK-MGT-RM@#

Description of Risk	WP	Risk Mitigation Measures
Task 1.1: Digestate was found to be toxic for microalgae. Likelihood: low, severity: medium	1	After identification of toxic compounds, strategies to reduce toxicity will be used (e.g. fed-batch feeding).
Task 1.4: Slow rate of CO_2 delivery into microalgae culture by the membrane carbonation system.	1	Operating pH of microalgae culture to be optimised to drive the forward equilibrium of CO_2 dissolution. Light intensity to be optimised to maximise photosynthesis.
Task 2.1: Algae create a fouling layer of "kamikaze" cells that block any harmful light. Likelihood: high, severity: high	2	Effective probe design utilising multiple spectroscopic methods.
Task 2.3: Plating medium favours another class of bacteria compared to liquid medium. Likelihood: medium, severity: medium.	2	16S rDNA will also be conducted on the liquid media and alternative plating media will be proposed.
Task 3.1: Different species exhibiting varying degree of cell-wall weakening during autolytic incubation. Likelihood: medium, severity: low.	3	Conditions for autolytic incubation will be made more extreme to induce cell-wall weakening (e.g. longer incubation time, N_2 sparging)
Task 3.2: Stirring forms emulsions resulting in significant solvent loss (<95% solvent recovery target). Likelihood: low, severity: medium	3	Milder mixing technologies and other techniques to minimize solvent trapping (e.g. pH adjustment) will be simultaneously investigated and combined.
Task 4.1: Low yield of suitable hydrocarbon chains from HEFA-SPK reactions with algal lipids. Likelihood: medium, severity: medium	4	Multiple strategies in hydrotreatment steps, using separate hydrodeoxygenation, isomerisation, cracking catalysts, and catalysts will be pursued.
Task 4.2: Unsuccessful integration of HEFA-SPK with H2 production. Likelihood: low, severity: medium.	4	Separate the two subprocesses to produce SPK, propane and H_2 independently; this will still allow the new route to be more self-sufficient.
Task 4.3: Algae TCR oil leads to low-quality fuel. Likelihood: medium, severity: medium	4	A more drastic set of temperature and pressure to be applied during hydrotreatment.
Task 4.4: Purification steps are too complex and costly. Likelihood: low, severity: high.	4	Change the hydrotreatment steps in Task 4.1 to minimise purification requirements.
Task 5.2: Lack of accurate available data for each stage of life cycle, particularly from real world SAF manufacturing facilities (due to limited access to commercial-scale plant or IP restrictions). Likelihood: medium, severity: low	5	SFS is currently partnered with the leading global SAF development company SkyNRG conducting a feasibility study for commercial scale SAF manufacturing in Ireland, providing direct access to accurate data in SAF manufacturing.
Task 5.2: Project carrying out activities which are not in-line with the RSB Certification. Likelihood: medium, severity: low.	5	Task 5.2 will start in M1 so help align project activities with RSB EU RED Certification standards from project commencement.

Task 6.1: Delay in the construction and build of pilot-scale cultivation and biorefinery facility. Likelihood: medium, severity: high.	6	A4F will recommend other workshops from their network for the build of photobioreactor. UCD will source equipment from other suppliers.
Task 6.2: Malfunctions in pilot facility delay lipid production. Likelihood; low, severity: high.	6	Prioritise repair before proceeding with WP. In the interim, purchase biomass from commercial suppliers.
Task 6.3: Scaled up production cannot reach the target 1 kg output of HEFA-SPKA.	6	Extend operational duration of pilot-scale facility in Task 6.2. Alternatively, purchase external biomass

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Table 3.1f: Summary of Staff Effort (in person months)

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	**Total per partner**
UCD	18	18	24	0	12	24	4	20	**120.0**
A4F	21	0	1	0	2	40	4	2	**70.0**
BIAE	0.5	0	0	0	0	9	0	0.5	**10.0**
IRIS	0	24	0	0	0	24	1	1	**50.0**
KIT	0	0	38	0	0	0	2	1	**41.0**
UCAM	0	0	0	49.7	0	49.7	0.7	0.7	**100.8**
UMSICHT	0	0	0	21	0	6.8	0.5	1	**29.3**
SFS	0	0	0	0	24	0	10	1	**35.0**
ESCI	0	0	0	0	0	0	24.5	2.5	**27.0**
Total per WP	**39.5**	**42.0**	**63.0**	**70.7**	**38.0**	**154.0**	**46.7**	**29.7**	**483.1**

Table 3.1g: 'Subcontracting Cost' Items

Item	Cost (€)	Description of Tasks and Justification
Partner 9/SFS		
Policy cross check	5,000	WP5, Task 5.2: Third party audit/verification of LCA

Table 3.1h: 'Purchase Cost' Items

Item	Cost (€)	Justification
Partner 1/UCD		
Goods & services	113,922	Purchase of materials, build, & assembly of pilot-scale biorefinery at BIAE site in Kilshane, IE (€70k), purchase consumables for experiments and pilot-facility operation (€40k) (WP1-3, 6), audit (€3,922).
Remaining Costs	74,800	**Total Purchase Costs: €188,722**
Partner 2/A4F		
Travel, subsistence	22,000	Multiple travels to pilot-scale demo site by A4F's team (PT - IE) (WP6)
Goods & services	66,500	Consumables (€48,500) (WP1,WP6), 'Solidwork' software (€18,000) (WP6)
Remaining Costs	31,500	**Total Purchase Costs: €120,000**
Partner 3/BIAE		
Goods & services	281,000	Purchase of materials, service cost to build & assembly pilot-scale cultivation system (1,000L photobioreactor, centrifuge, feeding system) at BIAE site, Kilshane, IE (WP6), based on A4F estimate.
Remaining Costs	4,000	**Total Purchase Costs: €285,000**
Partner 4/IRIS		
Goods & services	80,000	Purchase of optomechanical components for spectroscopy analysers (WP2)
Remaining Costs	28,000	**Total Purchase Costs: €108,000**
Partner 6/UCAM		
Travel, subsistence	25,000	Travel and conference at ca. €2,000/person/year for 3 persons (WP4, 6, 7)
Goods & services	78,000	Modification of HEFA-SPK rigs for high-pressure H_2 delivery (€45k), production of multiple batches of catalysts for bench and pilot experiments (€33k) (WP4, WP6)
Remaining Costs	32,800	**Total Purchase Costs: €135,800**
Partner 7/UMSICHT		
Goods & services	81,000	Purchase of catalysts (€51,000) (WP4, WP6), analysis of TCR-HDO products: (€20,000) (WP6), adaptation of TCR rig for algae (€10,000) (WP4, WP6)
Remaining Costs	27,000	**Total Purchase Costs: €108,000**
Partner 9/ESCI		
Travel, subsistence	11,700	Travel of camera crew for video production (WP7)
Goods & services	6,000	Room hire/catering for final conference (WP7)
Remaining Costs	28,000	**Total Purchase Costs: €45,700**

Table 3.1i: 'Other Costs Categories' Items

	Cost (€)	Justification
Partner 1/UCD		
Internally invoiced goods and services	16,000	Access cost to shared high-spec facilities at UCD for digestate, biomass and culture analysis, e.g. ICP, HPLC, GC, flow cytometer.

#§QUA-LIT-QL§# #§WRK-PLA-WP§#

3.2 Capacity of Participants and Consortium as a Whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

SusAlgaeFuel will foster an international, multidisciplinary and multisectoral partnership between three universities, one research organisation**,** and five SMEs across five European countries (Ireland, Germany, Spain, Portugal, UK) (**Figure 3.2a**), with inclusion of relevant industry stakeholders providing direct pathways to commercialisation. The consortium also counts on the participation of Boeing and CityJet in its Advisory Board.

University College Dublin (UCD) is Ireland's largest university with a global research portfolio in sustainability and circular bioeconomy**.** UCD has dedicated research support units for administration, finance, legal, HR, ethics, and IP management (NovaUCD), all of which will be availed for SusAlgaeFuel. Dr. Halim's Group is currently leading several projects to harness the power of microalgae in valorising waste products from agri-food sectors (dairy processing, brewery and AD operation). The flow cytometer methods described in WP2 and autolytic incubation proposed in WP3 have been directly derived from the group's findings. The Halim Group will transfer their technical know-hows in microalgae wastewater treatment, bacterial analysis and biorefinery processing to WPs1-3, WP6. The Group has access to core infrastructure in microscopy, flow cytometry, chromatography. Dr. Syron's Group has expertise in upscaling research for commercial deployment through the spin-out OxyMem. They will apply this knowledge to process integration in WP5.

Figure 3.2a: Partner's Locations

ALGAFUEL SA (A4F), Portugal, is a bioengineering company with over 20 years of experience in the Design, Build, Operation and Transfer (DBOT) of microalgae biotechnology. A4F has an extensive track record in microalgae, including the installation of the largest tubular photobioreactor units in Europe and successful coordination of FP7 & H2020 projects in microalgae biofuels (e.g. BIOFAT). A4F will apply their extensive technical know-hows in the production of microalgae at lab and pilot scales under autotrophic regimes to WP1 and WP6.

Bia Energy Ltd (BIAE), Ireland, is a commercial AD operator with the capacity to process 99 ktonne p.a of organic waste for the production of 4.8 MW of renewable electricity. Dr. Nolan will lead BIAE's participation and provide his expertise on AD systems optimisation. SusAlgaeFuel will benefit BIAE by providing a potential commercial outlet for their CO2-rich effluent (from biogas upgrading) and digestate liquid fraction, both of which are significant pain points in their operation. There is currently no suitable outlet for the CO_2 effluent in Ireland, while the significant volume of digestate generated from biogas production (ca. 100 ktonnes p.a.) requires considerable land bank to adhere to EU Nitrates Directive (91/676/EEC) and thus accrue significant cost to the business. BIAE will contribute anaerobic digestate (200 kg p.a.) and provide the site for the construction of pilot-scale algal production facilities.

IRIS Technology Solutions, S.L. (IRIS), Spain, is a manufacturer of real-time process monitoring solutions specialising in spectroscopy and artificial intelligence technology for engineering customised solutions for inline and online applications to transition food processing, chemical and biobased industries to Industry 4.0. IRIS has developed its own line of SWIR (shortwave infrared), hyperspectral imaging and Raman devices, in addition to data management and predictive software (Scadalytics). IRIS will use their multidisciplinary expertise in optical engineering, chemometrics, electronics, and data science for the development of smart sensors in WP2 and WP6.

Karlsruhe Institute for Technology (KIT) (Dr. Frey), Germany, has been involved in H2020 and national projects on cell-component recovery from microalgae (e.g. Sabana). The team will transfer their expertise in PEF treatment and microalgal biorefinery systems to WP3 in SusAlgaeFuel. The lab is fully equipped to accomplish the work, including bench-scale downstream processing auxiliaries (centrifuges up to 12 L capacity, disc separator STC 3-06-107, HPH and bead-milling, freeze dryer), and diverse PEF-facilities (0.5 L/h up to 1,000 L/h).

The University of Cambridge (UCAM) (Dr. Marek), UK, will apply their expertise in fuel synthesis and technologies on the development of algae HEFA-SPK route in WP4 and WP6. UCAM has the infrastructure needed to support WP4 activities and analysis for WP6, including two continuous high-pressure reactors and a batch reactor vessel operating up to 50 bar and at high temperatures, a separate rig for chemical looping production of H_2 and operation of up to 10 g of Fe-based catalyst, and material characterisation facilities (STEM, BET, XRD, TGA).

Fraunhofer Gesellschaft is Europe's largest research organisation (Germany). The **Umwelt-, Sicherheits- und Energietechnik (UMSICHT)** division (Dr. Apfelbacher) will apply their expertise in fuel transformation of lignocellulosic biomass by TCR-HDO (TOSYNFUEL project) on microalgae biomass residue in WP4 and WP6. UMSICHT houses specially built TCR reactors for hydrogenation at 400°C/200 bar at various scales (up to 500 kg/h).

Sustainable Flight Solutions Ltd (SFS), Ireland, was founded in 2020 to promote the decarbonisation of the aviation sector through the development of SAF markets. SFS is perfectly positioned to lead the sustainability assessments in WP5 and disseminate project results, with an extensive network in the aviation industry, comprised of SkyNRG, Avolon, Boeing, Orix Aviation, various Irish airlines and airport authorities.

European Science Communication Institute gGmbH (ESCI), Germany, is a not-for-profit organisation that empowers scientists to communicate with a non-expert audience using a message that not only tells, but also sells. They have accumulated significant experience in science communications, particularly in microalgae, through participation in various H2020/HE programme. They will lead the communication strategy of the project (WP7).

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